History of changes part A	
Generation of Part A.	20/08/2019
Breakdown of periodic deliverables in WP7 and modification of due dates to match the ones of the periodic reviews.	20/08/2019
Modification of due date of deliverables in WP5 and minor typos changes in the Gantt chart have been updated	20/08/2019
Addition of 3 new deliverables regarding the Communication Kit obligations due to the co-financing of Photonics21.	25/09/2019
ThingsO2 included as partner. ThingsO2 is now main beneficiary of deliverable D5.1.	26/09/2019
Task 7.3: “VIP parties” replaced by “VIP events”	30/09/2019
The role of MRA and TiO2 included in task T1.1 description.	30/09/2019
References to missing figures 27-30 included in the workplan description	30/09/2019
WP5/T5.2-T5.3/deliverables and MS5 shift from M30 to M28	04/10/2019
History of changes part B	
Generation of Part B.	21/08/2019
Table in section 3.4.1 has been corrected including LTP person-month and correcting errors in PMs distribution according to the budget in annex 2.	21/08/2019
Mention to emerging quantum-dot enhanced IR image sensors has been included in the SoA – Section 1.4 - Spectral imaging and sensing in industrial applications	21/08/2019
More detail justification of performance improvements for yield and wastage has been included in the tables of Section 1.3.1.4 corresponding to each use case.	21/08/2019
IMEC’s other direct cost has been modified according to Part A including missing large research infrastructure cost. Other direct cost of Senorics (Travel), and MRA (Total cost) have been corrected in the table in accordance to budget in annex 2. Jotis (Other goods) has been corrected in table including the missing registration fees (3,000€) concept according to budget in annex 2. Further explanation of SENORICS costs breakdown (Other goods) presented.	30/08/2019
Addition of communication requirements under section 2.2	25/09/2019
Update of linked third parties information in Sec 4.2 (CRS, SCM España, Mobiliario Royo)	26/09/2019
ThingsO2 included as partner	26/09/2019
TiO2 mentioned in Section 3.3	30/09/2019
Missing implementation Figures 27-31 included in Section 3.1.3	30/09/2019

Resources to be committed by L3Ps added in section 4.2	02/10/2019

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1 Excellence

MULTIPLE will bring together ***snapshot mosaic filters, OLED-based sensors,*** and state-of-the-art machine learning to deliver ***breakthrough and cost effective snapshot hyperspectral imaging*** and ***spectrometric solutions*** covering a broad spectral range and suited to actual industrial monitoring and control needs. MULTIPLE multimodal monitoring systems will be IoT native, exploiting ***open source cloud, big data, and deep learning*** technology. A fast orchestrated deployment of ***data-driven AI-based models*** will foster production optimization.

The affordability of monolithic uncooled CMOS and CCD sensors sensitive in the visible (VIS) and part of the near infrared (NIR) range has fuelled the spread of machine vision and photonic sensors throughout machinery and production lines. They are necessary elements of any automated production line we can imagine, monitoring a wide variety of magnitudes and providing feedback for optimising manufacturing processes and product quality. Europe has played a clearly leading role in the field with key suppliers of sensors (e.g. AMS), cameras (e.g. Basler, JAI, PhotonFocus), software (e.g. MVTec) and a vibrant network of SMEs and midcaps delivering turnkey solutions for process monitoring and control to virtually any manufacturing sector.

While hybrid and cooled sensor and FPA technologies (e.g. InGaAs, InSb, MCT) have been traditionally regarded as expensive and bulky, the high demand and technical advances on InGaAs FPAs have cut down their cost making them attractive for the industrial vision market. Relevant camera suppliers in this segment like PhotonFocus already market InGaAs cameras with tag prices under the 10,000€ barrier, with the expectation of further cuts in the short-medium term. Besides, new technologies like OLED sensors are set to make a breakthrough in terms of cost and performance and have already shown a big potential in chemometric applications.

The last years have seen a research boom in hyperspectral imaging (HSI) and spectroscopy measurement technologies, especially in a spectral range that spans the visible, short-wavelength infrared (SWIR), and mid-wavelength infrared (MWIR). This part of the optical spectrum has proved as a privileged proxy to material properties, especially suited to non-destructive monitoring. It avoids the damage and risks of ionising radiation, and also the environmental noise of thermal radiation. Compared to widespread panchromatic or trichromatic systems, spectrally resolved technology has demonstrated far higher accuracy and specificity, key to address more and more scenarios of industrial monitoring and control, across many different sectors.

This context explains the optimistic forecasts on the hyperspectral imaging systems market, projected to reach USD 21.33 bn by 2023 from USD 8.14 bn in 2017, at a strong CAGR of 17.5%[1]. Hyperspectral imaging -i.e. sensing with a crowd of finer spectral bands- has been associated to bulky and expensive spectrometers only capable of push-broom imaging, not suitable to take snapshots, while the term snapshot has been largely associated to a few spectral bands, with configurations other that Bayer-based or filter-based solutions both bulky and expensive.

Mosaic pixel-level filters by IMEC have set a breakthrough in the field, enabling snapshot hyperspectral imaging in the visible range first, and in the SWIR range more recently. This technology -as a coating process- is suitable for volume production and does not alter the compactness of imagers. That is, it is suited to deliver ***low SWaPC*** (size, weight, power and cost) solutions.

Moreover, the maturity of computational imaging for dual and multi-aperture optics -already empowering many smartphone cameras- brings a clear opportunity. Combined with the feasibility of cost effective snapshot HSI camera cores in the SWIR and VIS range, it sets the conditions to deliver novel snapshot HSI VIS/SWIR imagers with killing price and capabilities. The promise of bringing to the market broadband ***highly selective hyperspectral imaging*** covering what is likely the most important spectral range for process and product monitoring is only a collaboration away. A collaboration of few European organisations.

The use of low cost high performance ***OLED sensors***, together with advanced laser-based concepts, in the context of chemometric analysis offers clear opportunities to deliver novel on-line and in situ analysers in an even broader spectral range with a big impact on process and product quality.

The development of models for measurement and classification is still prone to handcrafted engineering workflows (costly and slow). Desktop software libraries and small sets of data are the main and most usual resources used for modelling, building, and testing. When faced to high dimensional data (e.g. hyperspectral imaging) this burden is likely to discourage the development effort. It is for a reason that most results reported on HSI have an academic source and few commercial solutions are available yet compared to the wide adoption of standard vision systems.

The recent success of ***data-driven*** approaches and ***deep neural networks*** -especially for image analysis- enables a faster development of models with better regression, classification, or segmentation accuracy. While the scarcity of suitable data for training has remained a barrier for adoption, new cloud-based architectures have demonstrated the capability to collect, store, and annotate big data, enabling the training of new models in a fraction of time.

[1] Markets&Markets. Hyperspectral Imaging System Market 2017-2023.

Moreover, the orchestration of large sets of data sensors in other sectors -like video surveillance for security or data analytics in omnichannel retail- are successfully exploiting ***cloud*** services ***and big data technology*** for data collection and analytics and large scale training and deployment of deep neural models.

Current monitoring systems mostly work in isolation to monitor or control a given specific operation. The combination of last edge technology and cloud-services for the ***orchestration of sensors*** (e.g. HSI sensors), big data collection and analysis, and large-scale training and deployment of deep neural models on the edge, enables the integral management of data and their use for process optimization and analytics.

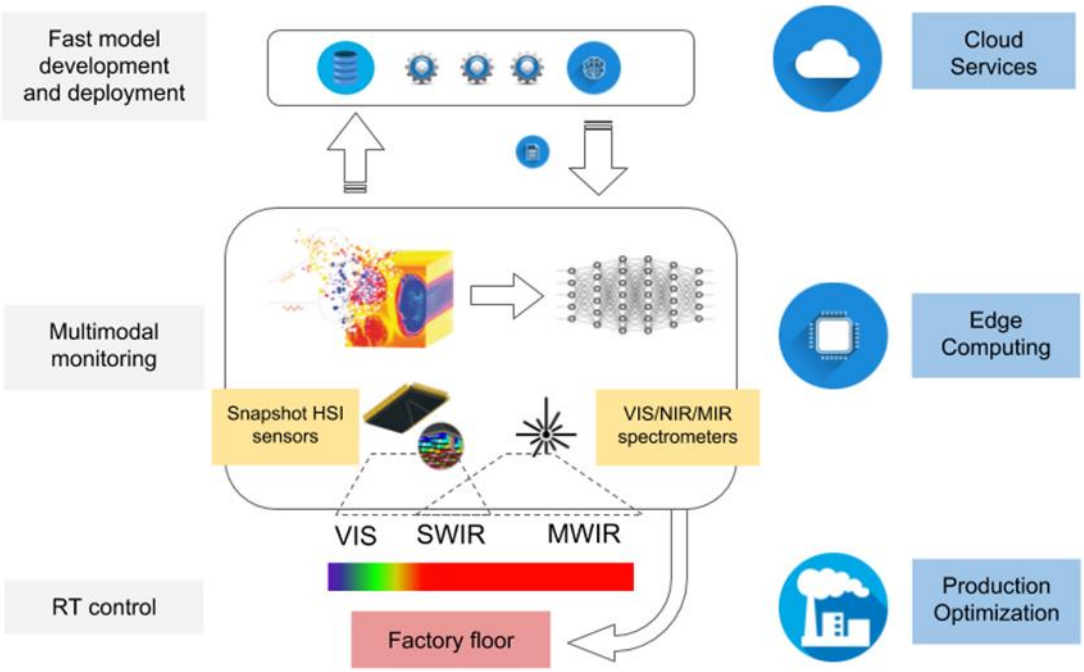


Figure 1. MULTIPLE concept.

MULTIPLE will deliver breakthrough and ***cost-effective snapshot HSI cameras*** in the broad ***VIS/SWIR range,*** complemented with cost effective ***OLED-based and laser-based chemometric analysers.*** On top of these sensors, MULTIPLE will develop novel multimodal monitoring systems that will be IoT native, by combining them with open source cloud, big data, and deep learning technology. An agile development and orchestration of complex AI-based models that optimize production will accelerate innovation schemes in relevant manufacturing scenarios.

1.1 Objectives

The main goal of MULTIPLE is the development of cost effective multimodal monitoring solutions with a breakthrough impact on production quality and efficiency. Key technology features pursued by MULTIPLE are a broader spectral range with snapshot hyperspectral capabilities, and an IoT native approach to the development and deployment of machine learning models (i.e. statistical and deep learning models) for overall monitoring and control of production lines.

With this aim, MULTIPLE will pursue the following technical goals:

1. The development of cost-effective HSI SWIR (0.9-1.7µm wavelength) camera cores ready for volume production, covering a missing offer highly demanded by the market.
2. The development of dual aperture snapshot HSI imagers combining snapshot CMOS and InGaAs hyperspectral sensors based on mosaic filters, delivering breakthrough snapshot hyperspectral imaging (HSI) capabilities in the whole VIS/SWIR range (0.4-1.7µm wavelength).
3. The development of novel spectrometric solutions for accurate highly specific chemometric analysis in an extended VIS/MWIR (0.4-5µm wavelength) range. Innovative concepts in OLED sensors, laser-based spectroscopy, and signal processing will be combined in MULTIPLE to deliver new spectrometric solutions with breakthrough sensitivity and performance.
4. The development of embedded deep models for regression, classification and real-time control, running on latest System-on-Chip (SoC) technology combining ARM processors and GPU/TPU. The aim is to provide onsite -in the factory floor- and online -during the manufacturing process- measurements of physical magnitudes (e.g. colour, temperature) and classification of different types of categories (e.g. material composition, quality grades) with high flexibility, accuracy, and reliability. Moreover, these observations will be used to monitor and control both specific process operations and production flows.
5. The development of an integral and scalable approach to the orchestration and big data collection of photonic monitoring systems under an IoT native approach. The aim is to leverage state-of-the-art open source cloud and

big data technologies to create a new set of cloud services that facilitate and boost the development, deployment and orchestration of new interrelated monitoring systems and complex decision support tools to optimise manufacturing production under multiple goals (e.g. maximum quality, lead-time, cost efficiency, waste reduction, energy savings).

6. The validation and demonstration of the proposed approach in very different manufacturing scenarios: woodworking, steelworking and food industry. Such scenarios are representative of typical schemes of both continuous processes happening along a complex production line, and multi-stage piecewise manufacturing processes. They reflect a wide set of magnitudes and features that make the case for multimodal monitoring.

Objective	Verification	Timing	Milestone
O1. Developing cost-effective snapshot HSI camera cores	• Suitable for machine vision volume market • VGA (640-512 px) resolution • 350 fps • up to 32 spectral bands • Spectral bandwidth (FWHM) <10nm • Tag price under 10,000€	M18	MS3
O2. Developing VIS/SWIR snapshot spectral imagers	• Dual aperture • VIS/SWIR range coverage (0.4-1.7 µm) • Up to 64 narrow bands • Tag price under 15,000€ • Spectral bandwidth (FWHM) <15nm • VGA resolution	M18	MS3
O3. Developing novel multimodal monitoring modules	• Colorimetry robust to lighting conditions • 2ΔE2000 colour differences (JND) • SWIR thermography independent on emissivity (2% accuracy) • Highly specific VIS/MWIR spectroscopic solutions combining versatile OLED sensors with laser-based approaches for chemometric analysis of gases and food and wood chemical constituents • Classification and regression of HSI images for quality grading and parameter estimation • Segmentation of HSI images for enhanced detection of defects and foreign bodies	M18	MS3
O4. Orchestration of monitoring systems	• Smart IoT monitoring/control devices in compliance with industry 4.0 and vision standards • Embedded implementation classification, regression, and segmentation • Cloud services for data ingest, aggregation, search, and visualization and fast model development and deployment	M24	MS4
O5. Integrated process optimization	• Production modelled and parameterised • Data-driven analytics and decision support system working • Fully traceable quality control throughout the production lifecycle • Process optimization in terms of quality, resource efficiency and speed	M30	MS5
O6. Demonstration in operational environment	• Steelworking: 15% Production performance, 10% overall efficiency, 10% quality improvement, 15% CO2 and NOx emissions reduction • Woodworking: 10% reduction on adhesive consumption, 15% shorten time to market, 10% resources and energy savings • Food: 100% quality control of production, 100% detection of foreign bodies, 100% detection of chocolate adulteration	M36	MS6

1.2 Relation to the work programme

MULTIPLE addresses the H2020-ICT-2019-05 call for the Photonics topic, *Application driven Photonics components*, focusing on the *Sensor-Based Optimization of Production Processes*. The alignment of the project with the goals of the call can be summarised around the following five points explicitly set in the topic description.

Broadband. MULTIPLE monitoring systems will work in a broad spectral range from 0.4 to 5 µm wavelength. Notably, MULTIPLE imagers will build on breakthrough snapshot hyperspectral sensors developed by IMEC in a

wide range that covers the full visible and SWIR ranges, from 0.4 to 1.7 µm wavelength. Mosaic filters in the VIS/SWIR range will provide breakthrough snapshot hyperspectral-imaging capabilities unique in the market. Cutting-edge OLED sensors and novel concepts on laser-based spectroscopy over an extended VIS/MWIR range will complete the offer of powerful spectroscopic highly specific capabilities, compliant with industrial operating conditions.

Multimodal. The information carried by photons in these spectral bands will be extracted and will cater to a variety of models that will estimate different parameters, measures, and quality grades. From hyperspectral estimation of mechanical properties of wood products, multi-wavelength high temperature thermography in the SWIR range to chemical composition estimation in the MWIR, surface and volume geometry or spectrally-resolved colorimetric assessment in the visible, among many other measurements and characterizations that are relevant throughout production lines and production stages.

Smart. The use of state-of-the-art machine learning technology -with an emphasis on deep convolutional networks- to build and train models that run embedded inside the monitoring systems ensures the highest possible device intelligence, and a timely feedback to act on the process, adjusting process parameters and production flows. The capability to handle data at scale -relying on elastic search and advanced visualization of big data- and to easily train and deploy models will enable new more efficient and powerful workflows involving development engineers of integrators and equipment suppliers, and production engineers of manufacturing end-users.

Cost-effective. MULTIPLE will bring cost effective and proven camera cores suitable for volume production that caters the mainstream machine vision market, led by PHF -a well-established and worldwide renowned supplier of cameras and camera cores for industrial applications. Camera cores with a maximum tag price of 10,000€ will ensure a fast adoption in a market-demanding capabilities only achievable with higher spectral specificity and coverage. The like can be told of OLED and laser-based spectroscopic solutions, built on inexpensive sensors. In order to make the final solutions truly affordable, cost effectiveness is enforced throughout the whole value chain. At the modelling level, MULTIPLE IoT native approach will bring services that cut down and scale development and deployment cost, typically the highest and most uncertain and risky component of any industrial innovation project budget. Changes in production (e.g. in materials or component features and quality grades) will be met in most cases by simple procedures to retrain and deploy models.

Process integrated. By explicitly addressing the orchestration of smart monitoring systems throughout the production line or chain, MULTIPLE ensures that optimization is pursued at the relevant production levels. The creation of complex interrelated models becomes feasible, by gaining awareness on both upstream and downstream conditions and issues. Moreover, this approach will facilitate the identification and correction of bottlenecks, issues and their causes, at a much larger scale than current industrial innovation practices.

1.3 Concept and methodology

MULTIPLE concept is based around three main pillars: i) Multimodal spectral sensors; ii) Orchestration of monitoring solutions; iii) Integrated process optimisation. MULTIPLE will be implemented and assessed in three different application domains in steelworking, woodprocessing and food, addressing product and process optimisation, in real manufacturing conditions.

1.3.1 Concept

MULTIPLE will combine the latest advances and concepts in snapshot hyperspectral imaging, OLED and laser-based spectroscopy, multimodal imaging, embedded processing, machine learning for classification and regression, and cloud-based model prototyping and deployment and analytics. The approach adopted will ensure the development and demonstration of a comprehensive set of novel monitoring solutions with high functionality and a cloud system for the orchestration of the whole monitoring network, fostering data-driven process optimization and innovation. The solutions proposed will feature:

- Novel industry-grade **hyperspectral imaging SWIR camera cores**. These camera cores will be cost effective with a tag price within reach of industry budgets, fostering innovations in the field of product and process monitoring.
- Cost-effective, integrated, and configurable dual aperture **snapshot hyperspectral imager** -spanning the visible, NIR and SWIR ranges- that will be stressed to provide multimodal imaging capabilities -spatially resolved in real time- in demanding industrial settings.
- Novel versatile and low-cost **OLED sensors and laser-based spectroscopy** will be applied over an extended range from VIS to MWIR for low cost and highly sensitive chemometric solutions, which will reinforce **multimodal measurement** capabilities, revealing spectral signatures of chemical constituents in different scenarios.
- **Real-time embedded edge computing**, ensuring timely feedback to control processes and production flows. SoC ARM and GPU/TPU processing platforms will be stressed to run classification and regression models

in real time.

- A **scalable orchestration of monitoring systems** based on cloud microservices, cutting down modelling and edge deployment effort and time and enabling process innovation and optimization at scale.

The MULTIPLE project will carry out four main blocks of activities that deal with:

i. **Multimodal spectral sensors**
ii. **Orchestration of monitoring solutions**
iii. **Integrated process optimisation**
iv. **Demonstration**

1.3.1.1 *Multimodal spectral sensors*

HSI SWIR camera cores

Spectral imaging is an imaging technique that captures and processes multiple narrow band images over a spectral range. It enables detailed analysis and identification of objects as they contain unique information at different wavelengths. There are three main methods of acquiring hyperspectral imaging data: spatial scanning, spectral scanning, and snapshot imaging. Snapshot sensors/cameras collect the entire 3D datacube in a single shot without scanning. As a result, they are free of scanning artefacts, far more robust and compact, and benefit from higher optical efficiency.

A prominent architecture with the potential to be compact is the spectrally resolving detector arrays (SRDA) also called snapshot mosaic cameras. IMEC's snapshot mosaic sensor technology is a unique compact and low cost solution, capable of real-time spectral imaging data acquisition.

After the commercial success of the snapshot HSI CMOS sensor in the visible range, IMEC has been working in translating the technology to the SWIR range by applying a tailored variant with InGaAs. As a result, IMEC has come with a cost effective and high performance FPA solution for HSI imaging in the SWIR range. MULTIPLE will integrate this SWIR HSI FPA in a new camera core following the strong standards of PHOTONFOCUS camera platforms and featuring all the preprocessing functionalities for noise reduction and correction and efficient data transmission that have made PHOTONFOCUS cameras a success in the market of industrial cameras.

MULTIPLE will integrate this sensor technology in a new cost effective and high throughput camera core featuring the high quality and robust image pre-processing and interconnectivity that can be found in any commercial camera core marketed by PHOTONFOCUS, one of the main world suppliers of industrial vision cameras. All the necessary developments will be put in place to ensure the required cost effectiveness, robustness and connectivity, in order to meet the demanding requirements of industrial machine vision volume market.

Dual aperture VIS/SWIR snapshot hyperspectral imaging

Over the last decades, computational imaging technology has become a key part of most consumer cameras in any format. Since the first models featuring multi-exposure HDR and stitching-based panoramas, more and more smartphone cameras are embracing dual aperture and even multi-aperture approaches to boost camera performance. This technology has been also used in multispectral imaging, mostly combining several sub-bands of the spectral range of CMOS sensors.

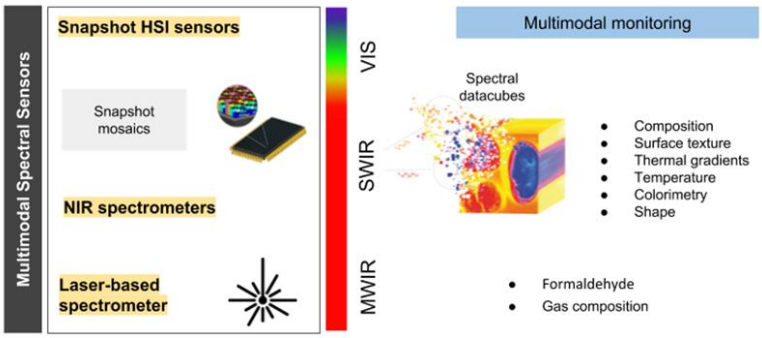


Figure 2. Multimodal spectral sensing.

Exploiting a dual aperture approach -one aperture for the visible range and the other for the SWIR range- and embedded capabilities for computational imaging, MULTIPLE will build a new camera capable of snapshot computational imaging in a broad spectral range, from 0.4 to 1.7 µm wavelength. Following this approach, optical performance -a key factor of HSI performance- may be easily and flexibly optimized for each aperture. Moreover, this will be possible with off-the-shelf and compact optical components, at a fairly low-cost. This is key to keep the final imager cost effective and compact, unlike bulky and expensive prism-based optics.

Therefore, MULTIPLE will bring a compact and cost effective broadband dual aperture camera, capable of snapshot hyperspectral imaging in the VIS/SWIR range.

Spectroscopic analysers

Spectrometric technology for chemometric analysis has gained popularity thanks to its non-destructive nature and

the capability to provide highly specific and accurate measurements, quickly and onsite. However, the SWIR and MWIR range are still synonyms of either sensitivity issues or expensive equipment. MULTIPLE will combine a recent breakthrough in tunable OLED sensors with highly specific laser-based spectroscopic schemes to a variety of chemometric analysis in the VIS/MWIR range (0.4 µm - 5 µm), from gas concentration measurement to the detection and concentration measurement of constituents in food and wood products.

Overall, MULTIPLE puts forward a complete set of cost effective and compact multimodal monitoring systems, capable to probe the very specific information carried by photons in different spectral bands spanning a broad range from 0.5µm to 5 µm wavelength.

1.3.1.2 *Orchestration of monitoring solutions*

In order to manage the big amount of data from the spectral sensors proposed and to stress the most of their potential, **MULTIPLE** will develop smart photonic -IoT native- monitoring devices. These **edge devices** will implement **novel real-time processing** pipelines based on Deep Learning, enabling processing of high dimensionality data like **hyperspectral datacubes** for different operations in the manufacturing industrial environment, with minimal engineering intervention.

Each monitoring system will be equipped with an AI capable edge computing device based on latest system-on-chip boards (e.g. SoC ARM/GPU/FPGA), providing functionalities for data collection, pre-processing, AI inference, control and ubiquitous communications.

Communication with other monitoring solutions, field devices or production management applications and services will be provided through standards defined in Industry 4.0 (RAMI4.0 model[2]). e.g. the adoption of OPC-UA[3] will allow machine-readable, semantically unambiguous self-description of properties and capabilities, together with secure machine-to-machine communication and data exchange.

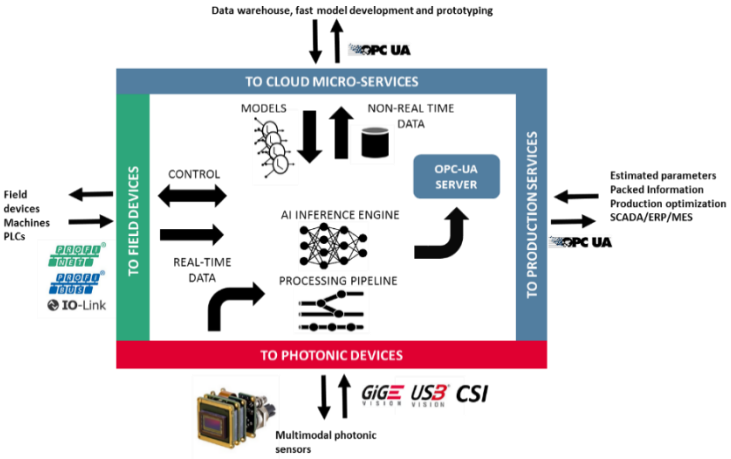


Figure 3. Smart photonic IoT MULTIPLE monitoring devices.

The embedded system will use the extracted information to make decisions, e.g. actuating over the field devices in closed-loop control, or to feed the production management services only with a meaningful and relevant stream of features. The core of the processing pipeline will rely on an AI inference engine enabling the deployment of high performance deep neural models using seamless cloud services. This approach will foster the re-configurability capabilities of the monitoring system reducing the engineering efforts when facing a new inspection problem, replacing the need of custom algorithm re-programming and the big challenge of manual feature extraction by simply training and deploying a new deep model.

Going beyond current machine vision and spectral monitoring solutions -which are focused on monitoring a specific operation at a specific point-, AI models will be trained to recognize anomalies in data, and continuously improve the performance. Moreover, the increased capability of data generation (e.g. hyperspectral imaging sensors) and deep learning requires mechanisms for supporting scalable data management (i.e. data ingest, analysis, and storage), demanding flexible cloud architectures for a fast development and retraining of models.

[2] https://www.plattform-i40.de/I40/Redaktion/EN/Downloads/Publikation/hm-2018-manufacturing.pdf?__blob=publicationFile&v=2
[3] https://opcfoundation.org/about/opc-technologies/opc-ua/

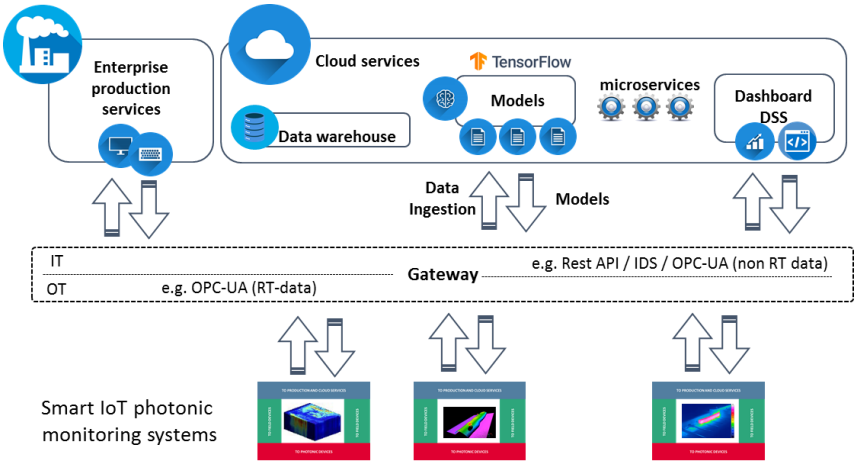


Figure 4. MULTIPLE cloud-based service oriented architecture supporting the orchestration of monitoring solutions and fast development of deep models.

Based on a micro-services architecture in the cloud and existing open source platforms, MULTIPLE pursues:

- Providing a **scalable data warehouse** to ingest, store, and manage (i.e. search and analysis) heterogeneous photonic multimodal data and other production data, and the capability for real-time detection of anomalies.
- Enabling the **fast development of deep models** and inter-related models combining data from different monitoring devices, and production floor observations (i.e. annotations).
- Providing the infrastructure **to seamless deploy** such deep learning **models in the edge computing devices** in an industrial environment for a granular monitoring and control of production.

1.3.1.3 *Integrated process optimisation*

MULTIPLE will establish an integrated production optimization framework allowing flexible control of the production line, applying real-time control at specific critical points (e.g. PID controller at machine level, automated production flow rerouting based on quality checks), as well as monitoring the overall production efficiency providing meaningful information (e.g. energy/material consumption, costs) for decision making and feedback to update production models and KPIs (Figure 5).

Relying on the information and data collected and stored in the scalable cloud data warehouse -raw and real-time data processed from the photonic monitoring solutions developed in MULTIPLE, together with other data coming from different sources already present in the factories (e.g. PLCs, SCADA, ERP)- and the power of the open source Elastic Search Engine and cloud micro-services, MULTIPLE will provide effective tools to analyse data and compute KPI and enable product quality and consumption traceability.

Such solution will facilitate to locate fault sources providing decision support feedback for adjustment and correction, reducing intervention times, and alerting production managers. The integrated system will apply big data analytics to have a better understanding of complex production processes involved in the production lines. This will improve the accuracy of the simplified models used for real time control and finding more accurate, but complex, deep models, or re-defining KPIs.

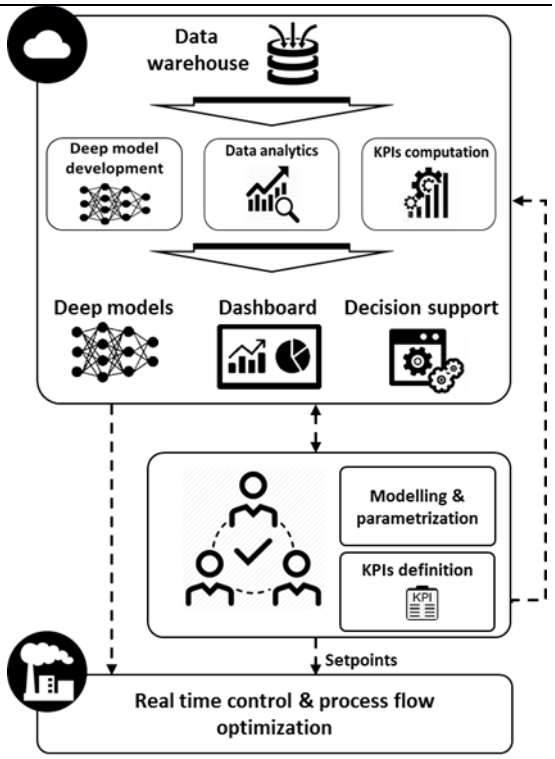


Figure 5. Integrated process optimization layout.

1.3.1.4 *Demonstration*

PRODUCTION OPTIMISATION OF STEELWORKING PROCESSES (LDL):

MULTIPLE will deploy and run a demonstration in a modern rolling mill production plant of LDL located at the Bayonne harbour (France). It produces a variety of steel plates including normalized fine-grain steels, weather-resistant and tempered steel used in heavy construction (e.g. wind farms, shipbuilding), mechanical engineering and pressure vessels applications. Currently the mill produces plates with 8-150 mm thickness, 1850-2500 mm width and up to a maximum length of 24 m. The process on the LDL factory has three steps:

i. *Reheating the feedstock*. Steel labs are reheated in a gas-fired furnace of 30m long where the temperature changes from room temperature up to around 1150ºC. This stage makes the steel softer and more malleable.

Reheating sequence requires different configurations depending on the specific steel quality and differs from one batch to another.

ii. *Reversible rolling mill stand*. The slabs are pushed in a stand that consists of two cylindrical steel rolls, where the steel is passed back and forth several times. In each rolling pass, the cross section of the steel slab is reduced, and the product is elongated by controlling the distance between the cylindrical steel rolls. The final thickness of the steel plate is proportional to the number of passes carried out during the rolling process that depends on the mechanical characteristics of the respective steel quality at the rolling temperature.

iii. *Cool down*. The steel plate is cooled to ambient temperature at a controlled cooling rates in order to ensure the final characteristics of the steel plate. In every step of the process, the quality of the feedstock, the raw material, is taken into account to achieve the required quality of the final steel product.

The reheating furnace represents approx. 70-80% of the total energy consumption of the plant, and the fuel consumption of the furnace up to 15% of the operational cost of a rolling process. Reduction of reheating costs is one of the major challenges in rolling mills. Currently, the reheating furnace in LDL is controlled based on predefined recipes defined by the operators (oxygen/fuel ratio, temperature ramps, holding time, etc.) which must respond to changes in product size and mix. Finding the optimal heating conditions is hard due to unexpected variabilities in the input material composition. Besides, reheating the steel to a uniform temperature is critical to that deformation does not occur. Variations of the steel slab temperature along the rolling mill stand due to production delays and/or low reheating temperature due to unexpected variations also requires dynamic adaptation of the loads and the rolling sequence.

MULTIPLE will improve the online monitoring capabilities inside the reheating furnace and during the reversible rolling mill. Novel accurate and robust multi-gas analyser will be deployed to provide in real time measurements of gas constituents at several locations inside the furnace and chimney. Hyperspectral SWIR imaging will provide temperature maps of the steel slab and furnace walls independent on emissivity even in the presence of dust and steam. These measurements, combined with other sensors data (e.g. steel chemical composition, geometry, fuel pressure and rate), will be used to optimize the operation point of the furnace and automatically adjust the reheating sequences and loads at the rolling mill, avoiding scraps and defected parts, and ensuring the end product quality (Figure 6).

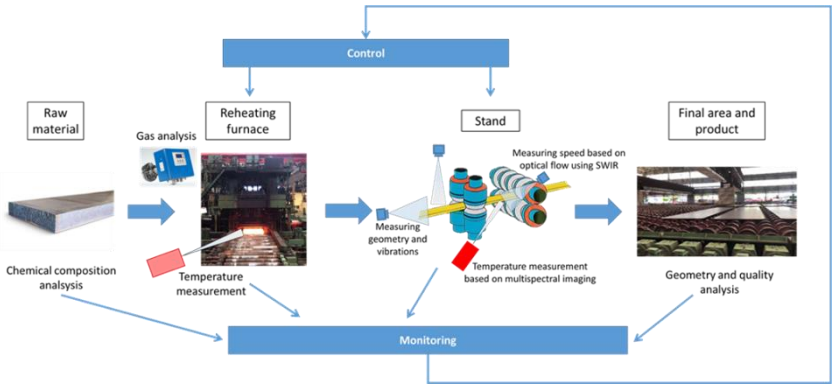


Figure 6. Rolling mill production line.

KPI	MULTIPLE improvement	Justification
Improved performance	Improvement in overall efficiency: 10%	Increase performance of the global plant thanks to an integrated optimization of the reheating and rolling processes[4].
Production performance	Reduction in delays due to low temperature of the steel: 15%	Improved heating strategy to get the right temperature of the rolling slabs by using multispectral SWIR imaging at different stages and integration with the control system
Energy and resource consumption	Energy savings in reheating furnace: 10%	Automatic set point calculation and real-time control for each production rate and quality mix. Online analysis of fuel content and integration with PLC. Use of gas analysers at the furnace outlet and chimney to establish optimum energy balance. Reduction of the excess air, prevention of leakages and heat losses from walls and doors with thermography.
Waste or by-products production	Quality improvement: 10%	Improved heating strategy of slabs leading to reduced scrap. Detection of detrimental defects with HSI cameras at the earliest stages before further processing and semi-product reallocation.

[4] Kilinç, E., Kaya, D. et al. "An Energy Efficiency Analysis of an Industrial Reheating Furnace and an Implementation of Efficiency Enhancements Methods". Energy Exploration & Exploitation 2014, 32(6), 989–1003.

CO2, NOx emissions	15% reduction	More efficient heating returns and improvements in stoichiometric ratio. Online analysis of fuel content and combustion with a multi-gas analyser and integration with the control system.

PRODUCTION OPTIMISATION FOR WOODWORKING AND FURNITURE (ROYO):

ROYO is a leading bathroom's furniture company with headquarters in Valencia (Spain), and presence in Poland, Mexico, the United States and India. Currently they produce more than one million pieces of furniture every year. The overall production process is summarised in the figure below:

Figure 7. Wood furniture manufacturing production line.

- Sizing: Input material (particle boards -chipboard, OSB, LSL-, fibre boards -MDF-, wood-based panels – plywood, LVL-) is cut to achieve the required shapes and sizes
- Drilling: Making blind and through holes to connect furniture parts
- Edge-Banding: The wood-based panel is covered by gluing a nobilitated layer (paper, plastic, etc.) increasing durability and giving aesthetic appearance
- Sanding: Belt-sanding is used to decorate the panel surface via selective abrasion which allows to obtain diverse patterns like honeycomb, etching, etc.
- Painting and lacquering: Applying coloured wood finishes that dry by solvent evaporation or a curing process
- Assembly and packaging: Each component is being assembled according to design. All joints are glued together, wiped dry, coated and inspected before being packed.

The vision of ROYO is to create products that can be used globally, but with local needs also considered. This implies the need to strive for flexible production and lot-size-one approach, which in turn calls for ensuring that production quality is constantly monitored in order to avoid defects that can be easily propagated throughout the process chain with consequent great damage in terms of: i) unwanted production delays reducing customers satisfaction, ii) raw material waste affecting the overall process environmental sustainability, iii) worker's time used unproductively with consequent economic damage for the company. In addition, one of the major concerns is to control the carcinogenic formaldehyde emissions during the manufacturing process and in the final product, mainly due to the use of adhesives, e.g. urea and melamine glues. Monitoring formaldehyde emissions inline is of high interest to reduce the time required for laboratory analytical methods, while ensuring a safety working environment and final products in compliance with the European regulation.

MULTIPLE will optimize the production by integrating a flexible and end-to-end quality control system, enabling the quality traceability from input material to the final product (e.g. applying VIS/SWIR HSI for estimating the moisture content, elasticity and density of the input material, analysing geometry defects, size and position of holes during drilling, or detecting excess of glue and applying colorimetry during painting and lacquering). The extracted information will be used to apply closed loop control strategies in some machines improving performance, and for re-routing defected components reducing material waste. In addition, MULTIPLE will deployed a real time inline gas analyser for measuring formaldehyde emissions.

KPI	MULTIPLE Improvement	Justification
Improved performance	Improvement in overall efficiency: 10%	Increase performance of the global plant thanks to an integrated end-to-end quality control system based on multi-spectral vision and deep neural networks.
Production performance	Shorten time to market: 15%	More accurate and flexible monitoring and quality control system able to cope with new product variations. Faster deployment of new models thanks to an edge/cloud infrastructure and integration of new monitoring nodes by plug-and-play.
Energy and resource consumption	Resource consumption: 10%	Optimizing and controlling the operation setpoints of the machines, reducing the use of material (e.g. polyurethane glue), while reducing risks for workers (formaldehyde emissions)
Waste or by-products production	Quality improvement: 10%	Avoiding scraps and re-routing defected products. Increase the overall defect detection efficiency with an improved multi-spectral vision system empowered by deep learning.

Formaldehyde emissions	10% reduction	More efficient application of adhesives. Integration of an online gas analyser for monitoring formaldehyde emissions and multi-spectral vision to control the proper adhesive application.

QUALITY CONTROL FOR FOOD INDUSTRY (JOTIS)

The third use-case of MULTIPLE is found in the dairy food industry. MULTIPLE will be deployed and demonstrated in a chocolate production line. JOTIS facilities in Athens occupy 30000sq. m. and have the most modern mechanical equipment in Europe, producing baby foods, powder mixes for baking and cooking, chocolates, syrups and chilled desserts. The company R&D and Total Quality Directorate incorporates 4 laboratories: Rapid QC analysis lab, Microbiological lab, Chemical lab and R&D lab. JOTIS aims to continuously create innovative products based on natural raw materials, and to ensure they are always of the highest quality. The process for the production of chocolate include the following steps:

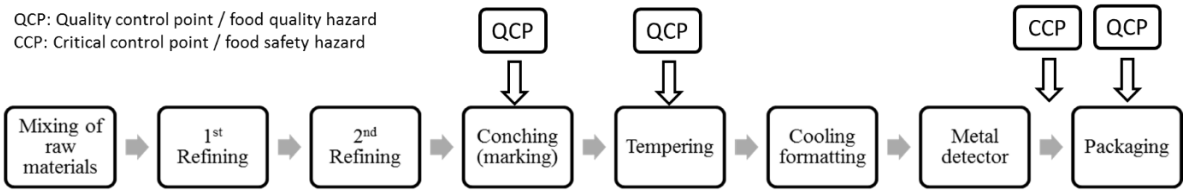


Figure 8. Steps in the chocolate production lines and current quality checks.

- Mixing blending raw materials
- Refining: the stage where the cocoa mass is pressed between cylinders in order to reduce the particle size.
- Conching: The process of stirring or mixing the chocolate for an extended period of time. It is the stage where the chocolate improves its flavour and texture:
  - Reduce the moisture content
  - Allow volatile oils and other ingredients to evaporate
  - Reduce the acidity of the chocolate through the evaporation of acetic acid
  - Round off the particulates within the chocolate through the use of friction
  - Develop the flavour more fully by allowing the flavour components to permeate the cocoa butter more fully
- Tempering: The stage at which the chocolate is cooled in order to form the desired crystal size and then heated back in order to maintain the formed crystals.
- Cooling-Formatting: The chocolate is moulded.
- Metal detector: The moulded chocolate goes through a metal detector for detection of any metal impurities

The main aim of JOTIS is to provide the highest end product quality and ensure that the product is within the specifications (e.g. fat, sugar content, color), independently on the variations of the raw material (e.g. cocoa) due to e.g. to different suppliers or poor growing weather conditions, while reducing waste. Key factors to be controlled that can cause quality losses are: fats, proteins, particle size, viscosity, tempering, colour of the final product, detection of unwanted bodies (metals mostly), homogeneity of the final product, e.g. in case the product contains nuts (even distribution of nuts in the chocolate bar).

MULTIPLE will deploy an integrated online solution to monitor and control different key parameters of the chocolate production at different stages, complementing the quality checks (QCs) currently done at laboratory, and optimizing the process parameters ensuring the final quality of the product independent of raw material variations. HSI and NIR spectroscopy will be applied to estimate e.g. fat content and type, protein content, crystallisation or final colour.

KPI	Multiple Improvement	Justification
Production performance	Increase overall efficiency: 10%	Overall process optimization, reliable and accurate control, faster decision making thanks to real-time and accurate information
Improved quality control performance	Increase the detection of products non-compliance with quality specifications: 20%	Continuously on-line quality checks. Integration of new on-line QC points based on HSI and NIR spectroscopy for faster decision making.
Waste or by-products	Reduction of waste production: 10%	On-line quality control and material re-routing. Integration of new on-line QC points based on HSI and NIR spectroscopy for faster decision making.

1.3.2 Positioning of the Project with Regards to Technology Readiness Level (TRL)

Technology	*Current TRL*	*Expected TRL*
SWIR snapshot hyperspectral imagers	*TRL 5/6*	*TRL 7/8*
IMEC has developed and demonstrated snapshot HSI in the SWIR range using an InGaAs sensor. A first camera core for demonstration has been built and is being tested in relevant conditions with real samples and hyperspectral image analysis software, making the case for a TRL 5/6. In MULTIPLE, a new industrial grade and cost effective read-out electronics will be developed for this sensor, providing robust advanced preprocessing and noise reduction. Therefore, an **InGaAs-based snapshot HSI SWIR camera** core will be developed and demonstrated in challenging tasks within three different industrial scenarios under operational environment conditions (TRL7). Since it will be fully integrated in PHOTONFOCUS industrial grade and widely supported imaging SDK, this camera core will be almost complete and qualified (TRL7/8).		
VIS/SWIR snapshot hyperspectral imagers	*TRL 5*	*TRL 7*
As CMOS-based HSI camera cores are already commercially available, the maturity of the InGaAs HSI core together with the need of integrating both sensors in a **dual aperture** approach limit the current maturity of the **VIS/SWIR snapshot HSI imager** to TRL5, as its interest and applicability in all the cases addressed has been validated in relevant environment with fairly close technologies. By the end of MULTIPLE, it will be demonstrated in real production plants under operational environment conditions (TRL7).		
OLED-based and laser-based spectroscopic analysers	*TRL 4/5*	*TRL 6/7*
As the operation of **OLED-based sensors** has been demonstrated and validated in lab conditions for the spectroscopic tasks proposed, and **laser-based approaches to spectroscopic analysis** that have demonstrated increased specificity and sensitivity, these technologies can be fairly regarded as validated in between a lab and a relevant environment (TRL4/5) for any of the application cases considered in MULTIPLE, in which other existing spectroscopic approaches have been at the least validated in the lab. MULTIPLE will bring this spectroscopic technologies to high performance and cost effective in situ spectroscopic solutions, demonstrated in a relevant environment, in some case under operational conditions (TRL6/7).		
HSI-ready cloud/edge computing and data-driven optimization	*TRL 5*	*TRL 7*
Edge-platforms featuring SoC ARM and GPU/TPU/FPGA and capable to run high performance deep learning models for images analysis are off-the-shelf components, and big data and cloud open source software technologies used for the creation of the data warehouse and the model prototyping and deployment pipelines (e.g. TensorfFlow, Elastic Search, Kibana) have been already used in production conditions in the context of multi-camera video surveillance. The maturity limit is set by the validation in relevant conditions of machine learning models for the analysis of HSI images in all the applications considered (TRL 5). By the end of MULTIPLE, a cloud/**edge computing and data-driven optimization platform capable of analysing HSI** datacubes and optimize production at different levels will be demonstrated under operational environment conditions (TRL 7).		

1.3.3 Linked national and international innovation activities

MULTIPLE is rooted in a strong background of research and innovation of the different partners of the consortium. Knowledge acquired in previous EU AM-based funded projects will be taken as a starting point:

Photonics & Embedded processing

MULTIPLE is built on previous technology developments on photonics and embedded processing researched on previous ICT-26 RIA projects (SEERS, I-ALLOW and MIREGAS). In this regard, cost-effective embedded imaging systems developed in these projects will be further developed and assessed in MULTIPLE prototype in real manufacturing conditions (TRL 7).

- **SEERS** (H2020-ICT-26-2014, GA 645114) led by AIMEN, proposes a breakthrough approach to multispectral imaging in a broad band of the infrared domain (0.7-14 µm wavelength) endowed with embedded image processing for image reconstruction, cognitive image fusion, video pre-processing and event-driven operation.
- **I-ALLOW** (H2020-ICT-26-2014, GA 645262) led by IMEC, aims the development of a cost-effective multi-spectral imaging and processing system for infrastructure monitoring in different light conditions (in day time, night time) and in all-weather conditions.
- **MIREGAS** (H2020-ICT-26-2014, GA 644192) partnered by AIROPTIC, addresses the development of a cost effective programmable multi-wavelength Mid-IR source for gas sensing.
- **MIRPHAB** (H2020-ICT-28-2015, GA 688265), partnered by IMEC, TEMATYS and EPIC, establish a pilot line to serve the growing needs of European industry in the field of analytical micro-sensors providing a reliable supply of mid-infrared (MIR) photonic components for companies incl. in particular SMEs already active in analytical MIR sensing.

- **EXIST** (H2020-EU.2.1.1.7. - ECSEL, GA 662222) led by IMEC, will research new technologies for CMOS image sensors that are needed in the next generation of several application domains (i.e. agri-food, manufacturing, healthcare, security and gas detection).
- **HYPERA** (H2020-SMEINST-2-2016-2017, GA 726572) led by IRIS, addresses the development of a novel and disruptive combination of Shortwave Infrared Hyperspectral Imaging (SWIR-HSI) in reflectance with Near Infrared Multispectral Imaging (NIR-MSI) in reflectance/transmittance and chemometric software for on-line foreign body detection and chemical composition analysis.
- **WIDELASE** (FP7-STREP-2012, GA 318798), partnered by AIROPTIC, aims the development of a new generation of lasers for sensing, detection and monitoring of hazardous organic substances.

Photonics & Data Analytics

- **FORTISSIMO 2** (H2020-FoF-09-2015, GA 680481) through CyPLAM experiment, partnered by AIMEN, addresses a Cloud-based-HPC optimisation tool applied to Additive Manufacturing, based on an image-based coaxial monitoring of the melt pool.
- **5G-SOLUTIONS** (H2020-ICT, GA 856691), partnered by IRIS will show how 5G allows to have data processing in the cloud for hyperspectral imaging data generated in the production plants of P&G and GLANBIA.
- **PHySIS** (H2020-COMPET-06-2014, GA 640174), partnered by IMEC will develop, test, and evaluate novel signal processing technologies for real-time processing of hyperspectral data cubes.
- **VISUM** (H2020-SMEINST, GA 731043), led by VISUM LTD (EX. IRIS), developed an IR spectroscopy cloud-enabled platform for the rapid, easy integration and customization of real-time monitoring systems for improved process control, product quality and safety in the food and pharmaceutical industries.

Digital Platforms & Orchestration

- **ZBREAK** (H2020- FOF-09-2017, GA 768869), led by AIMEN, addresses the development of a Predictive Digital Platform applied for improved maintenance models wrapped around physical systems for Zero-unexpected-Breakdowns and increased operating life of Factories.
- **BOOST** (H2020-ICT-2017-1, GA 780732), partnered by IMEC, aims the development of a Large Scale Pilot actions for demonstrating an open standardised and transformative shared data-driven Factory 4.0 model through 10 lighthouse factories.
- **AMABLE** (H2020-FOF-12-2017, GA 768775), partnered by AIMEN, aims the uptake of AM technologies by SMEs/mid-caps leading to the development of innovative business and service models and new value-chain models in a fully digital environment.

Photonics & Production Optimisation

- **MAShES** (FoF-01-2014, GA 637081) led by AIMEN proposes a compact, snapshot, and multispectral imaging system in the VIS/MWIR spectral range for embedded real-time control and high speed monitoring of laser-based manufacturing processes, focussing on an accurate measurement of temperature distribution, 3D seam profile and 2D melt pool geometry and surface texture dynamics, and process speed.
- **ProPAT** (SPIRE-01-2014, GA 637232) led by IRIS, aims the development of novel MEMS-NIR spectroscopic and granulometric analysers, as well as smart sensors for measuring composition, particle size, local bulk properties and process parameters for optimising industrial operations, integrate them into a versatile global control platform.
- **iCspec** (SPIRE-01-2014, GA 636930), partnered by AIROPTIC, addresses the development of compact, robust and maintenance-free sensors for fast in-line multi-species chemical composition measurements for process analytics of many technically relevant gases such as hydrocarbons.
- **REEMAIN** (FP7-2013-NMP-ICT-FOF, GA 608977), partnered by SCM, combines cutting edge knowledge from production processes, energy simulation software tools, energy and resource planning to develop and demonstrate a methodology and platform likely to boost the efficiency of both energy and material resources.
- **FACTORY ECOMATION** (FoF.NMP.2012-1, GA 314805), partnered by SCM, develops breakthrough innovations for cost-effective, highly productive, energy-efficient and near-zero-emissions production systems, by means of development of a comprehensive sensing, monitoring and data evaluation system within the factory.
- **TINTIN** (Spanish National project, 2015), partnered by RUSSULA and AIMEN, addresses the development of tools and technologies focused on control and autonomous reconfiguration of manufacturing systems involved in laminating hot product.

1.3.4 Methodology

MULTIPLE requires a multidisciplinary approach that involves activities which range from development of photonic solutions all the way to the pilot implementation and demonstration of the innovation action results. The project has a proposed duration of 36 months (3 years) and is divided in 5 phases covering the following activities:

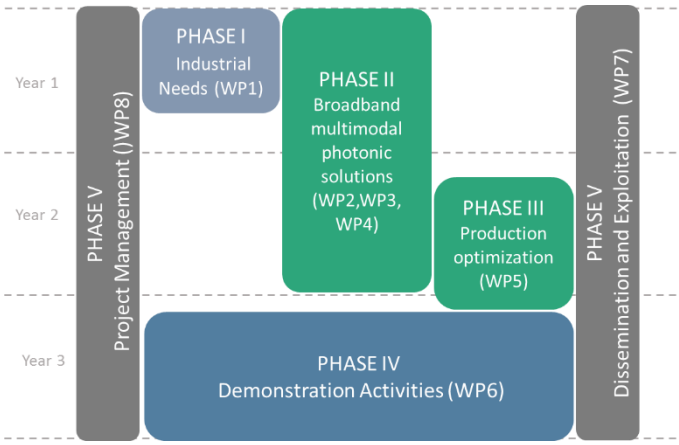


Figure 9. Overall MULTIPLE methodology.

PHASE I: Industrial needs. Applications driven monitoring requirements specifications for production optimization. Within these activities, which will be performed at the initial phase of the project the consortium will investigate all requirements from the selected industrial cases. The activities will lead to the extraction of solid specifications which will then be translated into tangible project results.

PHASE II: R&D activities focused on the development of multimodal spectral sensing prototypes matching the specifications and providing cost-effective solutions for monitoring relevant product and process parameters for production optimization. This will include the optoelectronic integration of state-of-art detectors, optical arrangements and embedded processing units, and the development and implementation of real-time multimodal processing algorithms based on AI models to extract meaningful information for the optimization of the processes. Scalable data storage scheme and micro-services supporting AI model development will be also implemented. At the end of PHASE II smart broadband multimodal photonic sensing solutions will be integrated and validated in real industrial settings.

PHASE III: R&D focused on the development of tools for production optimization. This phase will develop tools for exploiting the meaningful data obtained with the photonic sensing prototypes for providing feedback for process optimization and production management. At the end of this phase, the complete MULTIPLE concept will be ready and integrated in industrial settings.

PHASE IV: Complete demonstration of MULTIPLE approach in real operational conditions. The demonstration activities will be focused on the assessment of MULTIPLE for optimizing the production process in terms of speed, product quality and resource efficiency in three operational environments covering piecewise and continuous processes.

PHASE V: Project Management, Dissemination and Exploitation activities. This phase runs the whole duration of the project. The project management will keep track of all the involved activities and take actions when needed to ensure the smooth progress of R&D and demonstration activities. The project will run dissemination activities for making the developed technologies known to wider audiences. The project focuses on providing results that are ready for further market uptake. The consortium will undertake, from the beginning of the project, activities for including the results in the portfolio of the involved companies.

1.4 Ambition

1.4.1 Advances beyond the state of the art

Multimodal monitoring and process optimization in industry, a case for spectral sensing and imaging

Steelworking. Almost the 80% of the total energy consumption of the rolling mill plant is due to the reheating furnace. In situ and real-time monitoring of gas constituents inside the furnace offers the opportunity to optimise its efficiency through a closed loop control that minimizes the fuel waste and avoids unsafe CO levels and O2 levels that threat product quality. With this aim temperature measurements of the furnace and steel slab are also necessary to properly adapt the loads and heating sequences. Borescope thermal cameras offer spatially resolved temperature maps of the furnace and slab. The main limitation is the uncertainties in the measurements due to differences on material emissivity, presence of dust or steam, and the overall cost. UV (CCD), NIR and SWIR imagers have been applied to detect different defects and check quality in hot steel slabs[5,6,7,8]. MULTIPLE will develop a multi-gas analyser based on Mid-IR tunable laser spectroscopy for standoff detection of gas constituents inside the reheating furnace. Besides MULTIPLE will develop a cost-effective robust computer vision and thermography system based on snapshot SWIR

[5] A. O. M. Luiz et al., *Automatic detection of surface defects on rolled steel using Computer Vision and Artificial Neural Networks*, IECON 2010 - 36th Annual Conference on IEEE Industrial Electronics Society, Glendale, AZ, 2010, pp. 1081-1086.
[6] Alkapov et al. *Automatic Visible Defect Detection and Classification System Prototype Development for Iron-and-Steel Works*, GloSIC 2018.
[7] Maoxiang Chu, et al., *Multi-class classification for steel surface defects based on machine learning with quantile hyper-spheres*, Chemometrics and Intelligent Laboratory Systems, 2017.
[8] Sen Wang, et al., *Panoramic Crack Detection for Steel Beam Based on Structured Random Forests*, IEEE, vol. 6, pp. 16432-16444, 2018.

HSI will be able to estimate temperature maps independent on emissivity in an extended temperature range (400-1500ºC) and to estimate the speed of the steel slab using advanced optical flow techniques and detect surface defects. The use of SWIR will provide more texture information enhancing the robustness and accuracy. The following table provides a summary on these characteristics and the envisaged monitoring technology.

Magnitude/feature	Accuracy	Production stage(s)	Envisaged technology
Gas concentration of CO/CO2/O2 and NOx	CO, CO2, O2, NO and NO2 within ± 0.1 ppm on 1 meter basis and the accuracy of 2% of the measured value. Response time will be in order of 0.05 second	Reheating furnace (pre-heating, heating soak zone, chimney)	Laser-based IR Spectroscopy
Temperature	400-1500ºC with 2% accuracy	Reheating furnace Rolling mill	SWIR HIS
Speed of steel slab	0.05 m/s	Rolling mill	SWIR+ PWC-Net[9]
Oxidation, impurities and defects	LDL's panoply of defects	Reheating furnace Rolling mill	SWIR

Woodworking. Single batch, zero defect, and full quality control -including healthy related standards- require to monitor a number of chemical, mechanical, and aesthetic characteristics and magnitudes, from raw chipboards to final wood components. Notably, formaldehyde -a highly carcinogenic substance- is likely to surge from pressed wood products made with adhesives, e.g. resins or urea[10]. Class E1 rules (European standards EN717 and EN120: E1, E2, E3[11]) apply to panels for domestic use (furniture) requiring an average release of less than 2.5mg/m^{2}H and concentration lower than 0.1ppm. While MWIR spectrometry has demonstrated the capability to measure air concentration of formaldehyde, NIR spectroscopy and VIS-SWIR HSI have emerged as an operative approach for the on-line estimation of mechanical properties of wood (e.g. elasticity, density, moisture content), the detection of defects, or the assessment of colour. The following table provides a summary on these characteristics and the envisaged monitoring technology.

Magnitude/feature	Accuracy	Production stage(s)	Envisaged technology
Formaldehyde detection	0.01ppm	MDF production (Lacquering/Assembly)	Laser-based MIRS[12]
Elasticity	(MOE) 2.5 GPa	Sizing	VIS/SWIR HSI & NIR Spectroscopy[13,14]
Density	Ultralight/Light/Standard/Heavy	Sizing	VIS/SWIR HSI & NIR Spectroscopy[15,16]
Moisture	2%	Sizing	SWIR HSI[17]
Colour	2ΔE2000	Drilling / Lacquering	VIS HSI[18]
Defects on drills, surfaces, edges (excess of glue, irregularities)	ROYO's panoply of defect	Drilling Lacquering	VIS/SWIR HSI[19]

Food processing. In food production, quality control is a very critical process. Regarding food safety, adulteration or the presence of foreign bodies are very sensitive aspects. Otherwise, controlling the different parameters that affect

[9] Sun, Deqing, et al., *Pwc-net: CNNs for optical flow using pyramid, warping, and cost volume,* In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 8934-8943. 2018.
[10] George I. Mantanis, et al., *Adhesive systems used in the European particleboard, MDF and OSBindustries,* Wood Material Science & Engineering
[11] Harald Schwab, Rainer Marutzky, Bettina Meyer, "European Regulations for Formaldehyde"
[12] Mateusz Straszewski, et al., *Application of tunable laser formaldehyde analyzer for real time process control and emission monitoring in wood panel industry,* FLAIR 2018
[13] L. R. Schimleck et al., *Comparison of Methods for Estimating Mechanical Properties of Wood by NIR Spectroscopy,* Journal of Spectroscopy, vol. 2018, Article ID 4823285, 10 pages, 2018.
[14] Liang, H. et al., *Non-Destructive Methodology to Determine Modulus of Elasticity in Static Bending of Quercus mongolica Using Near-Infrared Spectroscopy*. Sensors 2018, 18, 1963.
[15] Fernandes, A, et al. (2012). *Measurement of intra-ring wood density by means of imaging VIS/NIR spectroscopy*, 67(1), pp. 59-65.
[16] Santos, A. et.al, *Estimation of wood basic density of Acacia melanoxylon (r. br.) by near infrared spectroscopy.* J. Near Infrared Spectrosc. 2012, 20, 267–274.
[17] Olena Myronycheva, et. al, *Hyperspectral Imaging Surface Analysis for Dried and Thermally Modified Wood: An Exploratory Study,* Journal of Spectroscopy, vol. 2018, Article ID 7423501, 10 pages, 2018
[18] Chlebda, Damian et al., *Assessment of hyperspectral imaging system for colour measurement.* Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy. 185. 10.1016/j.saa.2017.05.037.
[19] Ostertag, Edwin et al., (2009). *Hyperspectral imaging of wood surfaces: monitoring of the adhesive distribution.*

sensory quality are very important to ensure consumer satisfaction. Currently, quality control in a chocolate production line like the one at JOTIS is mostly statistical and done by analysing samples off line. Samples are collected during the production at critical control points and they are measured in the lab using analytical methods. Even after this control, about 1% of the total production is off spec (e.g. low gloss, undetected foreign bodies, uneven distribution of almonds). Monitoring the different parameters of interest will be deployed and tested in the successive steps of chocolate production from raw materials mixing (e.g. fat type and content but also possibly proteins and sugars), melting, chocolate tempering and final bars moulding. Further than existing literature on HSI, the ability to unveil not only the compositional distribution, but also that of the crystallinity (already shown locally by NIR) will be tested. Other parameters than can be calibrated using the deployed techniques will be tracked such as the homogeneity of the distribution of nuts, colour, or the presence of foreign bodies. The following table provides a summary on these characteristics and the envisaged monitoring technology. Although they will be tested in a chocolate processing and bar manufacturing line, they are generally relevant for other products too.

Magnitude/feature	Accuracy	Production stage(s)	Envisaged technology
Fat content and type	± 0.2% (on the % of fat)	Melting	NIR (in melt probe)[20,21]
Viscosity	± 400 cP at 40ºC	Conching	NIR[22] or MEMS viscosity sensor (in melt probe)
Crystallisation	Distribution and concentration of crystals	Tempering	NIR[23]/ HSI (possibly with novel polarisation enhancement)
Homogeneity and foreign body detection	Even distribution of almonds, Non currently detected foreign bodies (hair, nuts in case the chocolate is supposed to be nut free)	Final product conveying prior to packaging	HSI[24]
Colour and Gloss	2ΔE2000	Final product	VIS his

Spectral imaging and sensing in industrial applications

The affordability of CMOS technology and the advances on image processing algorithms during the 90's and early 2000's fuelled the widespread use of machine vision systems in industrial monitoring, real time control, and quality control applications. A variety of techniques (keypoint detection, photogrammetric measurement, laser scanning, or point cloud alignment, among many others) enabled advanced metrology and quality checking applications to flourish. CMOS and CCD technology also made possible the appearance of low cost spectrometers in the visible range.

However, a major limitation of CMOS technology is that it is not sensitive beyond the 1000 nm wavelength limit, when many interesting things can only be viewed in the next micron wavelength range. Alternative imagers based on hybrid technology like InGaAs present a much higher tag prize and do not cover the visible range. And the spectral range is getting more and more important as the capability to be sensitive to different material properties seems more and more within reach. Other emerging approaches based on quantum-dots (QD) [25,26,27] aim to open new opportunities in the visible and thermal infrared spectral regions. These devices are capable of providing low-cost high performance detection across the Vis-SWIR spectral range without the need of the hybridization step require by traditional SWIR detectors. However, from a performance standpoint the achievable quantum efficiencies in the SWIR range provided by first available cameras[28] (~15%) still far from those provided by InGaAs (>70%).

[20] https://www.fossanalytics.com/media/files/documents/brochuresanddatasheets/segments/cocao-and-chocolate/cocoa_and_chocolate_brochure_gb.pdf

[21] J. Čopíková et al., *Application of Near Infrared Spectroscopy in Chocolate Analysis,* Chemicke Listy 97(7):571-575, 2003

[22] Zeng et al, *In-line measurement of tempered cocoa butter and chocolate by means of near-infrared Spectroscopy,* Article in Journal of the American Oil Chemists' Society · June 1999

[23] Svenstrup et al., *Rapid instrumental methods and chemometrics for the determination of pre-crystallization in chocolate Gitte,* International Journal of Food Science and Technology 2005, 40, 953–962

[24] Zhang et al., *Multivariate Curve Resolution Applied to Hyperspectral Imaging Analysis of Chocolate Samples,* Applied Spectroscopy, Vol 69, Issue 8, 2015

[25] Hafiz, S.B., Scimeca, M., Sahu, A. et al. "Colloidal quantum dots for thermal infrared sensing and imaging", Nano Convergence (2019)

[26] Goossens, Stijn & Navickaite, Gabriele et al. "Image sensor array based on graphene-CMOS integration". Nature Photonics. 11 (2017).

[27] Ethan J. D. Klem, Chris Gregory, Dorota Temple, and Jay Lewis "PbS colloidal quantum dot photodiodes for low-cost SWIR sensing", Proc. SPIE 9451, Infrared Technology and Applications XLI, 945104 (4 June 2015);

[28] www.swirvisionsystems.com/

NIR spectroscopy is a well-established technique for inline composition monitoring. It is indeed highly sensitive to the chemical signature of a wide range of materials, especially organic ones. Using advanced data analysis approaches (chemometrics or even more sophisticated Artificial intelligence algorithms such as neural networks) to treat the resulting spectra, it can predict fine multiparametric information on the concentration of the substances of interest. It is also sensitive to the morphology of particles. As opposed to conventional QC tests, it does not require sample preparation and is virtually instantaneous, allowing saving both labour and process time compared to when resorting to conventional control procedures.

Hyperspectral imaging (HSI) has emerged during the last decade as a hot research topic, as it combines the power of digital imaging and spectroscopy. Every pixel in a hyperspectral image provides local spectral information across a large number of narrow bands. That is, instead of collecting a single spectrum, this technique records a hyperspectral image (HSI) of the sample, providing not only the local chemical composition, but also its distribution.

Until recently, HSI was mainly characterised by pushbroom scanning systems, while snapshot imagers were only available for multispectral imaging with a few spectral bands. While multispectral approaches were useful to gain contrast in certain situations and determine specific measurements (e.g. NDVI), the higher spectral resolution of HSI has allowed much finer analysis relevant in industrial operations. This has proved to be useful for the development of advanced monitoring and control methods in industrial processes, like food processing[29,30,31,32,33] or wood processing[34,35,36]. However, push broom technology presents a major issue: images are acquired sequentially -line by line- which takes time, and they need to be reconstructed before being processed. Therefore, it is prone to limitations and robustness issues regarding product speed, product vibrations or displacements in any direction other than the scanning one. This has motivated a number of works on the mitigation of spatial and spectral blur in the reconstructed images. However, denoising approaches only mitigate the problem and in real world cases line scanning means robustness issues, higher equipment complexity and cost and setup cost.

This is why so many research efforts in optics have been devoted to the problem of delivering snapshot spectral imaging systems with a significant increase in the number of bands[37].

In MULTIPLE, we propose an ensemble of new monitoring systems built on breakthrough sensors and snapshot mosaic filters that make a leap in spectral range, spectral sensitivity and selectivity. We will also address the development of different models (eg. chemometric models, deep learning for image analysis) to finally provide the decision support for process optimization. The big potential of this approach is comparable to the challenge posed by the volume and complexity of the data generated in the form of spectral datacubes. In this regard, we address the management of data as an orchestration problem and we will put in place embedded processing, cloud, and big data technology to provide an efficient and scalable solution.

Snapshot hyperspectral imaging

IMEC has developed a unique integrated hyperspectral filter/imager technology, where the spectral filters are monolithically deposited/integrated on top of CMOS image sensors at wafer level[38]. This is achieved using a set of CMOS compatible production steps, like deposition, patterning and etching, which allows pixel level accuracies in filter alignment. The spectral filters designs are based on the Fabry-Pérot principle, which enables the design and fabrication of ranges of optimized filters at various wavelengths with flexible/customizable layout (covering different groups of pixels or depositing filters per pixel) and spectral bandwidth.

[29] Mahesh, S., et al, *Hyperspectral imaging to classify and monitor quality of agricultural materials,* Journal of Stored Products Research 61 (2015): 17-26.

[30] Ravikanth, L. et al., *Extraction of spectral information from hyperspectral data and application of hyperspectral imaging for food and agricultural products,* Food and bioprocess technology 10, 2017

[31] Su, Wen-Hao, et al., *Non-destructive and rapid evaluation of staple foods quality by using spectroscopic techniques: a review,* Critical reviews in food science and nutrition 57, no. 5 (2017): 1039-1051.

[32] El-Mesery, et al., *Applications of Non-destructive Technologies for Agricultural and Food Products Quality Inspection,* Sensors 19, no. 4 (2019): 846.

[33] Su, Wen-Hao, et al., *Evaluation of spectral imaging for inspection of adulterants in terms of common wheat flour, cassava flour and corn flour in organic Avatar wheat (Triticum spp.) flour.* Journal of Food Engineering 200 (2017): 59-69.

[34] Tsuchikawa, S. et al., *A review of recent application of near infrared spectroscopy to wood science and technology,* Journal of Wood Science 61, no. 3 (2015)

[35] Mancini, M., et al., *Near infrared spectroscopy for the discrimination between different residues of the wood processing industry in the pellet sector,* Fuel 217 (2018): 650-655.

[36] Ma, Te, et al., *Non-destructive evaluation of wood stiffness and fiber coarseness, derived from SilviScan data, via near infrared hyperspectral imaging,* Journal of Near Infrared Spectroscopy 26, no. 6 (2018): 398-405.

[37] Hagen, Nathan A., and Michael W. Kudenov. *Review of snapshot spectral imaging technologies,* Optical Engineering 52, no. 9 (2013).

[38] Bert Geelen, Nicolaas Tack, Andy Lambrechts, *A compact snapshot multispectral imager with a monolithically integrated per-pixel filter mosaic,* Proc. SPIE 8974, Advanced Fabrication Technologies for Micro/Nano Optics and Photonics VII, 89740L (7 March 2014);

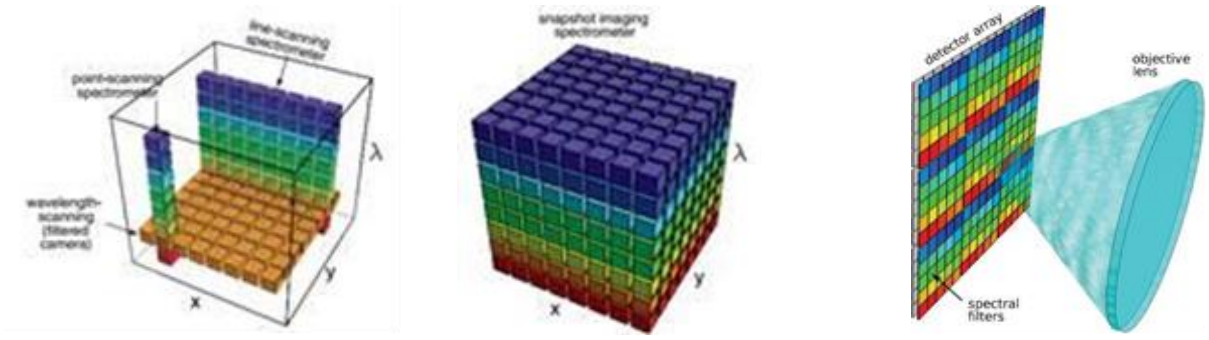


Figure 10. Snapshot multispectral data acquisition concepts.

Unlike grating or linear filters, in our approach, imagers can be covered with only selected spectral bands in a certain range instead of the complete spectral range. The result is a compact & fast hyperspectral imager made with low-cost CMOS process technology.

Similarly, for InGaAs based detectors, IMEC has developed a novel hybrid integration approach, the optical filters, processed on glass substrates, are integrated directly on top of the InGaAs detector with high accuracy and parallelism, and minimum gap between filters and Focal Plane Array to limit cross-talk. The high vertical and horizontal alignment accuracy of this method allows for different filter arrangements, covering group of pixels, like the line-scan implementation, or even per-pixel mosaic snapshot sensors[39].

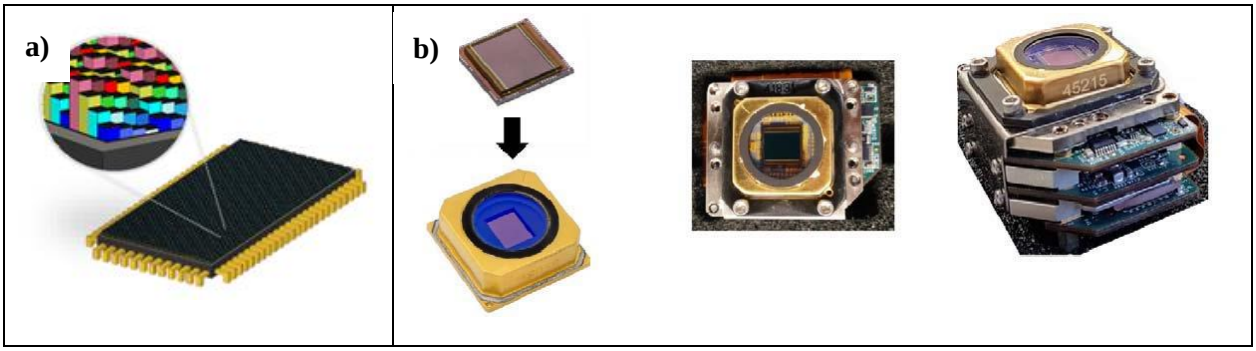


Figure 11. Monolithic integration of spectral filters on CMOS-based imager wafers (a); Hybrid integration of filters on glass substrates with InGaAs detector and readout electronics (b).

An overview of current prototypes based on this technology can be found in the table below.

[39] P. Gonzalez, et al., *An extremely compact and high-speed line-scan hyperspectral imager covering the SWIR range*, Proc. SPIE 10656 Image Sensing Technologies: Materials, Devices, Systems, and Applications V, 106560L (14 May 2018);

Sensor	SNm4x4 VIS	SNm4x4 REDNIR	SNm5x5 NIR	SNm2x2 RGB+NIR	LS150+	Cardinal640
Technology	CMOS	CMOS	CMOS	CMOS	CMOS	InGaAs
#Bands	16	16	25	4	150+	by specific filter
Spectral range (nm)	450-630 nm	600-875 nm	660-975 nm	RGB+NIR	470-900 nm	1000-1700nm
Detector resolution (pxl)	2048 x 1088	2048 x 1088	2048 x 1088	2048 x 1088	2048 x length of scan	640-512 px
Temporal resolution	340 fps	340 fps	340 fps	340 fps	1360 lines / sec	350 fps

Furthermore, all these sensors have also been prototyped by integrating them at camera level (as small as 4x4x4cm) and to further demonstrate the capabilities of each camera, a fully functional system level including illumination and acquisition software. The mosaic-based cameras can provide hyperspectral data at video rate, with a maximum speed of 340 fps, ideal for high-speed video data acquisition. This technology is being already commercialized by PHOTONFOCUS, with a mosaic snapshot imager based on a CMOSIS CMV 2000 image sensor. Compared to competing technologies, snapshot HSI is orders of magnitude cheaper and more compact.

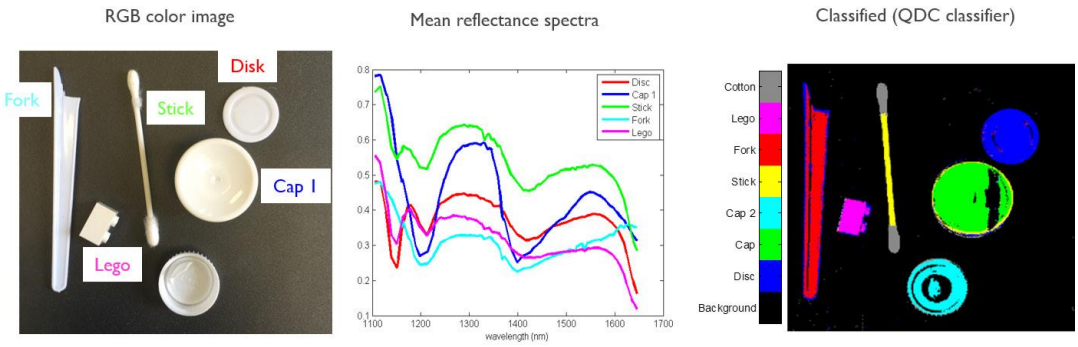


Figure 12. Classification example using IMEC SWIR Snapshot HSI sensor.

In MULTIPLE, the ambition is twofold: firstly, we will develop a cost effective Snapshot SWIR camera core based on InGaAs sensors suitable for industrial vision volume market and secondly we will combine it with an existing CMOS core to build a dual aperture integrated camera. This will push the state of the art in spectral sensors, bringing a unique snapshot HSI imager in a broad spectral range from VIS to SWIR (0.4µm to 1.7µm wavelength).

Dual-aperture spectral imaging

In traditional single-aperture spectral cameras, the final image is formed based on the interactions of light with optics, spectral filters (e.g. beam splitter, grating, interferometer) and the FPA. In this sense, the spectral range is limited by the sensibility of the detector technology, and the increased spectral resolution goes in detriment of the spatial resolution, dynamic range, sensitivity and SNR[40]. In contrast, multi-aperture imaging systems combine several optic and sensor configurations in the same device, acquiring an intermediate representations of the scene, that are combined using image processing and computational imaging to provide a final image with the information acquired from different configurations[41]. Thus, it enables the possibility to combine very different detector technologies in a single device, covering very different spectral bands. Interesting advantages of multi-aperture spectral imaging approaches are: reduction of the longitudinal chromatic aberration (defocus), possibility to improve the dynamic range, increased light throughput and SNR, compactness. On the other hand, this pose challenges regarding image registration and reconstruction.

[40] Aggarwal, M., & Ahuja, N. (2004). Split aperture imaging for high dynamic range. International Journal of Computer Vision, 58(1), 7-17
[41] Green, P. et al."Multi-aperture photography". In ACM Transactions on Graphics (TOG) 2007 (Vol. 26, No. 3, p. 68). ACM.

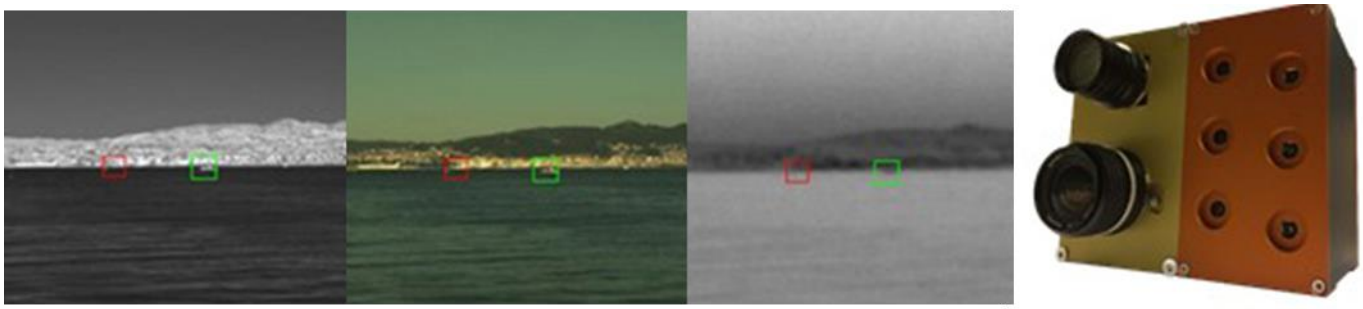


Figure 13. SEERS camera prototype with up to 8 apertures from VIS to LWIR. From left to right: NIR, VIS, LWIR registered image samples, overlaid with boat detections exploiting all the spectral bands.

SEERS project (GA 645114) coordinated by AIMEN, has recently developed modular, compact and cost-effective snapshot multi-spectral imaging cameras combining different detector technologies in the infrared (0.4-14). The solution combines monolithic uncooled detectors, CMOS, CMOS-based microbolometers and PbSe FPAs with a multi-aperture optical arrangement, and embedded processing capabilities to perform computational imaging and reconstruction in real-time[42].

MULTIPLE will develop a dual-band aperture camera integrating CMOS and InGaAs detector with filters-on-chip in a single extremely compact HSI machine vision device sensible from the VIS to SWIR range.

OLED-based spectrometers

SENORICS developed patented optical spectroscopic sensor solutions for detecting visible and near-infrared light which are low cost as they can be mass produced as being obtained through printing electronics and OLED[43] technologies. OLED technology uses an optical microcavity to increase the typically negligible external quantum efficiency in the spectral region of charge-transfer absorption by more than 40 times, yielding values over 20%. The broad spectral tunability is achieved via a simple variation of the cavity thickness, making this innovative device principle highly suitable for integrated low-cost spectroscopic photodetectors. SENORICS sensors capture light from 400nm to 1700nm, therefore can cover from the VIS to the SWIR spectrum within one integrated chip. The monolithic sensors are robust and can work without optical elements like gratings, they are customizable regarding pixel count, wavelengths covered, etc. Competing technologies are either too expensive (InGaAs) or not accurate enough (Si). MULTIPLE will apply OLED sensors to develop novel low cost spectroscopic analysers with maximum sensitivity and selectivity.

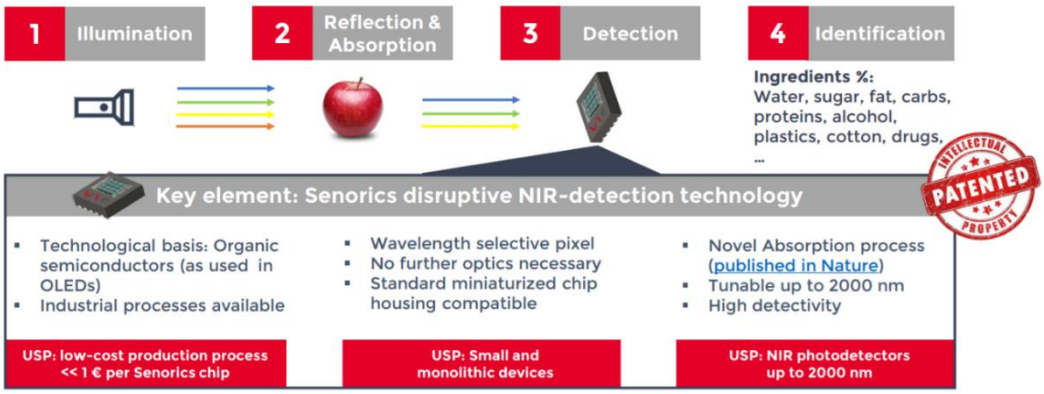


Figure 14. Senorics miniaturized spectrometer chips based on OLED technology.

Laser-based IR spectroscopy

Commercially available tunable laser analysers incorporating NIR and Mid-IR sources have been proven to be field-compatible for several years. AIROPTIC manufactures various types of in-situ and extractive laser based analysers of gases (O2, CO, CO2, NO, NO2, HCl, NH3, HF, H2O, CH4, C2H2, C2H4, C2H6, C3H8, C4H10 and many others) in multitude of various industrial applications. The general advantages of using a tuneable laser spectrometer for emission monitoring and process control in wood-based panel industry are the following:

- Short response time (a few seconds) vs. laboratory method.
- High sensitivity and dynamic range – covers range from single ppb to ppm level.
- High selectivity – measures accurately regardless of the gas matrix composition.
- Low maintenance – low cost of ownership.

[42] Méndez-Rial, R. et al. *A high-speed MWIR uncooled multi-aperture snapshot spectral imager for IR surveillance and monitoring*. In 2016 IEEE International Conference on Imaging Systems and Techniques (IST) (pp. 206-210).
[43] Siegmund, Bernhard et al. *Organic narrowband near-infrared photodetectors based on intermolecular charge-transfer absorption*. Nature Communications. 2017.

- Suitable for *in-situ* monitoring without a need for sample preparation or collection.
- Auto-calibration feature – built-in reference cell.

Today, almost all in-situ tunable laser-based analysers utilize DFB semiconductor lasers, which can only be tuned over a very limited wavelength range by temperature or current variation. Therefore, these devices are not suitable for the detection of multiple gas species. Semiconductor lasers with a wide tuning range in the Mid-IR are available, but they use free-space external cavities, which make the lasers bulky, costly and sensitive to shock and vibrations. This is a severe drawback in applications where several gas components have to be monitored simultaneously. Due to this limitation, it is presently necessary to install multiple tuneable laser analysers in parallel sample streams. This approach is inefficient, costly and requires installation of several flanges in the process, which often is not possible. To overcome this drawback, AIROPTIC has developed a multi-laser in-situ analyser platform -using extended Wavelength Modulation absorption Spectroscopy (WMS) -, which has the capability to analyse several components in one gas sample stream.

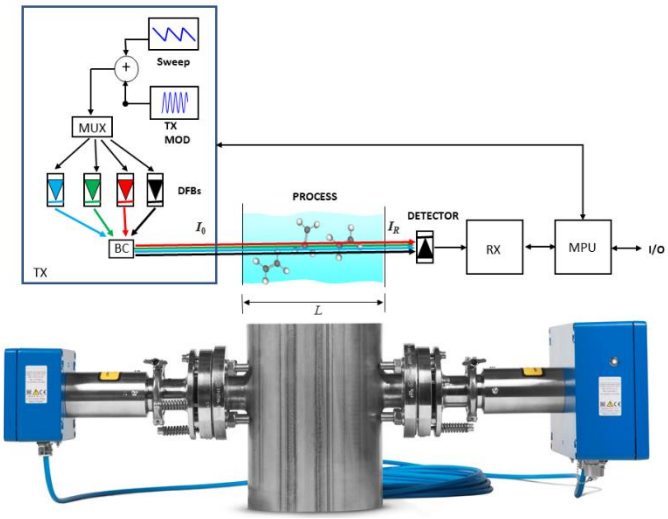


Figure 15. Proposal of the multi gas system for *insitu d*etection of several species in steel works.

In MULTIPLE, AIROPTIC will apply this principle can be used for multiple in-situ gas detection in steel works for optimization of combustion measuring CO/O2/CH4/CO2 as well as NOx. Based on a recent work in formaldehyde detection[44] , AIROPTIC will design a very sensitive extractive type analyser for end product qualification in furniture production, with a sensitivity below 1 part per-billion (<1 ppb) (Figure 15).

Machine learning for spectral and HSI analysis, moving to deep learning

To take all the power of onsite and on-line spectroscopy, complex and flexible analytic methods should be put in place. In practice, this has meant a rapidly increasing use of machine learning in combination with spectrometry during the last decade, revealing so useful in many industrial problems, like those addressed in MULTIPLE[45,46,47,48,49].

In recent years, the sweeping success of deep CNNs pipelines for image analysis has resulted in breakthrough advances in image segmentation[50], object recognition[51,52] and the estimation of a variety of parameters and semantic attributes relevant in very different application fields[53]. The automatic learning of the best features to solve a problem -without the classic need and burden of hand crafting their extraction- certainly offers very interesting opportunities in complex industrial applications. However, industrial vision systems are still lagging behind. In the best case, mainstream software packages are just starting to make available a very limited use of pre-trained CNNs, in their most recent versions[54].

In the context of HSI, the use of 3D-CNN networks has provided state-of-the-art results in segmentation tasks[55]. This type of architectures is compatible with the best performing deep models currently found in the PASCAL VOC challenge[56].

[44] Joanna J., *Messung der Formaldehydabgabe in der Holzwerkstoffproduktion in Echtzeit mithilfe eines Laseranalysator GasEye HCHO* XXXI-st International Scientific Conference WOOD MATERIAL OF THE XXI-ST CENTURY, Rogow, 12-14 November (2017)
[45] Su, W. et. al, *Non-destructive and rapid evaluation of staple foods quality by using spectroscopic techniques: a review*, Critical reviews in food science and nutrition 57, no. 5 (2017): 1039-1051.
[46] El-Mesery, Hany S., et al., *Applications of Non-destructive Technologies for Agricultural and Food Products Quality Inspection*, Sensors 19, no. 4 (2019): 846.
[47] You, Hanjong, et al., *Classification of food powders using handheld NIR spectrometer*, in 2017 9th ICUFN, pp. 732-734. IEEE, 2017.
[48] Gupta, O. et al., *Machine learning approaches for large scale classification of produce*, Scientific reports 8, no. 1 (2018): 5226.
[49] Gewali, U. et al., *Machine learning based hyperspectral image analysis: a survey*, arXiv preprint arXiv:1802.08701 (2018).
[50] http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=6
[51] http://image-net.org/challenges/LSVRC/2017/results
[52] http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=4
[53] Liu, W., et al., *A survey of deep neural network architectures and their applications*, Neurocomputing, 234, 11-26.
[54] https://www.mvtec.com/products/halcon/newest-features/
[55] Chen, Yushi, et. al., *Deep feature extraction and classification of hyperspectral images based on convolutional neural networks*, IEEE Transactions on Geoscience and Remote Sensing 54, no. 10 (2016): 6232-6251.
[56] http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=6

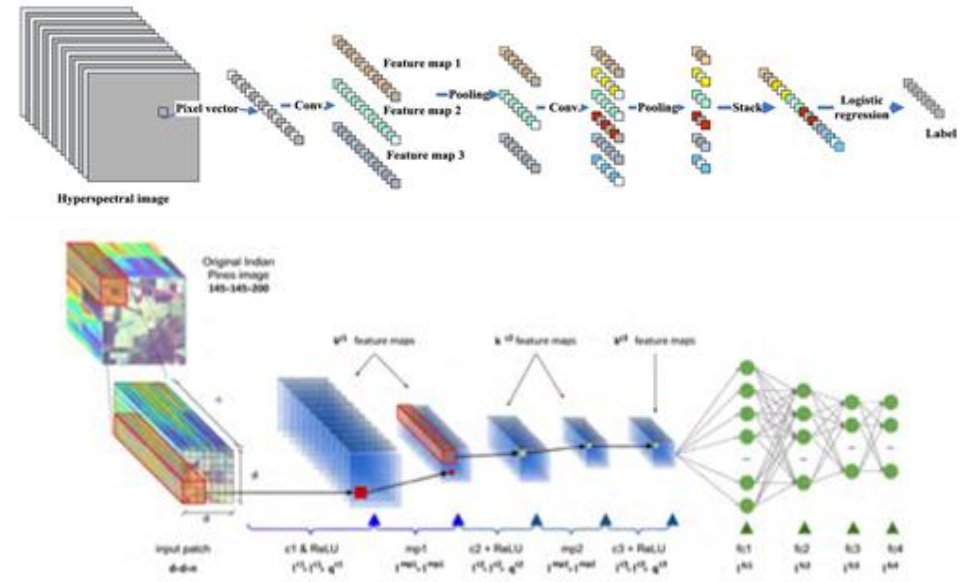


Figure 16. Examples of deep 3D CNN HIS processing architectures.

The main issue of deep learning approaches is that they require large datasets[57] annotated with the categories of interest or with a limited number of custom categories trained through transfer learning techniques[58]. The scarcity of HSI annotated data in most industrial applications is hence a clear barrier for adoption. However, cloud technology and IoT approaches bring a unique opportunity for the development of pipelines for automatic annotation of spectral images and measurements. Moreover, the use of visual semantic embeddings[59,60] and more elaborated zero shot and few shot learning schemes[61,62,63] has opened very interesting avenues to explore in industrial applications of spectral imaging.

In this regard, the deployment of orchestrated sets of multimodal monitoring systems with embedded processing and cloud services for the search and analytics of data may be expected to facilitate the creation of datasets annotated with relevant manufacturing parameters, removing these barriers.

And even though the most extended models require a large number of annotated images, costly and lengthy to gather and *do not offer a satisfactory solution for learning new concepts rapidly from little data*[64], the use of visual semantic embeddings[65,66] and more elaborated zero shot and few shot learning schemes[67,68,69] has opened very interesting avenues to explore in industrial applications of spectral imaging.

Based on the extensive expertise in industrial machine vision and recent developments of deep models by AIMEN and ABRAIA, MULTIPLE will test, select, and combine the best architectures of deep networks for the problems at hand, taking into account the specificities of the analysis required, together with the robustness and speed requirements. The data currently available in the factory will be used for the automatic and semi-automatic annotation of acquired HSI data, i.e. for the creation of large datasets.

Embedded processing for real time monitoring and control

Traditional machine vision systems have separate camera modules and processing units, e.g. industrial PC, connected with PLCs through fieldbuses for enabling automation and closed loop control. However, the rapid growth of data volume imposes increasingly challenging requirements on computational power and scalability, which together with the need of low latency for time critical process control, is moving the computational power as close as possible to the sensors edge. Current *edge* devices are able to collect and process data, interoperate with other devices, be controlled remotely, and make decisions that were impossible for a sensor alone[70].

[57] Deng, J. et al. “Imagenet: A large-scale hierarchical image database”. *CVPR 2009*
[58] Yosinski, J. et al. "How transferable are features in DNNs?" *Advances in neural information processing systems*. 2014.
[59] Frome, Andrea, et al. "Devise: A deep visual-semantic embedding model." Advances in neural information processing systems . 2013.
[60] Akata, Zeynep, et al. "Label-embedding for image classification." IEEE PAMI (2016): 1425-1438.
[61] Vinyals, O., et al. "Matching networks for one shot learning." *Advances in Neural Information Processing Systems*. 2016.
[62] Romera-Paredes, Bernardino, and Philip Torr. "An embarrassingly simple approach to zero-shot learning." ICML 2015.
[63] Xian, Yongqin, et al. "Zero-shot learning-A comprehensive evaluation of the good, the bad and the ugly." IEEE PAMI 2018
[64] Vinyals, O., et al. *Matching networks for one shot learning*, Advances in Neural Information Processing Systems. 2016.
[65] Frome, Andrea, et al. *Devise: A deep visual-semantic embedding model*, Advances in neural information processing systems. 2013.
[66] Akata, Zeynep, et al. *Label-embedding for image classification*, IEEE PAMI (2016): 1425-1438.
[67] Vinyals, O., et al. *Matching networks for one shot learning*, Advances in Neural Information Processing Systems. 2016.
[68] Romera-Paredes, Bernardino, et al., *An embarrassingly simple approach to zero-shot learning*, ICML 2015.
[69] Xian, Yongqin, et al. *Zero-shot learning-A comprehensive evaluation of the good, the bad and the ugly*, IEEE PAMI 2018
[70] Yuan Ai, et al., *Edge computing technologies for Internet of Things: a primer, Digital Communications and Networks*, Volume 4, Issue 2,

Embedded real-time multimodal analysis, image segmentation and classification algorithms for industrial applications, has been implemented in specific embedded boards (DSP, FPGA) with successful results[71]. However, the irruption of Deep Learning[72] and the increased amount and more in-depth data provided by spectral sensors is changing the field. Deep Learning provides more accurate and robust results regarding almost any application, whilst reducing the time to deploy or update the processing for new inspection tasks, i.e. by replacing the AI model with a "non-programming" approach.

The new heterogeneous architectures combining powerful CPUs with GPU/TPU/FPGA in a system-on-chip (SoC) [6] make possible the embedded deployment of these models. Even so, the high computational cost of CNNs usually requires accuracy trade-offs in order to grant is applicability in real-time. This is especially true when dealing with 3D data, e.g. hyperspectral imagery, where 3D CNN have shown to increased performance over solutions applying spectral analysis on a pixel by pixel basis[73,74,75]. Novel RT approaches apply dimensionality reduction pre-processing steps, whilst other use simplified network architectures to reduce complexity[76]. However, the development of novel efficient Deep Learning processing pipelines for embedded real-time hyperspectral data analysis is still needed.

Figure 17. SoC systems for IoT embedded devices demanding AI inference capabilities. From left to right: Nvidia Jetson Xavier[77], Xilinx Ultrascale+[78], Coral edge TPU SoC[79].

Building on the expertise of AIMEN in the development of computational imagers with embedded platforms[80,81,82,83], MULTIPLE will develop smart monitoring edge IoT solutions combining state of the art photonic sensors with the latest heterogeneous SoC boards able to deploy AI neural models. An off-the-shelf heterogeneous embedded SoC hardware architecture (e.g. Jetson Xavier) will be selected tailored to the needs of each specific sensor, finding an optimal trade-off among performance, power consumption and cost. MULTIPLE will develop novel real-time capable Deep Learning processing pipelines and artificial network architectures for hyperspectral image analysis, including efficient pre-processing, regression, segmentation, and classification tasks. The embedded system will allow concurrent processing, communication and real-time closed loop control.

<u>Cloud-based development and deployment of monitoring systems</u>

The performance of every machine learning model -from linear regression to CNNs- is bound by the training data, i.e. the dataset it is learning. While deep neural networks allow to model complex non-linear relations and are capable to capture high-level semantics at a new scale, they require large datasets for training.

Therefore, a proper and systematic gathering and management of data is essential for data-driven workflows built on state-of-the-art machine learning. In this regard, there are several aspects to address, namely:

- Search and analytics functions

2018, Pages 77-86.

[71] Bhowmik et al. *Embedded vision systems: A review of the literature,* in 14th Int. Symposium on Applied Reconfigurable Computing 2018.

[72] He, K., et al., *Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification,* in IEEE Int. Conf. Comput. Vis., pp. 1026–1034 (2015)

[73] Li, Ying et al., (2017). *Spectral–Spatial Classification of Hyperspectral Imagery with 3D Convolutional Neural Network.* Remote Sensing.

[74] Al-Sarayreh, Mahmoud et al., *Deep Spectral-spatial Features of Snapshot Hyperspectral Images for Red-meat Classification,* 2018 International Conference on Image and Vision Computing New Zealand (IVCNZ) (2018): 1-6

[75] He, Mingyi et al., (2017). *Multi-scale 3D deep convolutional neural network for hyperspectral image classification.* 3904-3908. 10.1109/ICIP.2017.8297014.

[76] Shuaifeng Z. et al., *Toward real-time 3D object recognition: A lightweight volumetric CNN framework using multitask learning,* Computers & Graphics, Volume 71, 2018, Pages 199-207

[77] https://developer.nvidia.com/embedded/buy/jetson-agx-xavier

[78] https://coral.withgoogle.com/products/dev-board/

[79] https://www.xilinx.com/products/technology/ultrascale-mpsoc.html

[80] Méndez-Rial, Roi, et al., *MWIR infrared gating imaging with uncooled PbSe FPAs for surveillance applications,* in Unconventional Optical Imaging, vol. 10677, p. 106772R. International Society for Optics and Photonics, 2018.

[81] https://www.clamir.com/wp-content/uploads/2017/07/CLAMIR-datasheet.pdf

[82] Garcia-Diaz, Anton, Roi Mendez-Rial, et al., *Embedded IR Multi Aperture Imaging for Surveillance Applications* (2017)

[83] Méndez-Rial, Roi, et al., *Embedded video rate super-resolution in the infrared with a low-cost multi-aperture camera,* in Unconventional Optical Imaging, vol. 10677, p. 1067705. International Society for Optics and Photonics, 2018.

- Data annotation and augmentation
- Model training
- Model deployment

An effective approach for search and analytics operations on big data sets should be scalable and efficient, capable to run in real time. Elastic Search, an open source[84] JSON-based search and analytics engine provides a distributed, multitenant-capable full-text search and has pushed the state-of-the-art bringing horizontal scalability and very high reliability. Complemented with Kibana[85] -a HTTP web interface for data visualization- and Beats and Logstash for data ingest, it renders a really powerful engine -completely open source- for data exploration and management (i.e. data search and analytics).

A common problem of deep learning is the need of powerful tools for model development, training, validation, and deployment of any model. In this regard, we will adopt TensorFlow[86] as end-to-end open source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and resources that lets easily build and deploy ML powered applications. It facilitates training and deploying models in the cloud, on-premise, in the browser, or in embedded devices. It features adapted versions of the framework (TensorFlow Lite, Tensorflow JS) that are adequate for edge-computing devices and allow running models trained in the cloud in embedded devices, overcoming library version compatibility issues.

Building on current cloud services for media asset management and image and video processing with deep learning models available at ABRAIA[87,88] and the extensive expertise of the team in industrial applications[89,90], MULTIPLE will build a new cloud service capable to manage hyperspectral images and spectral data besides other relevant information on the process to easily create and deploy deep models. It will leverage on Elastic Search, Kibana, and Tensorflow. Moreover, the service will facilitate the selection of the best deep learning architectures for each specific use case, taking also into account the restrictions of the embedded platform used.

Inspired by the success of the engineering workflow of traditional machine vision systems -with specific development environments[91] and vision libraries- , MULTIPLE will build a platform for the seamless development of deep learning models, harnessing the power of embedded computing platforms and cloud data and model management. The diversity and representativity of the three scenarios selected makes the case for a successful and comprehensive approach.

1.4.2 Innovation Potential

Innovation readiness:

The innovation resulting from MULTIPLE project comprises: different sensing VIS-SWIR hyperspectral solutions to provide multimodal monitoring capabilities (spatially resolved in real time) to be embedded into the production processes, including a cloud solution for orchestration of interrelated monitoring systems and complex decision support tools and the necessary deep models for regression, classification and real-time control, resulting into an integral photonic process monitoring, quality assurance and process optimization solution. These developments will enable the adoption of a real smart monitoring into EU manufacturing industry enabling highly efficient production lines and, thus, increasing the competitiveness of the European production industry and significantly contributing to its digitization. During the project, these solutions will be developed and demonstrated

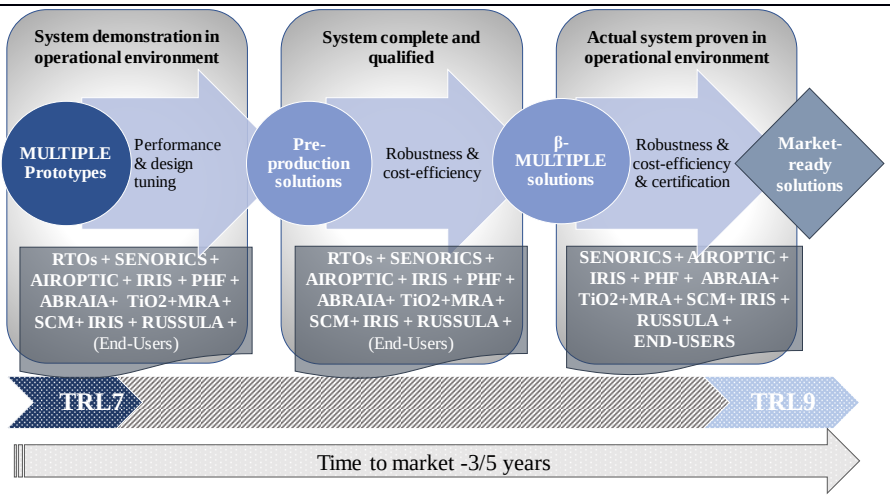


Figure 18. Evolution path beyond MULTIPLE project.

[84] https://github.com/elastic/elasticsearch
[85] https://github.com/elastic/kibana
[86] https://github.com/tensorflow/tensorflow
[87] https://abraia.me/docs/api/
[88] Jorge Rodríguez Araújo, et al., *Transfer learning for image tagging in fast fashion workflows*, WGML 2018
[89] Rodríguez-Araújo, Jorge, et al., *OpenLMD, multimodal monitoring and control of LMD processing*, In High-Power Laser Materials Processing: Applications, Diagnostics, and Systems VI, SPIE Photonics West, 2017.
[90] https://openlmd.github.io/
[91] https://www.mvtec.com/products/halcon/work-with-halcon/hdevelop/

in a relevant industrial environment for 3 different sectors (steel production, wood & furniture and food) and production concept (from continuous production to piecewise manufacturing) delivering highly efficient processes and manufactured products (TRL7). Namely, solutions will be demonstrated by assessing the compliance of the MULTIPLE-enabled manufacturing process and products with the end-user requirements (WP6). Once this phase is fulfilled, and ***beyond MULTIPLE project,*** an evolution of the solutions is devised based on Figure 18 dataflow.

At initial stages, contribution of RTOs will be required in combination with the effort of the Technology Providers (TPs) and support of technology integrators and users. In close-to-market stages, the main role will be given to industry partners and the solutions will be finally tuned for each targeted application.

All involved companies have a long track in the market and a deep knowledge and understanding of the innovation and commercialisation process as they have participated regularly in design and development of new products and services (detailed capabilities and expertise in section 4.1). In this sense, the involvement of market leaders and key players on providing solutions to industry like PHF, SCM, RUSSULA, IRIS or MRA & TiO2 (Grupo ALAVA) ensures a strong position for market uptake. Similarly, the early involvement of major manufacturers like ROYO, LDL or JOTIS also reinforces position in market uptake. Complementarily, EPIC and TEMATYS will bring in their expertise in technology transfer and networking capabilities to create awareness in target markets.

Activities to progress TRL to a market-ready status will require the mobilisation of financial resources. The initial approach of industrial partners is based on a mix comprising at least 15% own funds, ~50% in soft loans and bank loans (pre-production) and 35% from national innovation programmes. Synergies with ESIF will be sought, alignment with S3 of the relevant regions is ensured since advanced manufacturing and digital transformation are supported. Complementarily, support by EEN will be considered for advice on available funding and guidance or coaching. These options will be further defined in the Exploitation Plan, where a financing plan will be included.

Innovation management:

During the innovation process described, some risks may be eventually faced. Several measures have been considered:

Table 1. Market and Technical risks beyond MULTIPLE project

Risk	Mitigation measures
Unmet industrial needs	Industrial needs will be considered since the start and will go on during the whole project (WP1, WP5 & WP6). Participation of key industrial actors in the value chain in the project development (TPs and TUs) ensure the quality and pertinence of the input provided.
Endangered IP	A dedicated body, Innovation Board (IB) integrated by experts on business skills and IPR (Intellectual Property Rights) issues, will set IP protection/disclosure strategy.
IPR conflicts among partners	A clear framework regarding results ownership and access rights will be defined in the Consortium Agreement. The IB will oversee this type of situation to find suitable agreements.
Endangered freedom-to-operate	Regular surveillance will be made on IP landscape for early detection of possible conflicts (WP7) and, if needed decisions and agreements made (open innovation agreement, cross-licensing, etc.)
Endangered market position	Technology Watch activities on competing technologies will be done in WP7. Relevant findings will be positioned against MULTIPLE solutions and marketing strategy revised accordingly.
Availability of financial resources	In WP7, partners will work on defining a financial plan for the activities beyond the project to mobilise and ensure the necessary resources.
Modification of regulations and/ or standards	Regular surveillance on Regulations and Standards for early detection of possible conflicts with project developments. Specific activities in WP7 will address policy-makers while involvement of relevant market players (PHF, TiO2, MRA, SCM, IRIS, RUSSULA, LDL, ROYO, JOTIS, etc.) ensure that a communication flow can be opened with relevant policy-makers in different sectors and standardisation organisations, through the participation of EPIC.

Market Potential:

MULTIPLE scalable monitoring and optimisation solution, comprising several photonic sensing technologies from **OLED-based sensors and laser-based analysers to multimodal monitoring solutions for real-time (RT) monitoring through the implementation of RT embedded multimodal monitoring systems supported by cloud services for the overall orchestration of high dimensional monitoring systems and process optimisation decision supporting tools** for continuous production and piecewise manufacturing, both, tackling the customized demand of different representative cases of the wide manufacturing industry (steelworking, woodworking & furniture and food sector)**. MULTIPLE enables automated, zero-defects, flexible, rapid and cost-effective manufacturing** of products for the whole manufacturing sector highly positive impacting the **EU production industry competitiveness and its digitization.**

Market opportunity. Due to new horizons in manufacturing opened by Industry 4.0, phenomenal cost efficiency, better product quality and increased worker safety is expected, leading to a tremendous growth of the Industry 4.0 market, which is one of the fastest growing areas for digital technologies[92]. Digitisation of products and services can add more than €110bn of annual revenue for industry in EU until 2020. **North America and Europe region**s **would dominate the industry 4.0 market** by the 2018-2024 forecasted period[93]. However, the main risk for EU companies comes from China, which is one of the countries that stand to gain the most from automating and digitizing labour-intensive manufacturing processes. In order to keep EU manufacturing industry at the forefront of the technology and competitiveness it is crucial to keep the investment on R&D, specially focused on digitization.

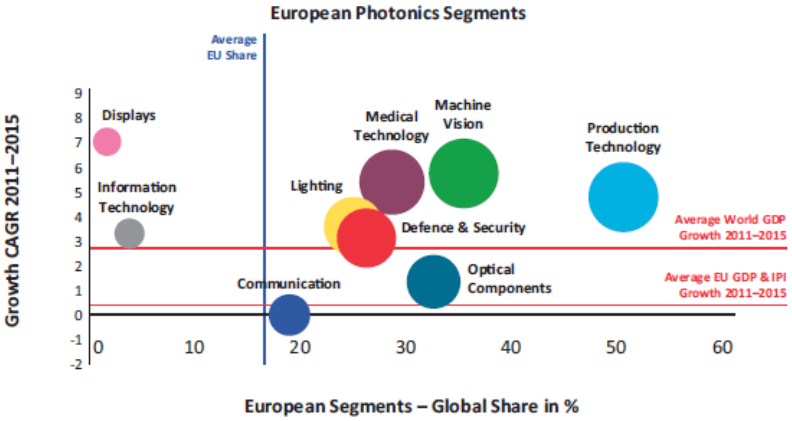


Figure 19. Average Growth rate of EU Photonics segments in the period 2011-2015 vs Average EU GDP &IPI Growth as well as vs Average World GDP Growth.

The photonics sector is of utmost importance in materializing the Industry 4.0 concept and the digitization of the EU manufacturing base. Photonic is important in itself; with a global market of €447bn in 2015 the sector is growing twice as fast as the global GDP. Seen on a European level photonics outgrows GDP by even 3.5 times. Europe is the second largest producer of photonics products behind China with around 300,000 people employed directly in the photonics sector. EU Photonics holds the 35% of the global market share for Optical Measurement & Image processing.[94].

The sub-segment of optical measurement includes a large variety of systems and components, mainly for industrial applications. The major products in terms of market volume are photonic sensors (binary sensors), spectrometers, semiconductor measurement systems, and geometrical measuring systems. Market growth during the last decade was supported by many sub-segments, including spectrometers[95].

As of 2017, the photonic sensors market was valued at USD10.5bn and is expected to reach a value of USD 23.18bn by 2023, registering a CAGR of about 14.11% during 2018-2023[96]. Photonic sensors are capable of providing highly accurate results in harsh environments, which cannot be achieved using conventional sensors. Thus, based on these factors the impact of this driver is expected to increase in future[97]. Specifically, silicon photonics is an evolving branch of photonics that offers a clear advantage over electric conductors used in semiconductors. These are used in high-speed transmission systems. The market is expected to invest heavily for the mass adoption of this technology, which will be a key aspect in the development of high-performance computing and improve performance and efficiency. Thus, the emergence of silicon-based photonics is driving the growth of the market[96].

The major products in terms of market volume are photonic sensors (binary sensors), spectrometers, semiconductor measurement systems, and geometrical measuring systems. Market growth during the last decade was supported by many sub-segments, i.e. spectrometers[98]. High price of spectrophotometer is a major restraining factor for the market growth[99]. MULTIPLE-enabled laser-based spectroscopy is built on uncooled inexpensive MWIR sensors, ensuring customised cost-effective solutions.

Laser-based gas analyzers have proven technologies that have different capabilities to cater to critical process industries. However, the market growth depends on key purchase influencers such as total cost of ownership, ease of operation and maintenance, and so forth. Given these conditions, laser-based spectroscopy analysers will expectedly gain the market share at the expense of traditional analyzers such as Zirconia and Paramagnetic based analyzers[100].

The sub-segment of imaging and image processing, the most important tasks of image processing systems in industrial manufacturing are production monitoring and control as well as quality assurance. The global machine vision market size was valued at USD9.95 bn in 2017. Increasing demand for quality inspection and automation in different industrial verticals is estimated to drive the market. These systems particularly assist in supervising work

[92] Markets&Markets. Industry 4.0 Market - Global Forecast to 2022
[93] Zion Market Research. Global Industry 4.0 Market Zion Market Research
[94] PHOTONICS21. Jobs and Growth in EU – Realizing the Potential of Photonics. PPP Impact report 2017
[95] PHOTONICS21. Market Research Study Photonics 2017
[96] Mordor Intelligence. Photonic Sensors Market. 2019-2024
[97] AlliedMarketResearch. Photonic Sensors Market.
[98] PHOTONICS21. Market Research Study Photonics 2017
[99] Coherent Markt Insights. Spectrophotometer Market. 2018-2026
[100] Frost&Sullivan. Frost Perspectives: Laser Based Gas Analyzers: An Emerging Technology Witnessing Significant Growth Prospect. 2017

environments. It offers features such as robotic guidance, automatic inspection, and process control in industrial applications[101]. In MULTIPLE, we propose an ensemble of new monitoring systems built on breakthrough OLED sensors and snapshot mosaic filters that make a leap in spectral range, spectral sensitivity and selectivity, and dynamic range monolithic uncooled CMOS cameras.

Hyperspectral imaging (HSI) combines the power of digital imaging and spectroscopy. As such, the use of hyperspectral cameras goes far beyond the standard camera-based vision systems more widely spread in the industry. Hyperspectral imaging offers several advantages, such as higher specificity and granularity, higher image acquisition speed as compared to other spectral imagers, and non-invasive imaging[102]. Technological advancements in the design of the sensors and cameras also support the growth of the hyperspectral imaging market. The manufacturing of the compact, lightweight, and low-cost peripheral devices that are used in industrial applications is propelling the market. However, a major hindrance to the growth of market is the relative complexity in analysing and processing excessive amounts of imaging data gathered using hyperspectral imaging system. Moreover, the high cost of cameras is another challenge for the market growth, which can be overcome by advancements in technologies[103]. The global hyperspectral imaging systems market was valued at USD 129.3 million in 2015 and is projected to reach USD232.8Mn by 2021, at a CAGR of 10.4% from 2016 to 2021[104]. Specifically, hyperspectral cameras segment led the global hyperspectral imaging systems market in 2015. This segment was valued at USD79.9Mn in 2015 and is projected to reach USD151.6Mn by 2021, at a CAGR of 11.3% during the forecast period[104].

The market shares of the Photonics producing countries have changed substantially during the last decade. While China gained market share, Japan, Europe, and Taiwan lost share. Europe lost market share during the last four years mainly due to three reasons; (i) the increased competition from China, (ii) the decrease of the value of the euro versus the currencies of most of the other Photonics producing countries (except Japan), (iii) the comparably low economic growth rate in Europe versus other regions. market share losses of Europe to China mainly concern photovoltaics, but also communication and lighting. Minor market share losses to China are also observed in production technology and for optical components. However, **Europe's highly dynamic photonics industry is already the world's innovation and market leader** in many of the products and services that will power the digital economy of the 21st century. Some 5,000 technology-intensive companies (many of them SMEs) directly employ over 300,000 people[105]. A global market share of 15.5% leaves Europe second only to China. Photonics is a fiercely competitive global market and expecting to triple annual European production to more than €200bn by 2030 is realistic – provided EU can stay at the forefront of photonics innovation. The photonic sensors are used in multiple applications, since these devices are able to **provide accurate cost-effective image sensing solutions** as compared to conventional sensors[106].

MULTIPLE solutions will **largely impact on several crucial manufacturing sectors for the EU economy**, starting from the Photonic sector and generating spill-over effects over the large manufacturing industry, with a first round of impact over the sector depicted in Table 2 and firstly validating and adopting MULTIPLE solutions.

The EU continuous processing industries as well as the piecewise manufacturing sector are facing the continuous struggle to keep a leading role in the worldwide competition. Innovation and competitiveness are therefore at the heart of the Commission's agenda and, as MULTIPLE stand on the brink of a new industrial revolution. MULTIPLE is committed to supporting the transformation of EU industry[107].

Continuous production system involves a continuous or almost continuous physical flow of materials. It makes use of special purpose machines and produces standardized items in large quantities (i.e **steel**, petrochemical, cement, sugar and fertilizer industries, etc.). Process industry is facing several challenges, including competitiveness, energy and resource efficiency, as well as safety that require a **radical change in the control and optimization of its own processes along the whole value chain**, from the extraction of natural resources to the end-of-life management of products. The implementation of virtual dynamic factories connected in real-time to plant operation is another concept that will radically change the way to operate plants by providing on-line information, control, monitoring and optimization of critical processes. A common challenge in industrial processes is the harsh environment, and various sensing strategies have been proposed to cope with this difficulty, including the use of remote sensors, robust embedded sensors, as well as disposable sensors. **In order to realize these changes, more precise information about processes is required, and therefore better sensing capabilities[108].**

MULTIPLE will tackle a representative use-case for continuous production onto a backbone of the EU

[101] GrandViewResearch. Machine Vision Market 2018-2025.
[102] Markets&Markets. Hyperspectral Imaging System Market 2017-2023.
[103] Global Market Insights. Hyperspectral Imaging market size 2018-2024
[104] MicroMarketMonitor. Hyperspectral imaging systems market. Global forecast to 2021.
[105] PHOTONICS21. Europe's age of light!. Photonics21 vision paper. 2017
[106] Allied Market Research. Photonic Sensors Market
[107] https://ec.europa.eu/growth/industry_en
[108] ACREO. Novel sensors and research results for industrial harsh environment processes. 2015

manufacturing industry, the Steelworking sector. With some 500 production sites across 23 Member States, employing around 330,000 workers, steel is not just a truly European industry but is also the basis of most major EU industries, notably construction and automobile. The EU is the second largest producer of steel in the world, although far behind China[109]. Today's increasing quality standards and higher energy costs demand more accurate and consistent temperature control in the steel making process[110]. Reducing energy consumption due to an optimal temperature delivery at the furnace stage will create important economical savings for manufacturers (energy constitutes a significant portion of the cost of steel production, from 20% to 40%[111][112]), greatly impacting the global industry competitiveness and generating a positive spillover effect on the downstream industries. Moreover, the multispectral data will allow obtaining the spectral signature of the rolled material, which will provide information of its composition and properties revealing the materials that compound the billet. This would provide the option to perform a quality control of future casts or infer certain properties of a casting from its spectrum only, heavily impacting over the clients' confidence onto the potential acquire products. Through-process quality control and process optimisation on rolling industry has been set a key need[113]. MULTIPLE will deploy its developed solution onto a rolling plant of a steel lamination factory as a proof-of-concept of the steelworking sector.

Moreover, **MULTIPLE** will demonstrate a representative use-case implantation for piecewise manufacturing onto a relevant EU manufacturing industry, **the wood and furniture manufacturing industry.** One quarter of the world's furniture is produced in the EU. In 2010, about 940,000 European workers were employed in approximately 130,000 firms. In the same year, the sector's production amounted to more than €83bn with a value added of nearly €29bn[114]. Germany, Italy, Poland and France ranked among the top 10 furniture manufacturers worldwide and held a combined share of 17% of world production and almost 60% of EU production. The furniture sector stands to benefit from the EU Flagship Initiative on Industrial Policy in the Globalization Age launched within the Europe 2020 strategy

Furniture retailers are increasingly aware of the complex consumer decision making process that makes up the buying journey when it comes to large ticket items[115]. While online continues to grow in importance (representing the 24% of total sales[116]), consumers combine online research with shop visits on their way from research and consideration through to buying. When talking about online selling channels, customers confidence is of utmost importance and thus, guaranteeing the whole finished product bunch quality is vital to be globally competitive. Moreover, in the bathroom furniture sector, it is all about holistic interior design demanding high-quality modular bathroom furniture[117] (increasing the retailer's pressure over finished products quality).

MULTIPLE will deploy an overall monitoring system tackling key bathroom furniture manufacturing parameters through the whole production process (from formaldehyde detection to elasticity, density, color and superficial defects), increasing product reliability and final quality and, thus, bringing to end-users adopting MULTIPLE solutions an advantage over their competitors.

Finally, **MULTIPLE will be demonstrated on a 3rd Use-Case onto the food sector, a strategic sector for EU and photonics**. Europe is the world's largest exporter of agricultural and food products, with the sector responsible for 7% of all jobs and 6% of European GDP[118]. From precision farming to food processing, Europe is the technology leader and will therefore play a central role in meeting EU challenges[119]. Photonics will help supply safe, nutritious and affordable food for all and establish a sustainable value chain from farm to fork. By using ever more precise sensors and measuring devices, farmers, food processors and ordinary consumers will be able to monitor and certify the safety, quality, content and even the origin of food – anytime and anywhere. Machine vision is to gain considerable momentum in the food sector over the coming years owing to its abilities such as improved detection of objects, enhanced analysis, monitoring tolerance, and accurate component measuring[101]. Within the whole food sector, MULTIPLE will demonstrate its advantages onto the chocolate manufacturing (which is a relevant sector). The global Cocoa & Chocolate market is valued at USD49 bn in 2018 is expected to reach USD68.7 bn by the end of 2025, growing at a CAGR of 4.2% during 2019-2025[120].

MULTIPLE will deploy an overall monitoring system tackling key chocolate manufacturing parameters through the whole production process (from fat content and type to viscosity, tempering gradient or chocolate crystallisation), increasing chocolate reliability and quality and, thus, enabling the production of a high-quality chocolate enabling

[109] European Commission. Industry in Europe. Facts&Figures on competitiveness & innovation. 2017
[110] Solutions for steel applications. Lumasensinc
[111] Saving One Barrel of Oil per Ton (SOBOT). American Iron and Steel Institute
[112] The State-of-the-Art Clean Technologies for Steelmaking Handbook, Asia Pacific Partnership for Clean Development and Climate
[113] Metal Magazine. Issue 02-2018. The Impact of Industry 4.0 on Long Rolling
[114] Centre for European Policy Studies. The EU furniture market situation and a possible furniture products initiative. 2014
[115] Furniture Retail Market Trends and Statistics. 2017
[116] Pragma Consulting. https://kinandcarta.com/
[117] Cambista. The bathroom industry is booming. 2018
[118] EC "The EU's common agricultural policy (CAP): For our food, for our countryside, for our environment". 2014
[119] PPP Photonics21. Europe´s age of light! How photonics will power growth and innovation
[120] MarketWatch. Global Cocoa & Chocolate Market. 2019

better production parameters control, detection of foreign bodies and enabling adaptation to different consumers tastes, which is the main growth driver for the sector (mainly from Asia Pacific countries)[121].

Implementation of highly advanced smart broadband multimodal photonic sensing solutions operating in the spectral range for improving production process through the monitoring of relevant process and product parameters is of utmost importance for EU industry. MULTIPLE approach focusses on **cost-effective** process-integrated solutions that are optimized in terms of **speed**, **quality**, and **resource efficiency** while addressing **embedded pre-processing** and suitably interpreting the acquired raw data for the **optimization of the processes**. Moreover, the overall multimodal orchestrated approach proposed **is a game-changer in several sectors where zero defect manufacturing and right-first-time production is needed for keeping competitive, heavily impacting on those sectors firstly tackled by MULTIPLE: i) Steelworking; ii) Woodworking & furniture, and; iii) Food sector.**

Table 2. Market size and growth forecast for the initial applications of MULTIPLE.

Market	Market Size	CAGR	Segments of interest
Steelworking	The global steel demand exceeded 1.71 million tons in 2016 and expected to reach **USD 1.01[122] trillion** by 2025. North America & Europe are expected to remain dominant yet mature steel-consuming regions,	**+2.6% CAGR** (2017 to 2025)[122]	Steel production and transformation
Woodworking & furniture	The global furniture market size is expected to reach USD **718.3 billion** by 2025[123]. APAC was the leading region for the global wooden furniture market in 2017	**+5% CAGR** (2018-2022)[124]	Furniture and wood piecewise manufacturing
Food	The global Food retail market will reach **USD 8.5 trillion** by 2020[125]. Europe is the second largest market for food retail due to the presence of the large number of branded stores and discounters such as Aldi, Carrefour, Auchan and Tesco which offer fresh and branded products.	**+6.1% CAGR** (2014-2020)	Food processor and retailers.

Complementarily to the analysis of the market opportunities and trends, an analysis of the IP landscape has been made on the main developments of the project, modular production modules, integration and digital infrastructure, and monitoring and control systems, in order to **guarantee the project´s results Freedom-to-Operate**. While patents from 3rd parties are quite significant in terms of application of NIR to wood monitoring, they are not directly relevant to the project as they do not deal with HSI and even less so with integrative solutions with novel sensors as planned in the project, in an orchestrated manner. The case of chocolate monitoring is also quite free from IP regarding NIR/HSI in general and even more with our advanced approach and the same for the steel sector. Main findings show that existing patents refer to inventions based in methods using **MULTIPLE solutions is strong and clearly enforced by already MULTIPLE partners protected IP** as shown on following table:

Table 3. Partners' FTO-relevant IPR

Patent Nª	Owners	Title	Key Protect
EP18198458	IRIS	An integrated optical system for the examination of sample materials	Illumination system maximising signal to noise ratio of NIR systems especially useful for handheld and in line integration
EP19382101	IRIS	Device for detecting foreign matter in an edible substance or in a container containing the edible substance and method employing the device	Overall imaging device that provides a cost-effective solution for target foreign body detection
EP2669660	AIROPTIC	method and apparatus for remote detection of alcohol vapors in the atmosphere	Signal processing schemes
N.A. ***(Under Aplicattion)***	SCM	Use of thermography in belt sanding quality control	Use of thermography in belt sanding quality control for the woodworking sector
US2018359438	IMEC	Hyperspectral imager for snapshot and line-scan modes	Sensor for acquiring an image of an object
US2018204863	IMEC	Image sensor, an imaging device, an imaging system and a method for spectral imaging	Image sensor for spectral imaging
US2009201507	AIROPTIC	Wavelength modulation spectroscopy method and system	A method and system for measuring the concentration of a gas component in a gas

[121] Chocolate Industry Analysis 2018 - Cost & Trends
[122] GrandViewResearch. Global Steel Market Size. 2017
[123] GrandViewResearch. Furniture Market Size. 2018
[124] Technavio. Global Wooden Furniture Market 2018-2022.
[125] Persistence Market Research. Food Retail Market. 2015

			sample
US2007176079	PHF	Optoelectronic detector with multiple readout nodes and its use thereof	Invention which discloses an optoelectronic detector for sensing light characteristics
US2003136915	PHF	Photodetector and method for detecting radiation	A photoelectric semiconductor light-detection system with programmable dynamic performance for radiation detection
PCT/EP2016/06925	SENORICS[126]	Method for detecting and converting infrared electromagnetic radiation	The invention relates to a method for the detection of infrared electromagnetic radiation, particularly an organic infrared detector for (near) infrared detection.
WO2016055047A1	SENORICS[126]	Apparatus for spectrometrically capturing light with a photodiode which is monolithically integrated in the layer structure of a wavelength-selective filter	Important for developing a new spectrometer based on OLED detectors
WO2016055048A1	SENORICS[126]	Optical filter element for devices for converting spectral information into location information	Important for developing a new spectrometer based on OLED detectors

MULTIPLE consortium brings needed and committed parties covering the whole solution value-chain (Figure 20):

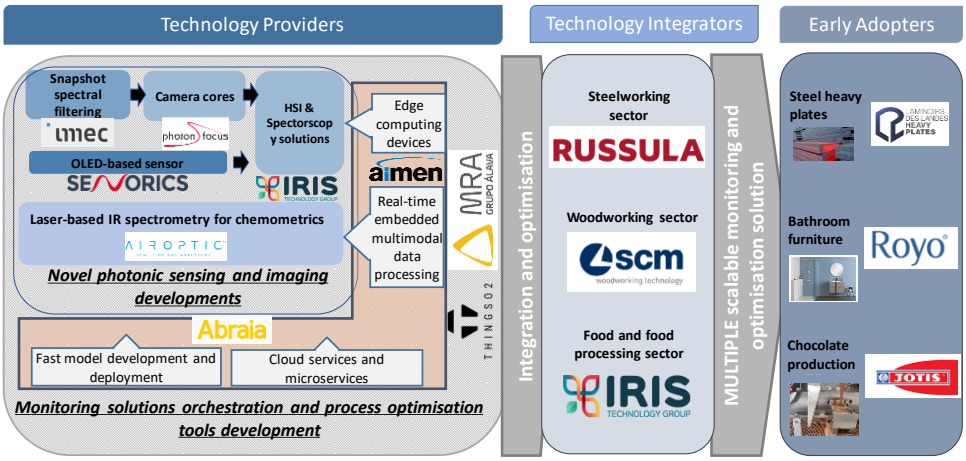


Figure 20. MULTIPLE value chain.

2 Impact

2.1 Expected Impacts

2.1.1 Expected Impacts according to the work programme

The MULTIPLE project expected impacts according to work programme topic "ICT-05-2019: Application driven Photonics components" and specifically to the Innovation Action "Sensor-Based Optimization of Production Processes" are presented in this section.

Increased competitiveness of the European production industry and significant contribution to the digitization of European industry

Strengthening market positioning and competitiveness of the MULTIPLE industrial partners. Firstly, industrial partners will be commercialising unique photonic solutions (SENORICS, IRIS, PHF, SCM, RUSSULA, AIROPTIC, ABRAIA and MRA) that can significantly improve competitiveness of relevant industries (steel production, wood & furniture and food) by, respectively, creating new market opportunities for integral photonic process monitoring, quality assurance and process optimization solutions, and by enabling the cost-efficient and flexible fabrication of better-performing products with reduced cost and resource consumption (materials and energy), both continuous and piecewise manufacturing processes. That is the case of industrial equipment manufacturers and integrators (SCM, RUSSULA and IRIS) and the End-Users (LDL, ROYO and JOTIS) who will benefit from market opportunities described in section 1.4.2 due to the production lines improvement depicted on section 1 (reduction of energy and raw materials consumption, higher production throughput due to diminished rejections and reworked pieces and reduced lead-time), leading to relevant increase on units sold and market share acquisition. A first market application for all industrial partners is depicted in the table below (estimations provided by the companies based on their own expansion plans, time frame 5 years, starting once the solutions are market-

[126] Patented by Technische Universität Dresden granting SENORICS exploitation exclusive license.

ready) demonstrating the impact of the project in the Consortium and clearly impacting over its competitiveness on the global market, increasing its turnover within 5 years of the MULTIPLE results commercialization estimated on almost €139M and 147 jobs generated:

Table 4. First Market uptake of MULTIPLE Results

	Product or service	Market(s) addressed	Estimated units sold	Turnover (€)	Job creation
IRIS	1. MULTIPLE scalable monitoring and optimisation solution integration and deployment for the food sector, as well as support for the wood one. 2. Turn key Line scan HSI or single band (Hypera systems) and NIR solutions for inline monitoring.	Food and food processing industries (beyond the project the pharmaceutical sector will be tackled) Food and woodworking and furniture sector.	82 327	1,148,000 € 1,635,000 €	6 job
PHF	Multimodal imaging solutions (CMOS cameras) for real-time monitoring	Manufacturing industries (steel, wood and food industries)	565	5,650,000	5 jobs
SCM	1. MULTIPLE scalable monitoring and optimisation solution integration and deployment for the wood and furniture sector 2. SCM machines incorporating photonic embedded quality solutions	Wood and furniture sector (beyond the project plastic, glass and other sectors will be tackled) Wood sector demanding drilling, edge-banding, sanding machinery or retrofitting existing machinery	88 1705	1,320,000 33,615,000	35 jobs
RUSSULA	MULTIPLE scalable monitoring and optimisation solution integration and deployment for the steelworking sector	Steelworking sector	72	1,152,000	3 jobs
AIROPTIC	1. Multicomponent in-situ analyser for optimization of combustion 2. Integrated control of wood production. 3. Multicomponent in-situ analyser for miscellaneous chemical processes. 4. Multicomponent extractive analyser for miscellaneous chemical processes.	1. Steel industry 2. Wood industry. 3. Chemical industry 4. Chemical industry	296 240 185 485	6,549,000 4,248,000 4,425,000 12,283,500	50 jobs
ABRAIA	Cloud micro services	Manufacturing industries (steel, wood and food industries)	207*	3,105,000	5 jobs
SENORICS	Improved OLED-based sensor to be integrated on VIS/NIR/SWIR spectroscopy solution	Spectroscopy manufacturers	830	830,000	5 jobs
ALAVA******	MULTIPLE scalable monitoring and optimisation solution	Monitoring systems integrators and providers	242	54,610,000	25 jobs
LDL	Improved heavy plates references optimally-produced	Offshore wind energy sector, naval, construction, industrial equipment, etc.	+5% turnover**	1,500,000	3 jobs

ROYO	Improved bathroom furniture references	Bathroom furniture market	+10% turnover***	2,500,000	6 jobs
JOTIS	Improved JOTIS chocolate references optimally-produced	Food retailers	+1,5% turnover****	5,325,000	4 jobs

** Number of cloud infrastructure adapted and service provided for a 4-years period.*
***+25,000 tonnes of heavy plates demand lead by cost competitive product (reduction of energy and raw material consumption and produced scrap and rejections).*
****+50,.000 units sold due to increase demand lead by production efficiency and cost reduction.*
*****+1,5% of company turnover increment due to the market share acquisition regarding high-quality produced chocolate.*
****** First market uptake expectations for the Grupo ALAVA companies together (MRA & TiO2), they both will be commercialising the MULTIPLE results, each of them attending different client and geographical markets (where they are already present) based on each own expansion strategy.*

Beyond this first application, the industrial partners will further exploit MULTIPLE results in the markets already identified in section 1.4.2. New applications will be considered on the basis of the MULTIPLE project results through the engagement of new technology adopters which benefits from the MULTIPLE results.

Reinforcing Europe's industrial competitiveness and leadership in photonics and beyond. Photonics play a dual role in the industrial sector: as a fast-growing and highly innovative sector itself, and as an enabling technology to help all other sectors of manufacturing compete successfully with their global rivals. In this framework, MULTIPLE addresses keystones of industrial competitiveness:

1. Improved business opportunities and value creation in Europe by reinforced cooperation along the value chain;
2. Measurable productivity and quality increase in manufacturing of parts, and sustained breakthrough innovations in high-added-value products fabricated in Europe in key sectors (steel production, wood & furniture and food) and helping in the overall industry digitisation process.

MULTIPLE will strengthen EU photonics manufacturing base. Nowadays EU has a leading position in photonic production technology (equipment for laser materials processing) and in optical measurement & image processing (mainly for industrial applications as components of monitoring and control systems), holding 50% and 35%, respectively, of global market[127]. However, this position is continuously at stake by the advance of China, US and Japan. Moreover, cost-efficiency, flexibility, high productivity and quality attained thanks to MULTIPLE solutions will allow **laser, optical measurement and image processing equipment, companies in Europe to compete glob ally through a breakthrough technology in manufacturing**, thus benefiting from business opportunities in fast growing markets as presented in section 1.4.2.

MULTIPLE has also the potential reinforcing competitiveness and leadership of other industries. Photonics **is a driver to competitiveness of at least 10% of the EU economy** since they are in the backbone of key manufacturing sectors (i.e., steel production, wood & furniture and food industries)[128]. Namely, laser-based IR, NIR spectrometry and multimodal imaging solutions are key photonic ingredients for the 4th industrial revolution by enabling automated, zero-defects, flexible, rapid and cost-effective fabrication. In this context, **broadening the spectrum of application laser, sensing and imaging measurement technologies are of paramount importance for EU industry** and to do so, Photonics community is focused on the development of advanced online sensors, 3D machine vision and imaging (process monitoring, quality control and high precision and interruption-free production process), IR (reducing cost-intensive downtime through less equipment breakdowns) and optical computing (AI and deep learning for optimisation process)[129]. **MULTIPLE will develop solutions for manufacturing process,** resulting in an **orchestrated high dimensional monitoring and process optimisation solution,** thus e**nabling onsite and online process control** (cost-effective spectral sensor solutions, deep models) **and quality assurance and process optimization** (cloud services boosting new interrelated monitoring systems and complex decision support tools). Moreover, such solutions will provide **cost efficiency** (downtime and inventory reduction, improved production quality and thus, reducing scrap and rejections), **waste reduction, energy savings and improved productivity** (zero-defects, reduced lead-time). Therefore, thanks to **MULTIPLE results, EU industry (**firstly validated on steelworking, wood & furniture or food industries, but easily transferable to other manufacturing sectors), will **gain a strong position to benefit from market opportunities like those considered in section 1.4.2**. In this context, **MULTIPLE has the potential for a relevant impact in EU manufacturing industry at large**.

[127] PPP Photonics21. Market Research Study PHOTONICS 2017.
[128] The leverage effect of Photonics Technologies: the European Perspective. Photonics21.
[129] PPP Photonics21. Europe's age of light! How photonics will power growth and innovation.

An estimation is shown in terms **growth and jobs** in the table on the right (including the impact over the Photonic sector itself), **due to the increased competitive gained through the adoption of MULTIPLE solutions**. Global market size based on the projections included in section 1.4.2 (as close-to-market value when MULTIPLE solutions are adopted). Attainable market estimation is based on the potential of MULTIPLE solution to enable EU industry to provide products addressing relevant market needs (section 1.4.2).

Target market	Global Market Value (M$)	Current EU share	Attainable additional market share (M$)		Job Creation
Photonic	928.510	17,0%	1,2%	11.142	37.069
Steel	1.010.000	18,5%	0,2%	2.020	36.334
Wood & Furniture	718.300	25,0%	0,5%	3.592	13.467
Food	8.500.000	17,3%	0,1%	4.250	16.642
			TOTAL:	**21.004**	**103.512**

*Ratio for Job caluclation based on turnover per employee in Europe (source: Eurostat)

Table 5. MULTIPLE impact on EU Industry Growth and Jobs

MULTIPLE will, therefore, significantly contribute to **safeguard value and job creation, especially high-skilled and high added-value jobs**, always linked to production activities **and support reshoring of EU industry**. The main drivers for reshoring are "production costs, quality and flexibility improvement, access to skills and knowledge, risk reduction and market access"[130,131], all of them directly linked to MULTIPLE objectives and impacts. Moreover, keeping EU manufacturing industry at the forefront thanks to photonics is key for Industry 4.0 approach and, thus, contributes to safeguard the future of the EU manufacturing sector (30M jobs directly and twice indirectly, and to 80% of total EU export), specially for SMEs (photonics sector is largely based on SMEs, more than 5,000 photonics SMEs in Europe[94]), therefore, creating proportionally more jobs than would typically be seen in a sector made up primarily of big companies.

2.1.2 Other Expected Impacts

Enhanced innovation capacity

MULTIPLE solutions and the knowledge and know-how developed during the project, **will set the foundations to further innovate in photonics, laser-based and IR spectral sensing, deep models and cloud computing**. Firstly, partners will gain capacities to:

- Develop spectral sensor solutions by novel hyperspectral imaging and laser based spectral sensing concepts.
- Develop embedded deep models for regression, classification and real-time control, on latest SoC technology.
- Develop a multiple-spectral-band photonic monitoring system to sense production process at different stages and in different scenarios: steelworking, woodworking and food industry.

All in an open innovation framework that will allow SMEs involved in the project (ABRAIA, IRIS, AIROPTIC, SENORICS and PHF) to operate within a streamlined, market-oriented approach supported by major industrial partners and RTOs. This idea is developed through specific technology transfer and training activities (WP7). The project will also impact the **innovation capacity beyond the consortium**:

- Availability to EU industry of novel photonic multispectral sensing, AI and cloud computing technologies, flexible and adaptable to their specific needs (materials and processes), and with an established methodology allowing rapid, robust, flexible, efficient and secure manufacturing process monitoring and optimisation. Enabled freedom-of-design will boost product innovation.
- Availability to EU academic and industry of unique knowledge, methodology and tools to streamline R&I activities in monitoring and production optimisation. The MULTIPLE technologies through multiple-spectral-band photonic monitoring and process optimisation models, will greatly improve the efficiency, productivity and sustainability of industrial processes and innovative products.

Technology transfer and outreach activities to make the project results available to EU industry and academia are considered in the dissemination and communication plan (section 2.2) to ensure this impact.

Better jobs and new skills

Jobs created in relation to photonics applications require a skilled workforce, normally with knowledge on physics and engineering, materials and manufacturing processes, design and modelling and ICT tools. This skilled workforce is open to most of the innovative technologies and products, as well as open to innovation activities in all companies throughout EU industry[132,133]. The transition to laser-based and IR spectral sensing, deep models, digital value chains and other technologies that underpin Industry 4.0 will require many new kinds of specialists covering everything from product design to product approval and certification. Replacing conventional machinery with intelligent systems will also require the training of workers affected by this transition[105].

[130] Emerging trends in global AM: challenges, opportunities and policy responses. United Nations Industrial Development Organization.
[131] Stentoft et al, 2016
[132] COM(2010) 682 final An Agenda for new skills and jobs: A European contribution towards full employment
[133] A Skilled Workforce for Strong, Sustainable and Balanced Growth. https://www.oecd.org/g20/summits/toronto/G20-Skills-Strategy.pdf

MULTIPLE will contribute to high quality job creation in the photonics and manufacturing industry and beyond in high-tech sectors oriented to the development of ICT tools (IoT and Cloud). In addition, in WP7 **training content and activities will be carried out** to ensure that developed knowledge and technologies are transferred to industrial partners and their workforce. Training activities and contents for students and/or post-doc and professionals will be developed and open seminars targeting stakeholders (industry, education, standardization bodies, certification entities, etc.) will also be organised. These activities are important to address the reality of an aging workforce that may be not so familiar with these technologies.

Promotion of women participation in photonic and manufacturing industry will also be attained. In line with Women in STEM (Science, Technology, Engineering and Mathematics) Initiative and She Figures 2018[134], participation of female scientists and engineers in this project have been promoted. Moreover, involvement of women in the R&I and industry decision- making roles will be encouraged considering gender aspects in any dissemination and communication activity, as well as by specific activities targeting women.

Leveraging private investment

MULTIPLE project has a relevant industrial core comprising 4 high-tech SMEs and 6 major industrial companies which are very much interested in the results to be obtained and therefore are committed to bring them to the market. Indeed, it is highly relevant the involvement of major manufacturers ROYO, JOTIS and LDL, in the project showing their interest and commitment with the project results for fully implement then within their manufacturing processes. Effort and investment will be needed to take project results from TRL7 to TRL9. Normally, this would require **an investment at least equal to the budget mobilised for the project**. It is also relevant that industrial partners participating in MULTIPLE project will be only partially funded by the EC (30% of their budget will consist on own funds). Considering markets' demand for photonic monitoring solutions and the competitiveness of MULTIPLE solutions, also relevant investments can be expected by equipment manufacturers aiming to integrate MULTIPLE solutions in their systems, as well as by End-Users implementing MULTIPLE multimodal solutions for monitoring and optimisation of their manufacturing production lines, increasing the private investment leverage by MULTIPLE.

Contribution to standards

Standards contribute to remove technical barriers to trade, leading to new markets and economic growth for industry[135]. They also facilitate technology transfer and they contribute to ensure safety of products. Interaction with standardisation organisations and contributions (under the leadership of EPIC and TEMATYS as industry/market representatives and experts, respectively) to the ongoing and near future standardisation are considered in WP7 as key action to ensure compatibility and interoperability with what already exists, thus booting market acceptance. A first assessment on current standardisation activities and committees was performed and, MUTLIPLE, aims to contribute to ongoing and future standardisation developments at CEN and ISO level. In this sense, MULTIPLE brings the ideal partner, EPIC, representing the whole EU photonics industry interests in standardisation (ISO, CEN/CENELEC[136]) and linking MULTIPLE outputs with standardisation activities. Main contribution is expected with current RAMI 4.0 (the reference model for Industry 4.0) architecture.

Environmental impact

MULTIPLE objectives and corresponding proposed work packages and tasks are aligned with EU environmental challenges, contributing to sustainable production:

Boosting resources (energy and materials) **efficiency, emissions and waste reduction and circular economy**. MULTIPLE will enable the introduction of integrated photonic monitoring of manufacturing processes into new applications in key EU industrial sectors (steelworking, woodworking or food industry), leading to smart industrial processes that will contribute significantly to more **efficient use of resources** and meeting environmental challenges[137 138]. Zero-defects manufacturing approach enabled by MULTIPLE through integration of intelligent, autonomous, self–adaptive, self-powered and cost-effective sensors and actuators for process monitoring, control and quality management and process adaption by self-learning, quality and process data-based modelling of process behaviour will lead to significant quality enhancement and resource efficiency[139].

Smart applications like MULTIPLE solutions will help to exploit **energy efficiency** potential as they can provide transparency about energy consumption patterns (through sensing) and help to optimize a company's energy management (through regression and deep models) contributing to stabilize the energy system by either consuming and storing energy (in case of energy surplus) or on the opposite by reducing their consumption (in case of energy

[134] http://publications.europa.eu/en/publication-detail/-/publication/c118ea10-58fa-4173-a2c4-65c746918c20
[135] JRC study: *How will standards facilitate new prod. systems in the context of EU innovation and comp. in 2025?*. EU, 2015
[136] ISO/TC39-Machine Tools; ISO/TC39 SC10-Safety; ISO/TC184 SC1- Physical device control
[137] A Photonics Private Public Partnership in Horizon2020. Photonics21.
[138] Photonics Roadmap 2018 update. PhotonicsNL, Dutch Optics Centre, Photon Delta and partners.
[139] Factories 4.0 and Beyond. EFFRA 2016.

shortage) on demand enhancing carbon footprint reduction[140]. **MULTIPLE will result in up to 10% reduction in energy consumption.**

MULTIPLE results will also address **GHG emissions** problem, via implementation of smart technologies like sensors and data analytics[141]. Sensing improve efficiency gains resulting in reduction of company electrical demand, and these savings result, likewise, in lowering carbon dioxide emissions[142]. **MULTIPLE will reduce by up to an 15% of CO_2 and NOx emissions**.

MULTIPLE photonics technologies will lead the change from worldwide trend of 'take-make-use-waste' model to **circular economy**. Advanced measuring will contribute to increased performance of production processes, **reduction of waste**, 'first time right' and inherently reusable production products[138] through ultrahigh-resolution cameras, 3D imaging, optical sensors or augmented reality. IR sensors will detect moisture content in chocolate processing ensuring quality and reducing scrap[139, 143]. **MULTIPLE will reduce by up to a 10% waste and scrap due to manufacturing process quality improvement**.

Societal impacts

Enhancing life quality of European citizens. MULTIPLE embraces the opportunity to *tackle climate change, one of the main concern of EU citizens* (92% of EU citizens see it as a serious problem and 74% a very serious problem[144]), through the reduction of energy demand, use of resources on key polluting sectors[145], i.e. industry (accounts for 16% of total EU GHG emissions) and energy supply (principal source of GHG emissions in the EU with a 25% of total). Moreover, the reduction of GHG emission to the atmosphere contributes to improve air quality within the EU and worldwide. Estimates of *the health impacts* attributable to exposure to air pollution indicate that PM2.5[146] concentrations in 2013 were responsible for about *467,000 premature deaths* due to long-term exposure in EU[147].

Furthermore, thanks to early anomaly detections, MULTIPLE will enhance the safety of the food chain by enabling a more comprehensive detection of foreign bodies (vs. only metals nowadays) in the food products, as well as possibly avoiding cases of adulteration.

Regions and countries benefiting from highly productive industrial capacities developed at DIMOFAC will **generate skilled jobs with decent salaries**, thus contributing to **European social cohesion**. Increased efficiency will give more people access to products at lower costs and therefore **support standards of living and well-being**.

Contribution to the attainment of the objectives of EU Policies

MULTIPLE fully responds to the key focus area in LEIT¨-ICT Programme, i.e. Digitising and transforming European Industry and services. Moreover, by contributing to previous mentioned challenge, MULTIPLE will also support the realisation of EU policies which are in the core of Horizon 2020 programme: Europe 2020 (intelligent, sustainable and inclusive growth), an EU Industry for Growth and Economic Recovery (promoting industry growth up to 20% of the GDP), Digitising EU Industry (digital technologies for a more competitive industry), Energy Package (energy efficiency to attain 20% by 2020), Roadmap to a Resource Efficient Europe (by 2020, waste is managed as a resource and market and policy incentives that reward business investments in efficiency are in place) KETs for growth and jobs (exploiting the leverage effect of photonics and advanced manufacturing in the growth of industry and in producing solutions to respond to societal challenges). Furthermore, by the alignment of MULTIPLE results with Green Photonics, contribution is expected to Raw Materials Initiative and Circular Economy Package (use of recyclable materials and less precious and critical materials, zero-defect manufacturing). In addition, MULTIPLE results will enable technologies to realise the Sustainable Development Goals (SDGs) defined by UN and adopted by the EU in terms of Clean energy (SDG 7), decent work and economic growth (SDG8), sustainable industrialisation and innovation (SDG 9), responsible consumption and production (SDG 12) and climate action (SDG 13).

2.1.3 Barriers and framework conditions

- ***Technology qualification***. Innovation leads to the development of new technologies that involve novel designs, concepts or applications not covered by existing rules, normative standards and industry practices. MULTIPLE solutions provide a, fast, accurate and robust process monitoring, including embedded processing and machine learning for process optimisation. Such solutions will be developed within the framework of needs and requirements defined by industry as set in WP1, 2, 3, 4 & 5 and compliance will be demonstrated in WP6.
- ***Regulations & standards***. Industry standards and requirements need to be met by MULTIPLE solution and

[140] Accelerating clean energy through Industry 4.0. Manufacturing the next revolution. UN Industrial Development Organization. 2017
[141] M Rokonuzzaman. Industry 4.0 to drive Sustainable Development Goals: Part II. 2018.
[142] Could IoT Help Combat Climate Change? TechNative. 2017.
[143] The health of photonics. An Institute of Physics Report. 2018.
[144] European Commission. Citizen support for climate action. https://ec.europa.eu/clima/citizens/support_en
[145] European Environment Agency
[146] Particular matter with a diameter of 2.5 µm or less
[147] Air quality in Europe — 2016 report. European Environment Agency report nº28/2016.

produced products, not to hamper the exploitation of project results. An early consideration of industry standards and requirements to be met by MULTIPLE, as well as by the resulting components, is included in the project (WP1). Relevant industrial partners facilitate this activity. Demonstration activities in WP6 will ensure that industry requirements are met. WP7 activity includes the interaction with standardisation organisations and contributions to standards.

- ***Obstacles to innovation deployment***. Availability of financial resources for investments in technology are limited, even more restricted in the case of SMEs. In the development of the MULTIPLE solutions, cost-efficiency is considered from the early idea and as part of the overall strategy in the work plan, concurrently considering productivity, investment and operation costs when assessing the technology path to be followed (benchmarking activities, T6.4, are planned assessing MULTIPLE-developed solutions performance when compared to market-ready solutions). In addition, easiness of deployment in terms of integration in existing manufacturing facilities has also been considered by developing easily plug-in solutions while working from an early stage with key technology vendor/provider (MRA & TiO2, Grupo ALAVA) and integrators (SCM, IRIS and RUSSULA). Additionally, the uptake of any new technology requires knowledge acquisition (involved production staff) and, therefore, MULTIPLE will release training materials to both partners´ staff and stakeholders (WP7).

2.2 Measures to maximise impact

2.2.1 Exploitation Plan

A draft of the Exploitation Plan of Project's results (EP) is presented in this section, to be further developed and updated during the project (WP7) considering the information gathered from research and validation activities, as well as inputs collected from all stakeholders in the framework of the dissemination and communication activities.

Key Exploitable Results in MULTIPLE. The innovations to be generated during the project have been identified and categorized into those related with: i) the knowledge and fundamental principles, ii) MULTIPLE solutions (system & subsystems); iii) improved processes enabled by the new manufacturing solutions implemented at the End-Users facilities.

i) Knowledge and fundamental principles:

- *Principles and fundamentals of optoelectronic systems based on spectral sensors*. Knowledge on development multimodal monitoring modules and performing multispectral measurements and parameters detection (including image processing and analysis algorithms for inline real-time (RT) control of manufacturing processes).
- *Principles and knowledge for orchestration of monitoring systems,* including data treatment and visualization and model development.
- *Knowledge on process optimization.* Know-how and expertise on manufacturing process modelling and parametrization, data analytics and decision support tools for quality control and self-optimisation.
- *Principles for integration and industrial deployment of monitoring and control manufacturing solutions*. Expertise in system integration and End-User cases process parameters and production lines configuration.

ii) MULTIPLE solutions:

- *MULTIPLE scalable monitoring and optimisation solution (Key Exploitable Result, KER)*: An **overall orchestrated optimisation solution for complex manufacturing chains based on novel spectral sensing solutions** for monitoring and control in manufacturing production lines. This overall solution integrates different stand-alone solutions (other exploitable results, OERs) for monitoring in industry such as: i) **multimodal imaging solutions** for real time and robust monitoring; ii) an **embedded monitoring system** relying on embedded deep models for regression, classification and real-time control, running on latest SoC technology combining ARM processors, GPUs and/or FPGAs (aiming onsite and online measurements of physical magnitudes and classification of different types of categories with high flexibility, accuracy, and reliability); **Novel and cost-effective spectral sensor solution**s on iii) **laser-based IR spectrometry** and iv) **NIR spectroscopy for chemometrics**; v) Orchestrated of the photonic monitoring systems in order to provide **decision support tools for optimization of the production processes** based on cloud and big data technologies providing cloud services and microservices.

iii) MULTIPLE improved processes: Only a few specific applications demonstrated in the project (from bathroom furniture to steel-roll heavy plates or food products) are considered, although on the basis of the solutions developed, many other productions lines for process manufacturing can be implemented.

Planned key exploitable results, exploitation team and expected key areas of application. Results related to *knowledge and fundamental principles* will be used in new research and innovation activities developed by the RTOs (AIMEN and IMEC) with participation of industrial partners. Complementarily, industrial partners will exploit these results as integrated knowledge in the MULTIPLE solutions.

- *Principles and fundamentals of optoelectronic systems based on spectral sensors and on system orchestration* will be used in R&I activities to further develop the MULTIPLE scalable monitoring solutions and novel spectral sensor solutions.
- *Knowledge on process optimisation and principles for integration and industrial deployment of monitoring and control manufacturing solutions* will be a key input in R&I activities to improve productivity of the different production lines to be deployed on further applications and scenarios.

Applicability of Key and Other Exploitable Results (KER and OER) of the project is depicted in the table below:

Table 6. MULTIPLE exploitable results

Project Result	Developed by	Exploited by	Targeted Industries
KER1: MULTIPLE scalable monitoring and optimisation solution	IMEC/PHF/SCM/ AIROPTIC/ SENORICS/ TiO2/ MRA/ABRAIA/ AIMEN/ IRIS/ RUSSULA	Lead User: MRA/TiO2 Early adopter: SCM/RUSSULA/IRIS	Industry technology integrators and manufacturing industries (woodworking, steelworking and food processing) involving both continuous process or piecewise manufacturing.
OER1: Snapshot VIS/SWIR HSI imagers	PHF/IMEC/ SENORICS	Lead User: PHF Early adopter: MRA/TiO2	Manufacturing integrators, machinery manufacturers and suppliers of turnkey OEM monitoring equipment.
OER2: Laser-based IR analysers	AIROPTIC/AIMEN	Lead User: AIROPTIC Early adopter: MRA/TiO2	Manufacturing integrators, machinery manufacturers and manufacturing industry (End-Users).
OER3: Multimodal monitoring systems	IRIS/AIMEN/ SENORICS	Lead User: IRIS Early adopter: MRA/TiO2	Manufacturing integrators, machinery manufacturers and manufacturing industry (End-Users).
OER4: Cloud and edge computing platform and service	ABRAIA / AIMEN	Lead User: ABRAIA Early adopter: MRA/TiO2	Manufacturing integrators, machinery manufacturers, suppliers of turnkey OEM monitoring equipment and manufacturing industry (End-Users)

Exploitation Strategy.

Consortium industrial partners have shown their commitment with the project (and their envisaged results) not as a one-off project but rather as the new path for growth and income generation. Two different roles have been identified according to their position within the exploitation strategy and the benefits they will achieve: Lead Users and Early Adopters. Under Table 7 and Table 8 is depicted for each industrial partner their particular Exploitation Strategy regarding their profile, positioning and the arising business opportunity.

Table 7. Exploitation by Lead Users

Partner	Profile	Exploitation Strategy	Business Opp.
AIROPTIC [SME, PL]	It designs, produces and markets of advanced laser based real time gas analyzers for industrial process control, environmental monitoring and safety and security applications. In 2013 we have introduced GasEye (high performance tunable diode laser gas analyzer) based on its proprietary laser-based technology. It covers the entire chain of the product development through the system production	***Improved product:*** Improved gas analyzers optimized for industrial continuous and piecewise manufacturing End-User cases and for orchestration within a multimodal scalable monitoring and optimization solution.	***Enter on new markets*** due to optimized solution for steelworking, working and food sectors and, also ***increase market*** share on those already present due to combined MIR and NIR sources will greatly expanding the market for laser based sensing
SENORICS [SME, GE]	Senorics develops customizable sensor solutions for VIS and NIR spectroscopy based on unique novel OLED cost-effective detectors and sensors with unique properties. SENORICS has been awarded the LOPEC Start-up competition: Most Impactful Technology	***Improved product:*** Improved OLED-based sensor to be integrated on a NIR spectroscopy.	***Enter on new markets*** due to novel and cost-effective solution for spectroscopy manufacuturers
PHF [SME, CH]	It is a renowned high-tech company designing and producing innovative and powerful CMOS Sensors and cameras as well as FPGA programming and software primarily for the Machine Vision Industry and interfacing industries. It's leading (CMOS) imagers	***New product:*** InGaAs camera for snapshot spectral imaging ***New product:*** Dual aperture VIS-SWIR HSI camera.	***Increase market share*** due to offering more competitive solutions in markets where the company is already a global leader.

	and cameras are used globally in 3D inspection and measurement, spectroscopy, process monitoring and defect analysis, motion analysis and quality inspection.		
ABRAIA [SME ES]	Start-up technological enterprise focused on cloud services and microservices applied to deep leranign for image processing.	***New service:*** Cloud services and microservices optimized for optical monitoring orchestrated systems data acquisition and management	***Enter on new markets*** due to an optimized solution for monitoring solutions for manufacturing industry (firstly for project sectors).
IRIS [SME ES]	Advanced engineering SME company specialised in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing	***New product.*** IRIS expects to extend its product line mainly comprising proprietary state-of-the-art IR analysers (integrating InGaAs array sensor) for at-line and in-line process monitoring. In particular IRIS is keen to set agreements with PhotonFocus for the integration of snapshot HSI to its solutions but also to expand the wave range of the Visum NIR products to cover VIS-SWIR in a single instrument ***New product:*** Extend NIR spectrometres family products incorporating OLED-based miniaturized detectors in the VIS-SWIR range. ***New product and service:*** MULTIPLE scalable monitoring and optimisation solution integration and deployment for the food sector and implementation and process optimisation services for customers	***Increase market*** share on food sector commercializing better performing solutions.
TiO2 [SME, ES] MRA [SME, PT]	They are the company leader in high-tech solutions. Their mission is to offer its clients a wide offer of products and systems based on cutting-edge technologies for related areas to Industry 4.0 Solutions, Sensors and Measurement, Communication, Predictive Maintenance, Monitoring, Test, Computer Vision or Thermography for multiple sectors	***New product.*** MULTIPLE scalable monitoring and optimisation solution for integrators and manufacturing industries	***Increase market*** share commercializing cutting edge technological and improved performance solution. MRA will focus on Portuguese market and clients, whether Spain and EU market will be attended through TiO2
SCM [LE IT]	A technological world leader in processing wood and other materials. SCM Group coordinates, supports and develops a system of industrial excellence in 3 large highly specialized production centres employing more than 4.000 workers and operating in all 5 continents. Since 1952, a leading producer of machinery and systems, as well as service provider, for the woodworking industry.	***New product and service:*** MULTIPLE scalable monitoring and optimisation solution integration and deployment for the woodworking sector and implementation and process optimisation services for customers.	***Increase market*** share on woodworking sector commercializing better performing solutions.
RUSSULA [LE ES]	Russula is a global engineering company, with more than 30 years of experience serving the largest steel producers in more than 30 countries. It develops engineering solutions to help steel producers around the world become high-performance businesses focusing on improving operational performance and production parameters.	***New product and service:*** MULTIPLE scalable monitoring and optimisation solution integration and deployment for the steelworking sector and implementation and process optimisation services for customers.	***Increase market*** share on steelworking sector commercializing better performing solutions.

Table 8. Exploitation by Early adopters

Partner	Profile	Exploitation Strategy	Business Opportunity
ROYO [LE ES]	Global leader on bathroom furniture production marketing on more than 60 countries, including Spain, Poland,	***Improved product:*** Improved bathroom furniture references due to improved piecewise production process reducing	***Increase market share*** due to offering more competitive solutions in markets where the

	México, USA or India. More than 1,000,000 units sold per year	rejected pieces, production time and cost and, thus, leading to an optimally-produced furniture piece.	company is already present
LDL [LE FR]	LDL stands for flexible production of heavy plates designed while also providing excellent customer service. LDL represents a new generation of steel-re-rollers due to the use of the highest achievable level of automation.	***Improved product:*** Improved heavy plates references optimally-produced due to improved continuous production process reducing rejected pieces, production time and cost.	***Increase market share*** due to offering more competitive heavy plates (and also guaranteeing higher quality) in markets where the company is already present and entering on new international markets.
JOTIS [LE GR]	Greek company founded on 1930 devoted to provide innovative food products based on natural raw materials, and to ensure they are always of the highest quality. JOTIS exports their products to the 5 continents becoming a global player.	***Improved product:*** Improved JOTIS chocolate references optimally-produced due to improved continuous production process guaranteeing optimum chocolate process parameters (temperature, particle size, etc.). Will be later on deployed in other food production lines.	***Increase market share*** due to offering high-quality chocolate-based products in markets where the company is already present and entering on new international markets.

Business supporting services-provider partners (EPIC and TEMATYS) will benefit from participation through the know-how and knowledge achieved and by better positioning themselves on the EU and international markets as key players in the advanced manufacturing sector, widening opportunities for offering services and gaining clients, market share and increasing turnover.

RTOs (AIMEN and IMEC) benefit by including the IP generated through the project in their IP portfolio to exploit it through licensing. In addition, knowledge gained will be used in new R&I activities and integrated in the education and training curricula.

Integrated Exploitation Strategy. The exploitation of project results will comprise the commercialisation of solutions and support services for full benefit from business opportunities by addressing target market segment's needs. Path to deliver innovation to market, as well as risks identified in this process and corresponding mitigation measures have been included in section 1.4.2, where market analysis and MULTIPLE positioning is presented.

The **MULTIPLE scalable global solution** will be ***commercialised by Grupo ALAVA (TiO2 &* MRA) as a global solution provider (whether each of them focused on different geographical markets)**. MRA & TiO2 will commercialise the whole system comprising all subsystems & technologies developed by partners and accounted as OERs under Table 6. Based on partners' experience, integration of subsystems is a not-desirable strategy for potential end-customers. To limit their risks, they are showing mainly ***interest into overall working systems***, and they prefer to work with a ***main-contractor*** for delivery of an overall working system, including all sub-systems and supporting tools. In this context, MRA or TiO2 act as ***system-integrator*** while the rest of technology developers act as ***suppliers*** (to be further analysed in the Exploitation Plan and agreements among partners fostered). As previously defined each project partner will be owner of project results developed (solely or jointly, according to the work plan and the Table 6) and the exploitation will be in accordance to that principle. MRA or TiO2 must acquire the products developed by project partners to commercialise/integrate the whole MULTIPLE scalable monitoring and optimization solution (guaranteeing no-IPR collision among partners).

MRA will be act as provider for the industry integrators*:* ***RUSSULA, SCM and IRIS***. When integrating the solution on real industry facilities, integrators will need to implement their knowledge on specific process optimisation and on controlling the processes. *It may be implanted in existing manufacturing facilities integrating current infrastructures or it will be implanted on totally new production lines.* Industry integrators (RUSSULA, SCM and IRIS), will acquire (at fair and beneficial conditions) the MULTIPLE solution and deploy and manage it to optimise manufacturer production lines increasing their technology portfolio and solutions to be offered to the industry.

IRIS will be focused on integration of monitoring and optimization solution and architecture for the Food and food processing sector within the project and on the pharmaceutical sector beyond it, whether **SCM** will tackle the MULTIPLE solution integration and implantation work for the woodworking and furniture manufacturing sector and on the plastics, stone or glass sectors beyond the project. In the other hand, **RUSSULA** will be focused on integration on steel working facilities aiming optimisation of the production process, which is its core business. Agreement between MRA & TiO2 and RUSSULA, SCM & IRIS are established for mutual prescription and boosting commercialization opportunities and clients reach. All industry integrators (RUSSULA, SCM and IRIS) will integrate and deploy the MULTIPLE overall solution at the End-Users (ROYO, LDL and JOTIS) ***facilities improving their current processes,*** increasing speed, quality and resource efficiency at their selected production lines. It is important to point out that End-Users are relevant players within their sectors and, thus, play a crucial role as early-adopters, smoothing the market uptake of MULTIPLE-developed solutions.

Lead-Users or Technology Developers (**SENORICS, AIROPTIC, ABRAIA, PHF)** will act as suppliers of their

generated results (imaging and sensing solutions and orchestration and optimisation tools ranging a wide spectrum of photonic technologies) to MULTIPLE overall solution to MULTIPLE integrators or to potential industry End-Users aiming to acquire specific simple and stand-alone solutions. **IRIS** will also commercialise the new NIR spectroscopy developed with the OLED-based sensor as a stand-alone solution.

Complementarily, industrial partners will be granted licences by the **RTOs** (AIMEN and IMEC) to use their generated IP in the exploitation of the project results in their applications, terms will include remuneration based on sales.

Business Model. A draft business model Canvas (Figure 21) has been developed and will be further refined (WP7) based on discussions with partners, key stakeholders and potential customers. 'Lean Start Up'[148] methodologies will be used to build the prototype business (including test and learn process) to ensure exploitation maximisation. On the Canvas, it is represented the common features for KER1exploitation. Specific business models for MULTIPLE subsystems, technologies and tools (OERs) fit in current partners business models avoiding the need to develop new business models.

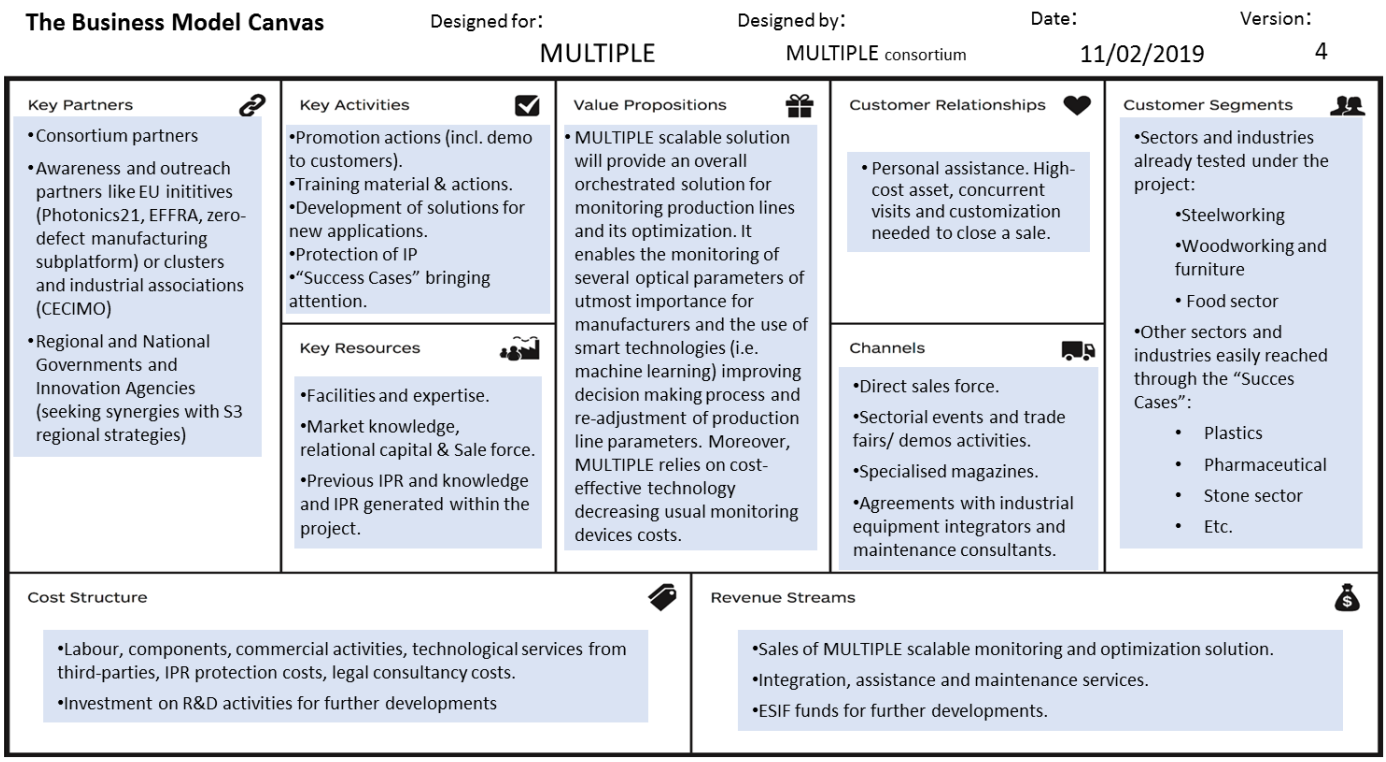


Figure 21. A draft Business Model Canvas for MULTIPLE scalable monitoring and optimisation solutions.

Consortium partners are committed to engage key stakeholders through all available channels. MULTIPLE model is a business-to-business (B2B) model of a medium cost and specialised equipment and thus, personal assistance and contact shall be made in order to materialise potential sales. In this sense, the direct sales force, highly experienced, of KER1 exploitator, MRA and TiO2, will benefit from Grupo ALAVA huge sales force and will also benefit from the industry integrators customer base. Grupo Alava is already commercialising photonic solutions from the technological partners of the consortium (PHF, IRIS, etc.). SCM, IRIS and RUSSULA (relevant players within their industrial domains) are a key asset due their strong relationship with industry and End-Users (potential clients). The involvement of different partners with a very different field of expertise, will deliver great opportunities for direct commercialisation through the exploitation of synergies and potential mutual prescription, benefiting from their marketing and sales network. Indeed, MULTIPLE benefits from having onboard a photonic market expect and commercialisation consultant, like TEMATYS, helping on defining the best suited exploitation strategy for the projects results. Moreover, relying on the integration of solutions from global key players like PHF will boost market acceptance and client willingness to pay. In addition, MULTIPLE will benefit from current partners positioning within the EU market and fostering synergies with related projects. MULTIPLE partners will take opportunity of reaching potential customers through sectorial events, trade fairs and demonstration activities (***showrooms, open-door days, etc***.) as well as specialised publications in that sense the involvement of a partner like EPIC will trigger dissemination and client confidence.

In addition to conventional and previously mentioned channels, professionals in the maintenance and asset

[148] The Lean Start Up, http://theleanstartup.com

management sectors, such as EFNMS[149] and GFMAM[150], will be also contacted. In most cases, it will be offered a revenue-sharing model (e.g. % in any sale they make) in return of gaining access to different markets. Federations, associations and clusters have a multiplier effect and leveraging penetration of MULTIPLE systems and solutions.

Financial projections. Based on mentioned strategy and partners´ experience, the initial selling-price for MULTIPLE solutions have been set (including subsystems and tools) with a sufficient profit level (ranging from 30% for KER1 to 35% for OER2&3, usually the profit level could be more than 40% in high technology level developments) representing a great business opportunity for all partners and remaining costs-competitive products (remaining largely cost-effective solutions when compared with market available solutions). Financial projections on the commercialisation of KER and OERs for 5 years (once commercialisation beings) is depicted under Figure 22. The number of units sold is in accordance with market evolution (already assessed on section 1.4.2). MULTIPLE is envisioning to generate €54.6M turnover and €17M of profit. The needed investment to further development steps (commercialisation roadmap) of MULTIPLE solutions would be about €7.4M (moving from TRL7 to 9 involves important qualification and marketing costs) and, thus, the Payback time will be within 4 years and an estimated ROI of 126,2%.

Commercialisation year (beyond the project)				Year 1			Year 2		
KEY EXPLOITABLE PROJECT RESULT	Partner	Item Price (€)	Profit/Item (%)	Sales (units)	Sales (€)	Profit (€)	Sales (units)	Sales (€)	Profit (€)
KER1: MULTIPLE scalable monitoring and optimisation solution	MRA/TiO2	160.000 €	30%	4	640.000 €	192.000 €	14	2.240.000 €	672.000 €
OER1: Snapshot VIS/SWIR HSI imagers	*PHF*	10.000 €	30%	15	150.000 €	45.000 €	30	300.000 €	90.000 €
OER2: Laser-based IR analysers	*AIROPTIC*	20.000 €	35%	10	200.000 €	70.000 €	20	400.000 €	140.000 €
OER3: Multimodal imaging systems	*IRIS*	5.000 €	35%	12	60.000 €	21.000 €	25	125.000 €	43.750 €
OER4: Cloud and edge computing platform and service	*ABRAIA*	15.000 €	30%	5	75.000 €	22.500 €	12	180.000 €	54.000 €
Total by year				46	1.125.000 €	350.500 €	101	3.245.000 €	999.750 €
Cummulative				46	1.125.000 €	350.500 €	147	4.370.000 €	1.350.250 €
Balance (Cummulative Profit - Initial investment)						- 7.049.500 €			- 6.049.750 €
ROI						-95%			-82%
Commercialisation year (beyond the project)				Year 3			Year 4		
KEY EXPLOITABLE PROJECT RESULT	Partner	Item Price (€)	Profit/Item (%)	Sales (units)	Sales (€)	Profit (€)	Sales (units)	Sales (€)	Profit (€)
KER1: MULTIPLE scalable monitoring and optimisation solution	MRA/TiO2	160.000 €	30%	39	6.240.000 €	1.872.000 €	70	11.200.000 €	3.360.000 €
OER1: Snapshot VIS/SWIR HSI imagers	*PHF*	10.000 €	30%	90	900.000 €	270.000 €	180	1.800.000 €	540.000 €
OER2: Laser-based IR analysers	*AIROPTIC*	20.000 €	35%	40	800.000 €	280.000 €	75	1.500.000 €	525.000 €
OER3: Multimodal imaging systems	*IRIS*	5.000 €	35%	50	250.000 €	87.500 €	90	450.000 €	157.500 €
OER4: Cloud and edge computing platform and service	*ABRAIA*	15.000 €	30%	35	525.000 €	157.500 €	55	825.000 €	247.500 €
Total by year				254	8.715.000 €	2.667.000 €	470	15.775.000 €	4.830.000 €
Cummulative				401	13.085.000 €	4.017.250 €	871	28.860.000 €	8.847.250 €
Balance (Cummulative Profit - Initial investment)						- 3.382.750 €			1.447.250 €
ROI						-46%			20%
Commercialisation year (beyond the project)				Year 5			MULTIPLE EXPLOITATION (in 5 years)		
KEY EXPLOITABLE PROJECT RESULT	Partner	Item Price (€)	Profit/Item (%)	Sales (units)	Sales (€)	Profit (€)	Sales (units)	Sales (€)	Profit (€)
KER1: MULTIPLE scalable monitoring and optimisation solution	MRA/TiO2	160.000 €	30%	115	18.400.000 €	5.520.000 €	242	38.720.000 €	11.616.000 €
OER1: Snapshot VIS/SWIR HSI imagers	*PHF*	10.000 €	30%	250	2.500.000 €	750.000 €	565	5.650.000 €	1.695.000 €
OER2: Laser-based IR analysers	*AIROPTIC*	20.000 €	35%	130	2.600.000 €	910.000 €	275	5.500.000 €	1.925.000 €
OER3: Multimodal imaging systems	*IRIS*	5.000 €	35%	150	750.000 €	262.500 €	327	1.635.000 €	572.250 €
OER4: Cloud and edge computing platform and service	*ABRAIA*	15.000 €	30%	100	1.500.000 €	450.000 €	207	3.105.000 €	931.500 €
Total by year				745	25.750.000 €	7.892.500 €	1616	54.610.000 €	16.739.750 €
Cummulative				1616	54.610.000 €	16.739.750 €			
Balance (Cummulative Profit - Initial investment)						9.339.750 €			9.339.750 €
ROI						126%			126,21%

Figure 22. MULTIPLE project exploitable results commercialisation financial projections.

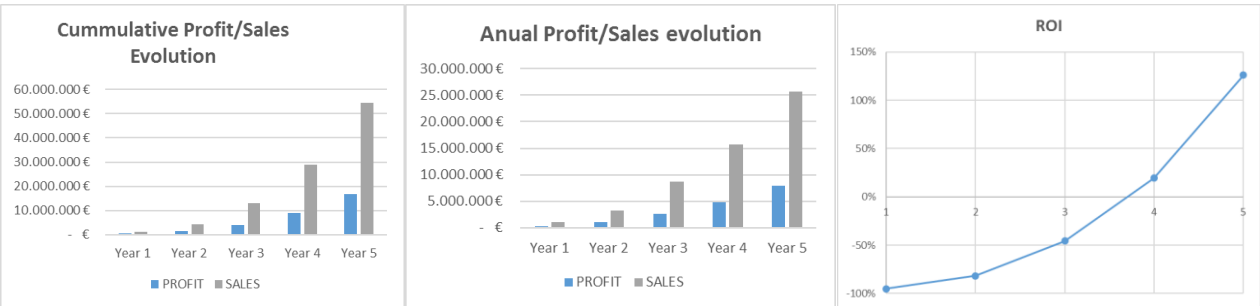


Figure 23. From left to right annual profit and turnover, cumulative profit and turnover evolution and yearly evolution of ROI.

Knowledge Management and Protection. The knowledge management will rely on the Project Steering Committee (PSC) and the Innovation Board (IB), more details in section 3.2. The participation of the relevant industrial complete value chain, represented in MULTIPLE consortium, provides an important perspective for this task. A Consortium Agreement (CA) will be elaborated to manage (amongst other things) issues related to IPR (ownership of results,

[149] http://www.efnms.eu/
[150] http://www.gfmam.org/

access rights or dissemination). The principles for knowledge management, to be included in the CA, are:

Provisions regarding Background. Further to what it is already established in articles 24 and 25 of Grant Agreement (GA), the following provisions have been defined to be included in the CA: (1) Access to Background for the project implementation will be on a royalty-free basis; (2) Access to Background for exploiting project's results will be on fair and reasonable conditions. Access rights to background for exploitation of project's results will be revised during the project, since this will depend on the technologies finally implemented in the results (several to be studied during the project, some to be dismissed). The review's conclusions will be included in the subsequent versions of the Exploitation Plan (EP) to be elaborated in WP7. If access rights to the partners' Background were eventually needed, they shall be granted and, if applicable, the final version of the EP will include the description and costs of these conditions. The most relevant knowledge and know-how necessary to achieve the project results are listed in table below well as the partners providing such background. A written agreement including all background needed or the project implementation and results exploitation will be included in the CA.

Table 9. Relevant partners´ background

Project Output	Background
Principles and fundamentals of optoelectronic systems based on spectral sensors	**IRIS** – Know-how in terms of monitoring systems, especially embedding optical systems and chemiometrics. Knowledge on optics used in the NIR systems implemented in the project **AIMEN –** Embedded RT image processing control & inspection **SENORICS-** OLED-based detectors and sensors developments for spectroscopy including filtering. **AIROPTIC –** know-how in gas analyzer design including signal processing, optics, spectroscopy, electronics, firmware, software. **IMEC –** Knowledge on snapshot spectral filtering design for hyperspectral images (Patent US2018359438). **PHF -** Expertise in CMOS image sensor design including operation modes such as ROI tracking, multi-exposure, non-destructive read out modes. Real time image pre-processing technologies in FPGA's or DSP's. **SCM-** Use of thermography in belt sanding quality control
Principles and knowledge for orchestration of monitoring systems	**ABRAIA** – Expertise on developing and deployment of cloud services for media asset management and image and video processing with deep learning models **AIMEN** – Real-time capable Deep Learning processing pipelines and artificial network architectures for machine vision **PHF** – Knowledge on integration different sensing technologies within their CMOs camara solution for multiple applications. Software-defined camera platforms and image pre-processing IP's. **IRIS** – Knowledge on integration of Process Analytical Technology (PAT) **TiO2 & MRA** – Expertise on IoT solutions development and photonic devices integration and orchestration for industry applications.
Knowledge on process optimization	**IRIS** – Knowledge on artificial intelligence techniques application for industrial monitoring. **SCM -** Patent pending for an own sanding diagnosis systems. **AIMEN -**Use of deep learning and principal component analysis for predictive control. **ABRAIA** – Knowledge on builing platform for the seamless development of deep learning models and cloud data and model management **TiO2 -** Elastic Search engine to provide data visualization and analytics as core of the business intelligence
Principles for integration and industrial deployment of monitoring and control manufacturing solutions	**AIMEN -** Expertise in development of real-time closed-loop control based on high speed SWIR/MWIR sensors, in an embedded configuration using FPGA. **AIROPTIC -** Expertise in manufacturing of laser-based spectrometers for industrial applications in power, wood, chemical, petrochemical, steel working and food industries. **IRIS** – Knowledge on integration monitoring systems in production lines, as well as of chemiometrics (cost effective imaging system for target foreign body detection). **SCM** – Knowledge and patents on integration of vision scanning systems for drilling within the woodworking sector. Own solution allowing to analyze the panels after the edging process to detect and identify defects so as to increase the quality of the finished product. **RUSSULA** – Knowledge on steelworking sector production processes optimisation and equipment integration. **LDL** – Production processes requirements and quality control specifications for continuous steel heavy plates production. **ROYO** – Production processes requirements and quality control specifications for bathroom furniture piecewise production. **JOTIS -** Production processes requirements and quality control specifications for chocolate continuous production according to EU food regulations.

Provisions regarding Project's Results.

The following procedures shall be considered in the CA further to the GA provisions in Articles 26 to 31: (1) *Protection*. Partners will have a proactive policy to protect IP arising from this project as it is envisaged in Table 10. Software developments (software), when not considered in patents, will be protected by copyright. Moreover, industrial design and trademark schemes will also be used, when necessary, to complement product protection in the market. Copyright will protect any written material produced. Know-how generated, when it cannot be protected by any of the mentioned mechanisms, will be then protected by trade secret implementing the necessary measures to ensure such protection. For joint applications agreements will be signed beforehand to manage the protective measures. (2) *Dissemination*: The Consortium will have a proactive policy to disseminate the project results (see 2.3.2 and WP7), with prior notice of any dissemination activity to all partners with sufficient information on the results to be disseminated. The research partners will aim for fast and open access publication and are provided with dedicated budgets to publish according to the "Gold Model" standard, (publication with immediate, free of charge open access). If for certain reasons the "Gold Model" cannot apply (e.g. dissemination budget reserved for open access journals exhausted), the open access will be given according to the "Green Model" (e.g. the publication is made available in a repository located in project website), possibly after an embargo period. Open Access to research data will be provided according to the Data Management Plan. (3) *Access Rights to Results*: access for carrying out the project work will be on a royalty-free basis, while for exploitation will be granted on fair and reasonable conditions. Access Rights for internal research, education or training will be granted on a royalty-free basis. Other provisions on access rights will be defined in the CA.

Table 10. MULTIPLE Results Protection strategy

Project Outcome	Protection Strategy
KER1: MULTIPLE scalable monitoring and optimization solution.	Patent + trade mark
OER1: Snaptshot VIS/SWIR HIS imagers	Patent + industrial design + trade mark
OER2: Laser-based IR analysers	Patent + industrial design + trade mark
OER3: Mutlimodal imaging systems	Patent + industrial design + trade mark
OER4: Cloud and edge computing platform and service	Copyright + trade mark or trade secret

Open Access and Open Research Data: MULTIPLE consortium will ensure open access to peer review scientific publications. Furthermore, the MULTIPLE consortium will seek for opportunities to provide open access to not peer-reviewed scientific publications, such as monographs, conference proceedings, etc. In addition, MULTIPLE Project will participate in the Open Research Data Pilot according to Article 29.3 of the GA. The project will collect different types of data: product & process parameters, industrial specifications, designs, testing procedures and results, models and modelling procedures, KPI, market data, life cycle cost produced data or personal data. The research data to be opened include KPI validation data, market data, results presented in the deposited scientific publications, known as "underlying data". The quality and performance data of the end-users and research data related to industry results will be considered confidential as otherwise their competitive position for exploiting such results will be endangered. Personal data of people involved in project activities will be managed according to EU/ national Data Regulations. Peer-review publications, open research data and cost analysis data will be made accessible for at least 5 years after the end of the project within partners' own internal servers, the project website and/or on a public repository (preferred option will be ZENODO´s repository through the creation of a dedicated community for the MULTIPLE project). Data Management Plan (DMP) will be further elaborated during the project (D7.2), including detailing data that will be generated, whether and how it will be exploited or made accessible (following H2020 guidelines[151,152]).

2.2.2 Plan for Dissemination and Communication

A first draft of the Dissemination and Communication Plan (D&CP) is presented, and it will be further developed during the project (WP7). While the Dissemination is focussed on the public disclosure of MULTIPLE results to specific target groups, Communication aimed on promoting the project itself and its impacts.

(a). Dissemination

[151] Guidelines on FAIR Data Management in Horizon 2020, v3.0, 7/16
[152] Guidelines Rules on Open Access to Scientific Publications and Open Access to Research Data in H2020, v3.2., 3/17

Dissemination objectives. The main objectives of the Dissemination are: (O1) to raise awareness and interest of potential users on the project results; (O2) to potentiate interaction with stakeholders and potential users to obtain key feedback to enhance exploitation opportunities; (O3) transfer of knowledge among the partners (O4) effective acquisition of new skills by users; (O5) ensure a broad applicability of the project results taking into consideration regulations and standards; (O6) to foster MULTIPLE technology solutions wider acceptance by EU manufacturing industry.

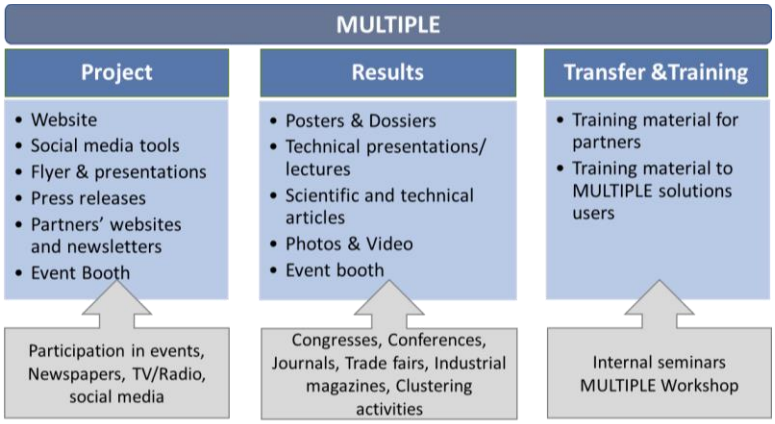


Figure 24. MULTIPLE dissemination & communication toolkit.

Dissemination milestones.

In the very initial phase of the project, dissemination will be focused in presenting the project to raise awareness among stakeholders (all target audiences). Following the generation of research output dissemination will be more focused to the scientific community, standardization organisations and key industry representatives. As the project progresses, dissemination will focus on presenting results related to MULTIPLE solutions (the whole solution and each of the subsystems and special tools) through specific outreach activities to all audiences. By the end of the project, the focus will also be oriented towards relevant industrial demonstration.

Although during the whole project transfer activities among the partners will occur, specific actions are planned by the end of the project based on all the results obtained. Similarly, training activities will also be specifically addressed at the end of the project and beyond.

Dissemination activities, audience, messages, timing and KPIs are shown on the following table:

Table 11. MULTIPLE Dissemination plan

Activities[153]	Audience[154]	Key message	Time planning	KPI dissemination
Project website & Social Media, MULTIPLE community, press releases, media appearance **(OP)**	**ALL** (O1)	Proj. obj., activities, partners, public docs, videos	M36	+5,000 web visits, +200 followers
Project in partners' websites and/or social media **(OP)**	**IND** (O1)	Project objectives	M36	~2,000 persons reached
Project brochure/leaflet/didactic materials (**PD**)	**ALL** (O1)	Proj. obj., activities, partners, public docs.	M36	~3,000 readers
Internal seminars (**PD**)	**PP** (O3)	Technical background, keys to use MULTIPLE solutions	With project meetings	At least 2 seminars
Online/offline/in-house training **(E)**	**IND, HLE** (O4)	Results, knowledge and fundamentals.	Beyond the project	~20 university students/ PhD ~25 trainees
MULTIPLE workshops **(E)**	**IND, SH, SO**(O1,2,5)	Results, main process features	M36 onwards	~ 100 attendees
Scientific Articles Physics/ Optics Journals: Applied Optics, Journal of Applied physics, Journal of Optical Society of America, Optics Express, ACS Photonics, Nano Letters, Applied Physics Letters, Journ. of Soft Eng. & Applications, etc. **(PP)**	**SC** (O1)	Results, specifically knowledge and fundamentals	M12 onwards	8+ scientific articles ~2000 readers
Presentations/lectures/posters in photonics congresses/ conferences: ISA, OSA Advanced photonics, CLEO, Photonic West, SPP Conference, Sensor+Test, ACHEMA, etc. **(PD)**	**SC** (O1&2)	Project contents, developments and results.	2021/2022 and beyond	2+ PPTP & 3+ posters ~400 researchers

[153] Activity type: Project Documentation (PD), Project Publication (PP), Online Presence (OP), Events (E)
[154] Audience: project partners (PP), Industry (IND), Scientific Community (SC), Stakeholders (SH), Standardization Organisation (SO), High-level education (HLE).

Networking and Clustering EU-funded projects and initiatives in photonics and in monitoring and optimisation **(E)**	**SC** (O1&2)	Project contents, developments and results.	M12 onwards	2+ clustering activities 10+ project contacted.
Organisation of parallel project events (Photonics West, Photonics Europe, OFC, PIC international, VISION, etc.)	**SC** (O1&2), **IND** (O1, 2 &6)	Project contents, developments, results of the solution	2021/2022 and beyond the project	2+ events reaching 150+ researchers & industrials
Technical articles in Photonics and Industry Journals: Photonics Spectra, Optik&Photonik, EuroPhotonics, Spectroscopy online, Industrial Photonics, Journal of Lasers, Optics & Photonics, CIRP Journal of Manufacturing Science and Technology, Journal of Applied Metalworking, European Journal of Wood and Wood Products, Journal of Food Processing, etc. **(PP)**	**SC** (O1) **IND** (O1&6)	Project contents, developments, main process features and results.	M12 onwards	3+ articles ~2000 readers
Open Access Repository **(PP)**	**SC, SO, HLE** (O5&O6)	Project public datasets, results and information	M12 onwards	~25 downloads
Participation in industrial events, fairs and workshops: Image Sensors Europe. LASER World of PHOTONICS, Photonics West, LIGNA Trade Fair, FoodTec, Advanced Factories, VISION, Photonics West, etc.) **(E)**	**IND, SH** (O1,2&6)	Broad MULTIPLE solutions. Presentation to industry and stakeholders. Booths presenting posters and project dossiers	2021/2022 and beyond the project	Participation on 6+ industrial events/fairs/workshops. ~5,000 attendees
Liaison and collaboration with standardization organisations (ISO, CENELEC and IEC) **(PD)**	**SO** (O5)	Information sharing and workshop participation	M12 onwards	1+ liaison with relevant ISO/CEN-CENELEC TC
Presentations in events promoted by EU Initiatives, platforms, and association (Photonics21, EFFRA I4MS, SAE, Manufuture, EFIC, EUROFER, ORGALIM. EFFoST, PFP, SECPHO, etc.) **(E)**	**IND, SH** (O1,2&6)	Project contents, developments and results.	M12 onwards	4+ presentations 2+ posters. ~700 attendees
Promote MULTIPLE results through Enterprise Europe Network **(PP)**	**IND** (O1, 2 &6)	Results and main features of MULTIPLE	M22 onwards	2+ dossiers

(b). Communication

Communication Objectives (why communicate?):

The main objectives of the Communication are: (O1) to involve young minds into the development of photonic sensing solutions and technologies and, specifically, related to the obtaining of monitoring and optimisation solutions for the manufacturing sector, and their importance and impact onto the economy and society; (O2) to help on promoting gender equality and integrating the gender dimension in research and innovation activities as well as promoting new gender-neutral career opportunities derived from project results; (O3) to achieve societal endorsement linked to research on new and key technologies due to their potential for contributing to overcome main societal challenges; (O4) to involve industry potentially open to the adoption of MULTIPLE technology and solutions and achieve their support to the project and to future initiatives linked with developing and improving photonic monitoring solutions for production lines approach; (O5) to contribute to the mobilisation of the European Research Area for the adoption of MULTIPLE-generated knowledge and the development of new technologies and uses enabling the full potential of the MULTIPLE technology solutions; (O6) to raise awareness among EC, Public Authorities and Policy Makers to foster cooperation in spreading the benefits of digital and intelligent manufacturing and contributing to regulatory process.

Communication Strategy (how to communicate?). MULTIPLE will deploy a ***360 degrees Communication Strategy***, combining online and offline channels, content marketing strategies & tools, growth hacking techniques, analytics tools, media relations, advertising campaigns, agreements with top events and associations and work with stakeholders. This 360° approach is visualized in Figure 25.

To reach the communication objectives, at qualitative and quantitative level, MUTLIPLE consortium will focus on low-cost alternatives to traditional marketing, e.g. using social media, viral marketing instead of buying advertising through more traditional media.

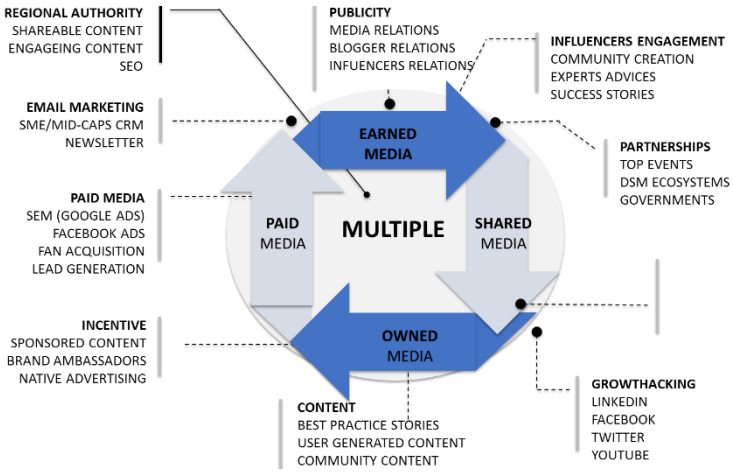


Figure 25. MULTIPLE Communication Strategy.

Communication Material. Communication materials (presentations, brochures, newsletters, didactic-oriented materials, etc.), regularly updated, will be developed to disseminate the project, including a Communication Kit (narrative text, photographs, slides and any other suitable communication material, complemented with copyright licenses for the EC and Photonics21). MULTIPLE will be the ***project brand***.

Communication plan (what, when, to whom and where communicate?).

Table 12. MULTIPLE communication plan

Audience	Key message	Activities	Time	KPIs
Young Minds	Impact on youth daily life	Short video for schools, visits, didactic materials, etc.	M8 onwards	Reaching at least 200 students
Public in general	Impact on job creation, energy efficiency, environ. & life quality	Press releases, articles, presentation on Radio & TV, web & docs, social media	M1 onwards	+30,000 people reached
Women	Career opportunities in manufacturing derived from project results	Participation in the EU Platform of Women in Science, EU Assoc. of Women in STEM, etc.	M12 onwards	Involvement in at least 3 activities
Stakeholders	Project results and its impact beyond manufacturing industry.	Authorities and Ind. events, including when possible Standardiz.&Certification.	M12 onwards	+3 presentations.
European Research Area	The MULTIPLE project results and its impact	Project information in CORDIS and own & external means	M12 onwards	+3 CORDIS/ EC publications
EC, Authorities & Policy Makers	Impacts on industry and citizens and existing limiting barriers	Presentation, Ad-hoc briefings & fact sheets in EC, Auth. & Policy Makers events	M1 onwards	+2 policy & engaging events

Communication Requirements

This project is funded by one of the calls under the Photonics Public Private Partnership (PPP). All communication activities related to the project will acknowledge the context of the Photonics PPP, for example by stating that the project is an initiative of the Photonics Public Private Partnership.

Specifically, for workshops, press releases, presentations etc, the EU emblem and Photonics21 logo will be displayed prominently together with the text "Photonics Public Private Partnership". The link www.photonics21.org will also be included. When communicating on Twitter or other social media about project activities, #Photonics will be included together with @Photonics21 and @PhotonicsEU.

The consortium will issue a press release at project launch, additional press releases whenever the project has reached a significant milestone or exceptional scientific, economic or societal impact is expected, and a final press release at project end.

A communication kit about the project (as a minimum a narrative text, photographs, slides possibly complemented by a video and any other suitable communication material, accompanied with copyright licences for the European Commission and for Photonics21) will be prepared and delivered *at the end of project month 3 at the latest*. The communication kit will target audiences beyond the project's own community, including the broader public and potential end-users, and focus on expected outcomes and related socio-economic benefits for the EU. The communication kit will be updated *at mid-term and at project end* to reflect project progress achieved.

The communication activities will involve public relations experts to ensure the pertinence of messages to target audiences and high outreach at both national and EU level through the use of the relevant national and EU-wide media channels (e.g. newspapers and broadcast).

Project communications such as project press releases, workshops announcements, websites, brochures will respect the principle of fair visibility for all partners. If the logo of any individual beneficiary is included, then the logos of all beneficiaries will be included.

PHOTONICS PUBLIC PRIVATE PARTNERSHIP

3 Implementation

3.1 Work plan

3.1.1 Overall structure of the work plan

The work plan has been designed to develop and deploy a multimodal spectral sensors and orchestrated deep models for integrated process optimisation. MULTIPLE is organised in 8 Work Packages (WP), being WP1 to WP6 technical WPs. In addition, WP7 is dedicated to dissemination and exploitation and WP8 is dedicated to Project Management. The strategy proposed to achieve the main goals of the project is based on the completion of the aforementioned eight work packages.

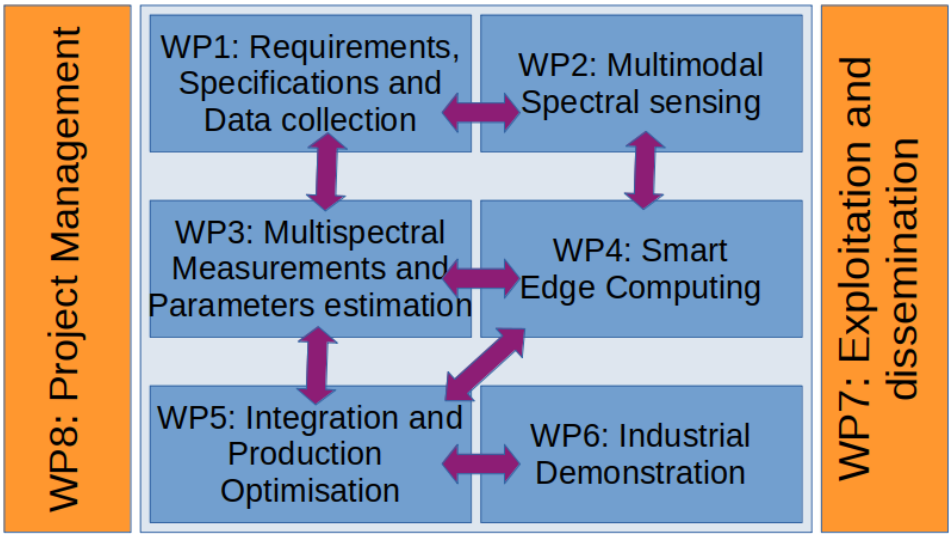


Figure 26. Interdependences of the Work Packages.

WP1, Requirements, specifications and data collection: This WP focuses on develop a detailed definition of the requirements and specifications and will elaborate a multimodal data collection for development and benchmarking. It is structured in three different tasks. *T1.1, Industrial use case requirements and specifications*, will provide a detailed definition of the end-users requirements. *T1.2, System architecture*, will provide the definition of the overall MULTIPLE system architecture. *T1.3, Data collection*, will collect multimodal imaging and spectral data for the development of the MULTIPLE system.

WP2, Multimodal spectral sensing: WP2 focuses on the development of a set of multimodal sensing solutions to monitor and estimate relevant product and process parameters. It is structured in four different tasks. *T2.1, Snapshot spectral imagers*, will develop custom mosaic filters and mask for a VGA InGaAs detector. *T2.2, InGaAs-based camera cores*, will design the InGaAs image sensor readout module electronics to be compatible with PHF's camera core. *T2.3, Dual Band VIS-SWIR HSI imager*, will develop a dual-aperture HSI broadband camera covering the VIS and SWIR range. *T2.4, Spectrometers*, will develop cost-effective versatile spectrometers tailored to the three demonstration applications.

WP3, Multispectral measurements and parameters estimation: This WP focuses on the development of algorithms and models for product and process parameters based on the data provided by WP2. It is structured in four different tasks. *T3.1, Chemometric models for spectroscopy*, will develop neural network and analytical models for robust estimation of gas concentration ratios for different compounds using the laser-based IR spectroscopy supplied by T2.3. *T3.2, Thermal modelling and colorimetry*, will provide the measurements of emissivity independent thermal imaging inside the furnace of the steelworks and colorimetry in the food and woodworking cases. *T3.3, HSI segmentation and classification*, will develop algorithms and models for segmentation and classification of HSI suitable for real-time processing. *T3.4, HSI regression*, will develop regression models exploiting VIS-SWIR HSI datacubes.

WP4, Smart edge computing: This WP focuses on the development of smart edge computing devices with AI embedded processing capabilities. It is structured in four different tasks. *T4.1, Edge computing devices*, will develop edge computing devices for managing the HSI and spectrometers developed in WP2. *T4.2, Data warehouse*, will provide the cloud data warehouse suited to the spectral data. *T4.3, Real-time embedded multimodal data processing*,

will provide smart standalone IoT photonic monitoring and control solutions. *T4.4, Fast model prototyping and deployment platform*, will provide a cloud deployment platform for fast and scalable model prototyping that will ensure smooth model compatibility from cloud to edge computing devices, addressing too the implementation of a graphical interface for data annotation.

WP5: Integration and production optimisation: WP5 is a transversal work package focused on the system integration of MULTIPLE, and the production optimisation. It is structured in three different tasks. *T5.1, Multimodal monitoring system integration*, is a long-term task devoted to the integration of the different subsystems that compounds MULTIPLE. *T5.2, On-line optimisation*, will develop close-loop control for process optimization in steelworking and woodworking, and on-line quality control for chocolate production and woodworking. . *T5.3, Data-driven production optimisation*, will develop the reporting and decision system to support Business Intelligence based on KPIs and data analytics, process flow control and anomaly detection.

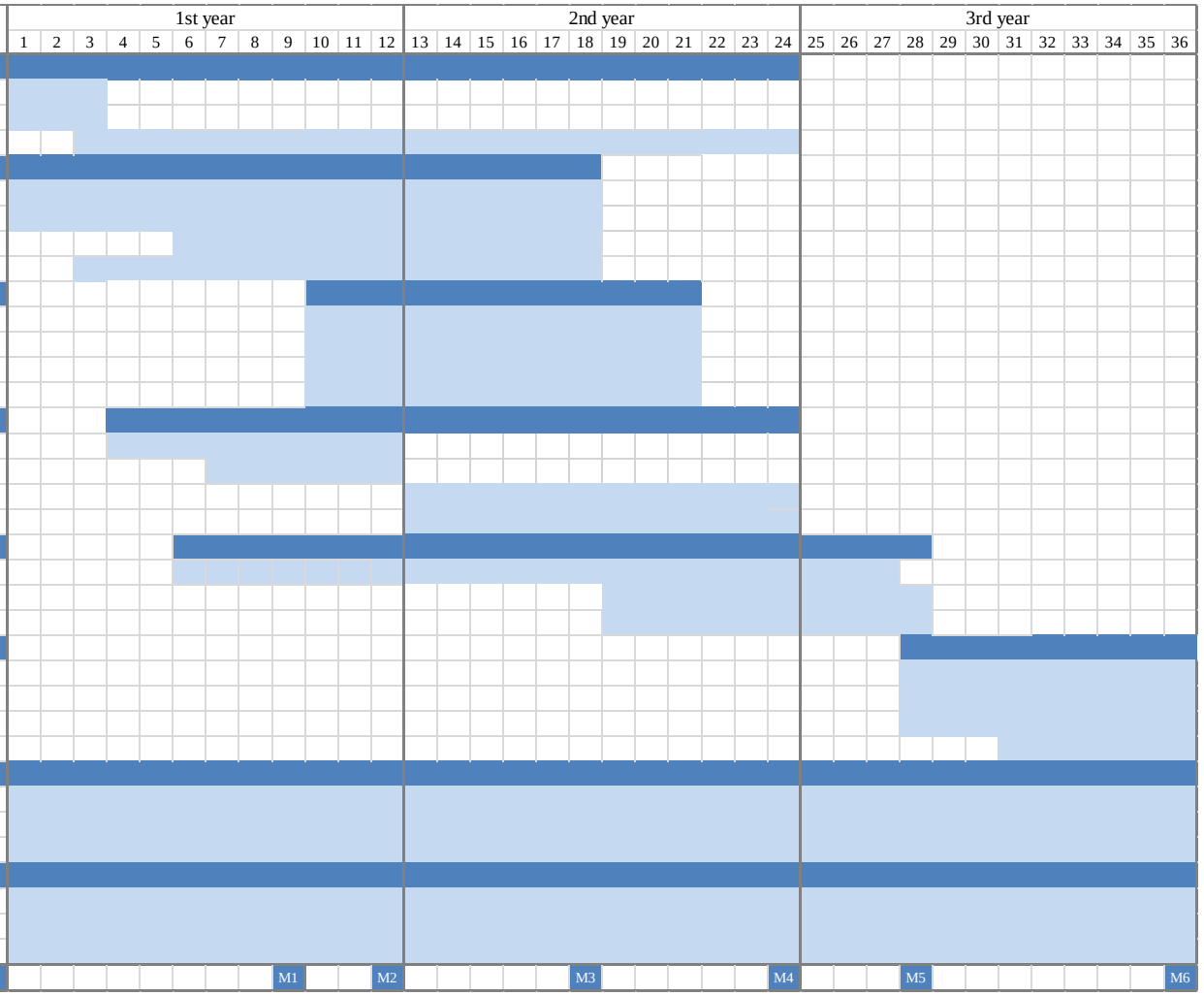
WP6: Industrial demonstration: This WP is the final technical WP. The one who involves the complete project evaluation and demonstration in real industry tasks. The execution of this work package will be focused on checking the capabilities of the MULTIPLE system in three different relevant scenarios such as woodworking, food processing and steel-working, where the Technology Readiness Levels (TRL) expected in MULTIPLE will be checked. In addition, this WP involves the benchmarking task.

WP7: Exploitation and dissemination: WP7 involves the development of the different dissemination, training and knowledge management activities that will be identified and integrated into a Dissemination and Communication Plan and Exploitation plan. Besides, this work package also involves the development of the technology transfer activities.

WP8: Project management: WP8 involves the management activities (overall project management, legal, financial, commercial and policy issues) and to manage the technical (time, resource, competence and budget risks) and strategies tasks.

3.1.2 Work plan timing

	Description	Lead	Start	End
WP 1	Requirements, specifications and data collection	RUSSULA	1	24
T1.1	Industrial use case requirements and specifications	ROYO	1	3
T1.2	System architecture	RUSSULA	1	3
T1.3	Data collection	SCM	3	24
WP 2	Multimodal spectral sensing	IMEC	1	18
T2.1	Snapshot spectral filtering	IMEC	1	18
T2.1	InGaAs-based camera cores	PHF	1	18
T2.3	Dual band VIS-SWIR HSI imager	PHF	6	18
T2.4	Spectrometers	AIROPTIC	3	18
WP 3	Multispectral measurements and parameters estimation	IRIS	10	21
T3.1	Chemometrics models for spectroscopy	AIROPTIC	10	21
T3.2	Thermal modelling and colorimetry	AIMEN	10	21
T3.3	HSI segmentation and classification	AIMEN	10	21
T3.4	HSI regression	IRIS	10	21
WP 4	Smart edge computing	AIMEN	4	24
T4.1	Edge computing devices	AIMEN	4	12
T4.2	Data warehouse	ABRAIA	7	12
T4.3	Real-time embedded multimodal data processing	AIMEN	13	24
T4.4	Fast model prototyping and deployment platform	ABRAIA	13	24
WP 5	Integration and production optimization	RUSSULA	6	28
T5.1	Multimodal monitoring system integration	IRIS	6	27
T5.2	On-line optimization	RUSSULA	19	28
T5.3	Data-driven production optimization	TiO2	19	28
WP 6	Industrial demonstration	SCM	28	36
T6.1	Rolling mill production line demo	LDL	28	36
T6.2	Furniture manufacturing demo	ROYO	28	36
T6.3	Chocolate production line demo	JOTIS	28	36
T6.4	Benchmarking	SCM	31	36
WP 7	Exploitation and dissemination	EPIC	1	36
T7.1	Dissemination and communication	EPIC	1	36
T7.2	Exploitation and IP management	TEMATYS	1	36
T7.3	Technology transfer activities	EPIC	1	36
WP 8	Project management	AIMEN	1	36
T8.1	Project coordination	AIMEN	1	36
T8.2	Project monitoring and risk management	AIMEN	1	36
T8.3	Strategic management of the project	AIMEN	1	36
	Milestones:			



3.1.3 List of Figures

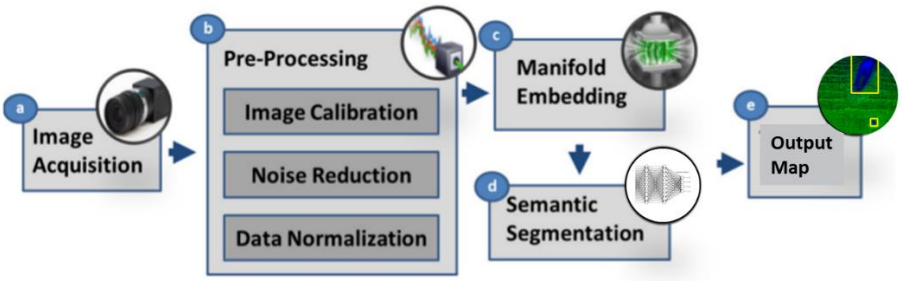


Figure 27. Real-time semantic segmentation processing pipeline.

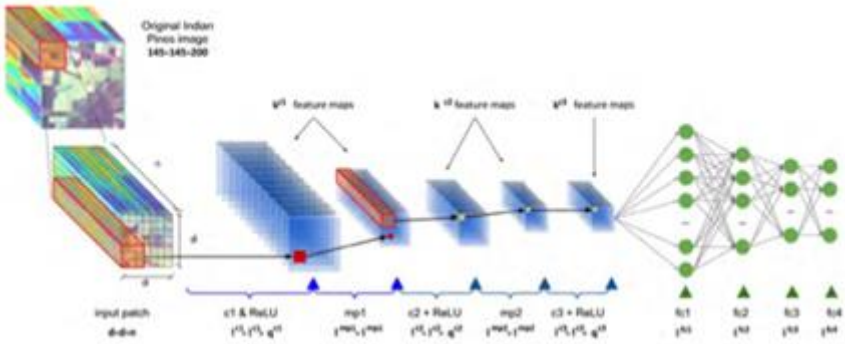


Figure 28. Example of 3D-CNN hyperspectral image pixel regression architecture

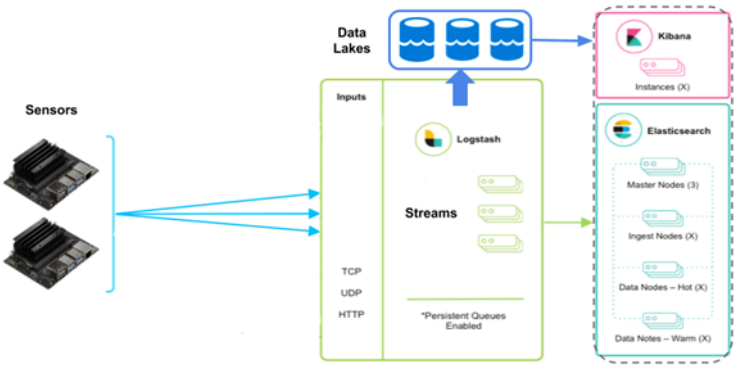


Figure 29. Basic concept of scalable data warehouse based on state-of-the-art open-source technology

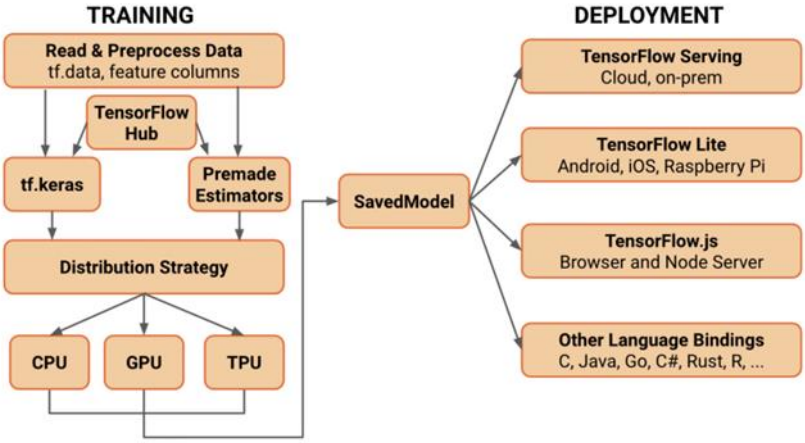


Figure 30. Tensorflow 2.0 schematic framework overview for training and deployment.

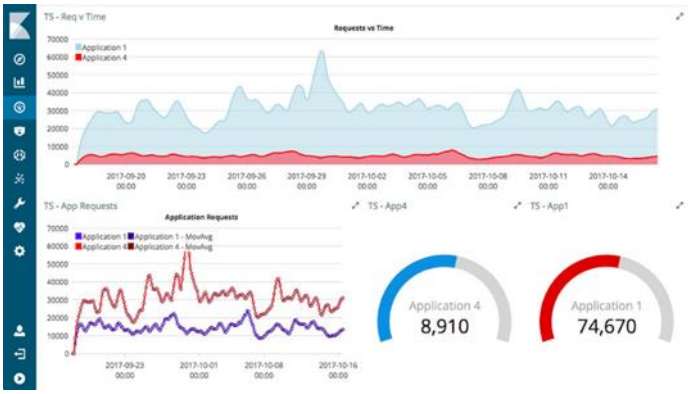


Figure 31. Open source Kibana allows supporting easily configurable and intuitive dashboards for the visualization of data and KPIs

3.2 Management structure and procedures

3.2.1 Management structure

a. Project Steering Committee (PSC). The PSC consists of one senior staff representing each partner of the project and is chaired by the Coordinator. The PSC is the highest decision-making body of the Consortium and will be responsible for determining the long-term strategy and orientation of MULTIPLE and taking major strategic decisions on matters affecting the project (changes to Annexes 1 and 2 of the Grant Agreement, IPR issues, suspension or termination, etc.) or the consortium (change of coordinator, entry or withdrawal of a partner). The PSC will meet at least twice every year.

b. *Project Coordinator (PC).* AIMEN will act as the PC and will be the legal entity acting as the intermediary between the consortium and the EC/Agency. The experience and commitment of AIMEN will ensure the success of project management. AIMEN has been involved in 54 international R&D&I projects, 34 of them under the EU R&D Framework Programmes, acting as coordinator in 16 of such projects. The PC will be responsible for the management of technical, financial and legal issues of the project according to the provisions in the GA, the CA and the decisions of the PSC. Namely, the PC will coordinate the execution of the activities in the work plan and the allocation of resources, while monitoring the effective and efficient implementation of the project against the plan. The PC will also be responsible for the compliance with contractual obligations with the EC/Agency, including elaboration and submission of reports and deliverables (including financial statements and related certifications). Moreover, the PC will manage the communication flow within the consortium and with any external organisation. In addition, the PC will perform functions such as EC financial contribution management, arranging meetings and provision of minutes or maintenance of project procedures. Moreover, AIMEN will appoint a Scientific Manager and an Administrative Manager that will comprise the Management Team that will be responsible for the day-to-day management of the project. Within AIMEN, support will be provided from its own Project Management Office, which deals with the financial, legal and administrative aspects of all R&D&I projects.

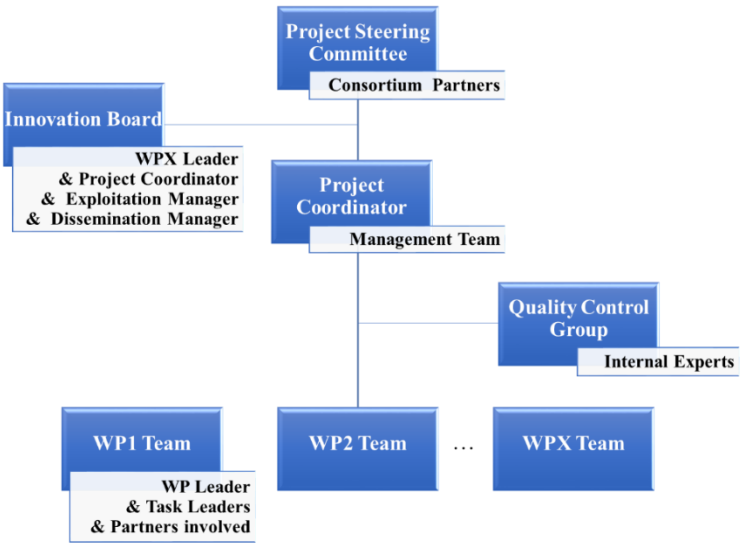


Figure 32. MULTIPLE Management Structure

c. *Work Package Teams (WPT).* In the WPT, representatives of all partners involved in a Work Package (WP) will discuss, coordinate and take decisions to effectively carry out the WP activities according to the established work plan and the decisions of the PSC. Each WPT will be led by a Work Package Leader (WPL) who is responsible for coordinating the activities conducted within the WP, while monitoring the effective and efficient implementation against the work plan. In addition, the WPL will be responsible for informing the PC on the tasks progress and on any relevant issue affecting the WP, as well as for the communication flow within the WP. Complementarily, Task Leaders (TL) will be appointed to ensure expected outputs from the tasks are coherent, can be met in time (effectiveness), and within funding constraints (efficacy). The TL will inform the WPL about the tasks progress. Each WPT shall meet at least every three months, beginning at the WP start date according to the work plan. The corresponding WPL will chair these meetings.

d. *Innovation Board (IB)*. The IB will work on smoothing the pathway to innovation by identifying any possible barriers and proposing actions to the PSC in order to maximise the opportunities for the exploitation of the outcomes of MULTIPLE project across the value chain. The IB will also conduct the efforts in relation to the activities to be developed in WP7.

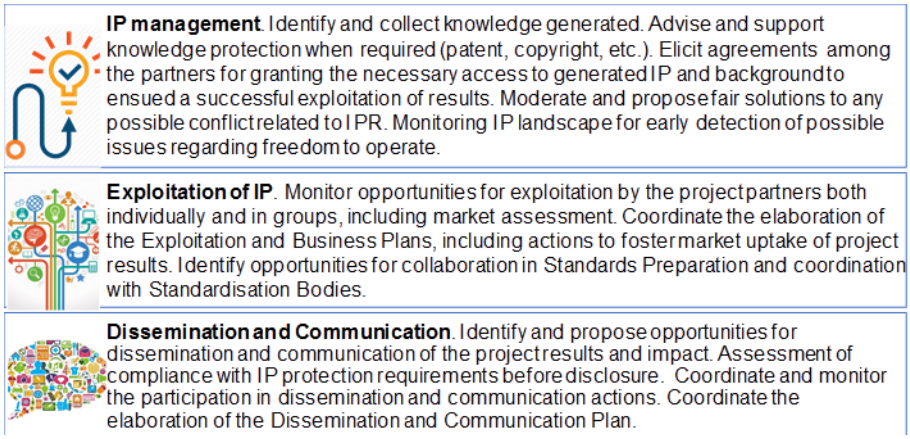


Figure 33. Innovation Board activities

The IB shall consist of the WP7 Leader -who will chair the IB-, the PC (as representative of the PSC), the Exploitation Manager (representative of the leader of Task 7.2). These representatives will be appointed based on their skills and expertise on business development, IPR issues and dissemination and communication activities. The IB will be in charge of activities summarized on Figure 33.

The IB will continuously collaborate with all partners, establishing dedicated task groups when necessary (i.e task groups on business plan elaboration for different project results). Collaboration with the QCG to ensure quality of publications is expected. Collaboration with experts from the Common Exploitation Booster will be foster if opportunity arises.

The IB and task groups will primarily work on a remote basis and will organise face-to-face workshops or seminars, in parallel to PSC meetings, to engage all partners in their activities and to promote discussions among them on relevant issues (IP protection strategy, dissemination or exploitation strategy, etc.).

e. Quality Control Group (QCG). The QCG will support the PC in monitoring and ensuring the quality of the project outputs, i.e. deliverables and publications. The QCG will comprise up to 5 members- selected within the partners' staff and appointed by the PSC- with adequate background and experience to judge scientific excellence and quality of such outputs. In case the QCG does not find deliverables or publications on an appropriate quality level, recommendations for adjustments will be given to the responsible of the output. In case of a conflict between them, decision shall be made by the PSC. The work of the QCG will be organized on a remotely basis.

3.2.2 Management procedures

An efficient set-up of management, monitoring and control procedures has been defined comprising: *Consortium cooperation*, *Communication*, *Monitoring and reporting* and *Decision making mechanism*.

<u>*Decision making mechanism.*</u>

Primarily, decisions in the consortium bodies (PSC, WPT, IB) should be made by consensus. If discussions do not lead to an agreement and consensus, the case will be decided upon voting. Each consortium body member will have one vote, 2/3 majority of the votes cast will be sufficient.

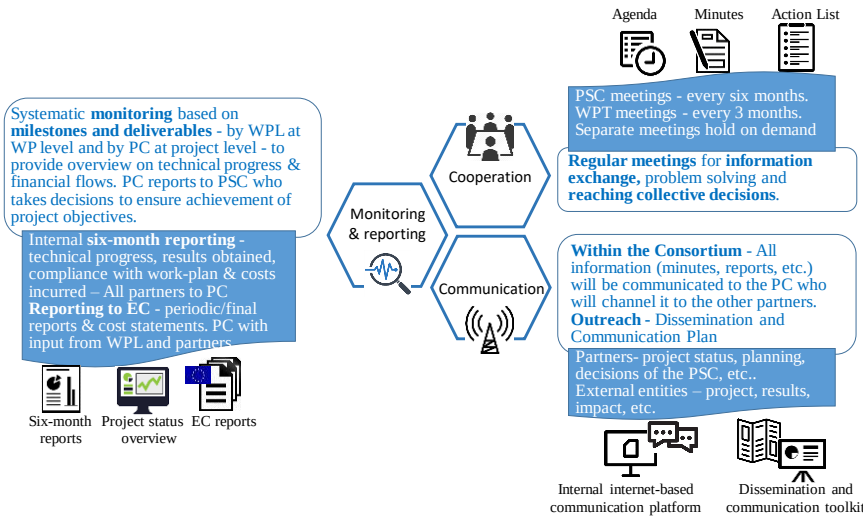


Figure 34. Management procedures

In order to enable an agile decision-making process ready to cope with a wide range of issues, the decision-making process has been established. In the unlikely case of irreconcilable differences that cannot be resolved by the PSC, the issue will be addressed following the procedure defined in the CA under Settlement of Conflicts provisions.

3.2.3 Appropriateness of the management structure and procedures

The management structure and procedures have been defined to guarantee the participation of all partners in strategic decisions through the PSC while maintaining sufficient agility in the day-to-day activities by adequate delegation of operative decisions, always in the strategic framework defined by PSC. Additionally, management activities have been distributed considering the different levels in the structure, empowering the WPL to assume a leading role in the technical management that will effectively support the PC in charge of the overall coordination. Quality of project

outputs will be ensured thanks to the QCG. Furthermore, in order to foster the innovation activities a dedicated body, the IB, has been created dealing with all issues related to protection, dissemination and exploitation of project results.

Further to the management structure and procedures herein defined, a CA will be agreed among the partners setting the provision regarding project governance, IPR management, confidentiality or management of EU Grant among others.

3.2.4 Innovation Management

Innovation management comprises both to obtain the necessary information on and from the market and to effectively use this information in the project to achieve the proposed objectives, including the successful exploitation of the results.

To keep an active surveillance of the market one body has been defined, the IB. Additionally, the IB will be in charge of managing the knowledge created in the project, identifying the barriers in the market up-take of project results (IPR issues, standards, market assessment) and defining the necessary actions to overcome them and successfully bring results to market, including Exploitation and Dissemination and communication Plan.

The PSC adequately supported and advised by the IB is entitled to make a decision on modifications of the project plan, allocation of resources or IPR issues.

Furthermore, activities above mentioned have been considered in the project work plan, specifically under the WP7.

3.3 Consortium as a whole

The MULTIPLE consortium has been built upon the application-driven approach to ensure the wide industrial impact of the MULTIPLE results. The Figure 35 shows the collaborative project approach among countries and organizations (SMEs, Large Enterprises, RTOs and EU associations and consultants) ensuring the exploitation and deployment of the project results at the EU at large. The project is organized upon the common roots: novel photonic sensing and imaging solutions development, monitoring solutions orchestration and cloud services and process simulation tools and, finally deployment and validation at manufacturing facilities at the steelworking, woodworking and food sectors.

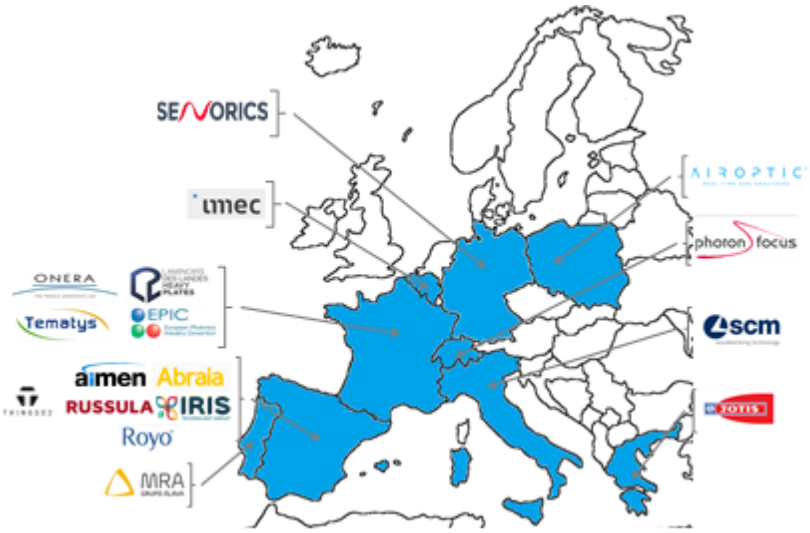


Figure 35. MULTIPLE consortium

The partners compose an inter-complemented consortium, capable of achieving the project objectives and results addressed in the project, with a certain overlapping of competences and expertise. ***MULTIPLE will bring an opportunity to put together all this expertise for the deployment of photonic monitoring and optimisation solutions at EU manufacturing industries, greatly improving its competitiveness.***

SENORICS, IMEC, AIROPTIC and PHF have specific knowledge for developing novel multimodal hyperspectral sensing solutions. IRIS provides expertise and knowledge on photonic solutions integration. Upon this ground knowledge, AIMEN and ABRAIA will complement photonic monitoring solutions through orchestration integrating smart edge computing and real-time embedded multimodal data processing and fast models and cloud services, respectively. MRA & TiO2 (Grupo ALAVA) as photonic devices integrator and commercialisation for industry implantation will integrate all MULTIPLE photonic developments into a scalable and orchestrated monitoring and optimising solution. Process optimization and integration between different technologies for industrial deployment relies on the role of experienced partners like IRIS, SCM and RUSSULA. The presence of three End-Users, highly relevant onto their markets (LDL at Steelworking, ROYO at woodworking and JOTIS at food sectors) ensures a wider uptake of the project results by industry. It is also relevant the implication of the photonics community through EPIC, selected partner to coordinate the dissemination, communication and training activities. TEMATYS will lead exploitation and IP management of project results. Moreover, the consortium gathers relevant connections and links with the major photonic stakeholders, including standardization and certification committees and the relevant European forums and platforms (Photonics 21, EFFRA, LIM, Photonics West, etc.) to ensure the project results widespread.

4 Members of the consortium

4.1 Participants

Partner 1	Asociacion de Investigacion Metalurgica del Noroeste – (AIMEN), Spain

Acronym of the project: MULTIPLE		
Participant Nº: 1	**AIMEN**	aimen

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Description of the entity

AIMEN is a Non-Profit association constituted by 100 companies, which supplies technological support to more than 400 companies dedicated to industrial or commercial activity related to metallurgy, automotive sector, civil construction, mechanics, shipbuilding, chemicals, foundry, machinery, ceramics, food, cement, wood and its by-products, plastics, engineering, assembly and others. AIMEN promotes and undertakes research, as well as improves manufacturing technologies in these sectors.

Located in the Northwest of Spain, AIMEN is a highly-professionalized organisation with more than 230 employees specialised in Materials Science, Manufacturing Process Engineering, Control Engineering, and Laser Processing. Its mission is to improve competitiveness and technology know-how of Manufacturing Industry.

In this project, AIMEN will contribute with its competence in the field of monitoring and embedded control systems design, Real-Time process control, hardware design, software processing, online control systems design and data analytics of process data in manufacturing. Its interdisciplinary team of engineers and physicists provides competencies in the fields of sensors design, embedded control and software processing.

AIMEN has active presence in the main European and National technology platforms and initiatives related with photonics and manufacturing technologies, specifically in PHOTONICS21 (related to the Photonic21 cPPP), EFFRA (related to Factories of the Future PPP), and EU Platforms like EUCOG. AIMEN has an active role in these initiatives and has contributed to the definition of R&D&I priorities at European or national level, including the RIS3-Galicia strategy. AIMEN is also present in other national and international expert panels, consulting committees, groups and societies like SECPhO (Southern European Cluster in Photonics and Optics as well in the Institute of Electrical and Electronics Engineers (IEEE), International Institute of Welding (IIW), EARTO, the European Laser Institute (ELI), and several sectorial industry clusters (automotive, shipbuilding, metalworking).

Moreover, AIMEN is leading a Regional Digital Innovation Hub on digital transformation for manufacturing: http://www.galacticadih.eu/

Headquarters: Laser Applications Centre

Polígono Industrial de Cataboi SUR-PPI-2 (Sector 2) Parcela 3

36418 O Porriño (Pontevedra)

Telf: +34 986 344 000

www.aimen.es

Role in the project and in the WPs

AIMEN is the project coordinator and WP4 leader devoted to the development of Smart edge computing. In WP1 will help the end-users to define the industrial requirements and specifications for the industrial demonstrations (Task 1.1), will participate in the definition of the MULTIPLE system architecture (Task 1.2), and will define data needs and receive data collected on Task 1.3. AIMEN will be involved on the development of models for the multispectral measurement and parameters estimations, leading task T3.2 (Thermal modelling and colorimetry) and T3.3 (Geometry and kinematics) a collaborating with IRIS on T3.1 (chemometrics) and T3.4 (Quality grading). AIMEN in WP5 will collaborate to the End-Users demonstrators process optimization and on WP6 MULTIPLE implantation and validation at real industrial facilities. Finally, AIMEN will collaborate in the overall Dissemination & Communication strategy and on the Exploitation and technology transfer activities in the WP7.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Roi Méndez (male),

He received the degrees of MSc in telecom engineering and MSc in applied mathematics from University of Vigo (UV), Spain, in 2008 and 2009. He obtained his Ph.D. in signal and image processing in 2012, UV. He was a postdoctoral researcher at the Fraunhofer Institute for High Frequency Physics and Radar Techniques (FHR), Germany (2012-2014), and in the Signal Processing for Communication Croup (GPSC) at UV (2014-1016). He has held visiting research positions at Geoscience Research Center (GFZ), Potsdam, Germany (2011) and University of Texas at Austin (2015). He is currently with the Robotics and Control Unit in AIMEN. His interests include image and signal processing, hyperspectral imaging, and real time processing.

Adrián Pallas (male),

Adrián Pallas Fernández (M): M.Sc. in Computer Vision at University Rey Juan Carlos (URJC) in 2017. Obtained his B.Sc. in Computer Science at University of A Coruña (UDC) in 2015, while coursing his M.Sc. he started working at Geomni (Verysk Analytics) in detection and 3D reconstruction of buildings on large scale aerial images with 3D vision techniques. In November 2016, he joined the Robotics and Control department of AIMEN Technology Center as R&D Computer Vision Engineer, where he is researching on Computer Vision and Machine Learning for industrial and environmental applications. He is taking part in several national and EU projects focused in 3D Vision, Multispectral Imaging and Deep Learning.

Manuel Álvarez (male),

M.Sc. in Engineering, graduated in Electronics Engineering in 2006 at Vigo University. Professional development at Vigo University (focused on mobile robots applications), in automation sector, and in R&D center. Since 2010 he is employed at the AIMEN Technology Centre. Extensive experience in the field of PLC's, robotic and vision systems, process control, system integration, automation processes, application programming and development of industrial projects. At this time, he is involved in several R&D projects related to robotics, machine vision and process automation. He has participated in several Spanish and European projects involving vision systems and integration of robotized processes, dealing with robotics, hardware design, and software development. In recent years he has presented several papers in international conferences (IECON, ETFA, etc.) focused on machine vision, robotics applications and automation systems.

Additional personnel of the Robotics & Control, Advanced Manufacturing Processes and Flexible Production Processes and Systems will collaborate in the project.

Personnel of the International Project Management Office of AIMEN will give support for project management, as well as from the Innovation Management area and the marketing department.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

***Publications*:**

- Garcia-Diaz, A., Mendez-Rial, R., Preciado, M., Souto, Á., & Harvey, A. (2017, June). Embedded IR multi aperture imaging for surveillance applications. In Applied Industrial Optics: Spectroscopy, Imaging and Metrology (pp. ATh1A-2). Optical Society of America.
- Souto-López, Á., García-Díaz, A., & Méndez-Rial, R. (2018, May). Temperature imaging in the MWIR range independent of emissivity. In Optical Sensing and Detection V (Vol. 10680, p. 1068005). International Society for Optics and Photonics.
- Méndez-Rial, R., López, Á. S., & García-Díaz, A. (2018, May). Embedded video rate super-resolution in the infrared with a low-cost multi-aperture camera. In Unconventional Optical Imaging (Vol. 10677, p. 1067705). International Society for Optics and Photonics.
- R. Méndez-Rial, V. Panadeiro, C. González, A. Pallas. Real-time arc welding process monitoring using Deep Learning on an IoT OPC UA device. Machine Learning Prague Conference. February 2019.
- Lapido, Y. L., Rodriguez-Araújo, J., García-Díaz, A., Castro, G., Vidal, F., Romero, P., & Vergara, G. (2015, July). Cognitive high-speed defect detection and classification in MWIR images of laser welding. In Industrial

Laser Applications Symposium (ILAS 2015) (Vol. 9657, p. 96570B). International Society for Optics and Photonics.

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**SEERS**	H2020-ICT-26-2014	Snapshot spectral imagers for cost effective IR Surveillance. Developing of a highly competitive system for IR multispectral sensing and intelligent surveillance of a wide variety of critical infrastructures.	Coordinator
2	**MAShES**	H2020-FoF-01-2014	Multimodal spectrAl control of laSer processing with cognitivE abilities. Development of a breakthrough compact imaging system for Real-Time closed-loop control of laser processing, built on a novel multispectral optics and multisensory arrangement in the VIS-MWIR spectrum.	Coordinator
3	**Z-BREAK**	H2020-FOF-09-2017	Strategies and Predictive Maintenance models wrapped around physical systems for Zero-unexpected-Breakdowns and increased operating life of Factories.	Coordinator
4	**PHENOMENON**	H2020-ICT-30-2017	Laser Manufacturing of 3D nanostructured optics using Advanced Photochemistry. Development and validation of optical structures by direct laser writing.	Coordinator
5	**LASHARE**	FP7	Development of a robust assessment framework for innovative laser equipment paving the way for new manufacturing applications.	Partner-Development of an IR monitor system for RT-closed loop control by measuring the thermal emission of the weld pool.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The AIMEN team gathers experience in: data gathering, embedded processing, pattern recognition, machine learning and condition monitoring. To accomplish with project activities and tasks, the most relevant technical AIMEN capacities are related with sensing technologies and Real-Time embedded processing. AIMEN owns a laboratory of CPS and embedded systems for electronics design (FPGA, GPU, ARM) and software development (VHDL, CUDA, C++) applied to condition monitoring, real-time signal processing and pattern recognition.

Robotics and Control Unit provides high-end knowledge in the design and development of factory automation solutions applied to different manufacturing systems. In the case of the Robotics and Control Unit the main infrastructures available for the proper development of this project are:

- A Laboratory for Industrial Robotics with several manipulators (from ABB, KUKA, MOTOMAN) that could be easily integrated with different systems and tools to test different pilot applications into laboratory environment.
- A laboratory equipment for the development of innovative sensors and embedded systems.
- Machine vision laboratory including different vision systems and vision software.

Partner 2 Abraia Software S.L.– (ABRAIA), Spain

Acronym of the project: MULTIPLE		
Participant Nº: 2	**ABRAIA**	Abraia

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Description of the entity

ABRAIA is a recently founded SME that develops tools and offers cloud services involving imaging, artificial intelligence, and visual perception technology. It also provides consultancy services to design and deploy image and video creation, processing, and optimization pipelines.

The company is currently working for INDITEX -the largest apparel company in the world-, monitoring, improving, and automating their media processing and delivery workflows in zara.com, dealing with a flow of several hundreds of thousands of images and thousands of videos a year.

ABRAIA offers a SaaS (Software as a Service) that is available through API Rest as well as plugins for CMS (Content Management Systems). ABRAIA's cloud services transform and manage digital media assets, using own developed deep learning and visual perception models as core technology.

ABRAIA has received the IEBT recognition (technology-based initiative) by the Galician government.

Role in the project and in the WPs

ABRAIA contribution will be mostly done in WP4 and WP5, being in charge of the cloud infrastructure and services that support the Data Warehouse and the fast prototyping and deployment of deep neural models for HSI image and data analysis.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Antón García Díaz (male), Founder.

PhD. Currently researching novel solutions to cloud-based image and media processing, using deep learning models. Before founding ABRAIA, he spent over a decade researching novel imaging concepts and visual perception models for computer and machine vision applications. In this role, he has developed and deployed novel commercial products in industrial applications. He participated in H2020 and FP7 projects involving imaging and vision both as coordinator (MAShES, SEERS, CARLoS) and member of the research team (FORTISSIMO 2, LASHARE, AMABLE). Active in the dissemination of technology trends and research results among professionals (e.g. program Committee member of OSA Applied and Industrial Optics topical meeting 2018 and 2019, collaborator of freecodecamp.org). He has authored papers in major conferences and journals in recent years (e.g. IEEE PAMI, SPIE Photonics West; invited speaker to Imaging and Applied Optics OSA conference 2017 and invited talk paper at SPIE Photonics Europe 2018). Reviewer for different peer-reviewed journals and conferences (e.g. Pattern Recognition Letters, Journal of Vision, Pattern Analysis and Applications).

Jorge Rodríguez Araújo (male), Founder.

M.Sc. in Industrial Electronics Engineering. Currently developing and deploying cloud services for media optimization and management. As an engineer, he has played a main role in the development of embedded imaging and image analysis products for innovative industrial applications, in close collaboration with industrial partners. He has been eager contributor to open source software in the field (e.g. openlmd) . Before starting ABRAIA, he has been involved in national and European research projects (MAShES, FORTISSIMO, LASHARE, ALAS, REPTILE) dealing with software development for cloud-based image processing, and machine learning for closed loop process control and process automation.

Additional engineering personnel of ABRAIA will collaborate in the project dealing with tasks related to HSI image analysis using deep models, and web services development.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

***Publications*:**

- Leboran, Victor, Anton Garcia-Diaz, Xosé R. Fdez-Vidal, and Xosé M. Pardo. "Dynamic whitening saliency." IEEE transactions on pattern analysis and machine intelligence 39, no. 5 (2017): 893-907.
- Jorge Rodríguez Araújo, Antón Garcia Díaz. “Transfer learning for image tagging in fast fashion workflows” WGML 2018
- Rodríguez-Araújo, Jorge, and Antón García-Díaz. "OpenLMD, multimodal monitoring and control of LMD processing." In High-Power Laser Materials Processing: Applications, Diagnostics, and Systems VI, vol. 10097, p. 100970B. International Society for Optics and Photonics, 2017
- Méndez-Rial, Roi, Álvaro Souto López, and Antón García-Díaz. "Embedded video rate super-resolution in the infrared with a low-cost multi-aperture camera." In Unconventional Optical Imaging, vol. 10677, p. 1067705. International Society for Optics and Photonics, 2018.
- SaaS for on-the-fly and cloud-based image and video analysis, transformation and optimization based on visual perception models. (https://abraia.me/docs/), and API REST and connectors to our services with different CMS platforms (Wordpress, Prestashop).

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

- ABRAIA is currently doing research in cloud-based deep learning for image and media analysis and processing applied to the management of digital assets (e.g. improved flexible search, automatic image and video tagging). In this regard, we have different own developed prototypes for metric-based ranking, learning and recognition of visual semantic embeddings, and object segmentation.
- The team at ABRAIA has been recently leading the development of open source solutions for embedded and cloud-based process monitoring for laser processing applications **https://openlmd.github.io/**

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Algorithms and cloud services and microservices for the management of media and the automatic processing of images and videos using deep learning models. Image and video management and optimization technology to support web and cloud collaborative workflows.

Partner 3 Iris Technology Solutions, S. L. – (IRIS), Spain

Acronym of the project: MULTIPLE		
Participant Nº: 3	**IRIS**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

IRIS Technology Group (http://www.iristechnologygroup.com/) is an advanced engineering Group that was established in Barcelona in 2007. Since 2011, it also operates from Dublin (Ireland) whereas the Barcelona office is referred to as IRIS Technology Solutions S.L. It specializes in the manufacture and integration of Process Analytical Technology (PAT)-based real-time quality monitoring solutions for the process industries and IoT systems for smart manufacturing, predominantly targeted at the food, pharmaceutical and chemical industries. In particular, our product line comprises proprietary state-of-the-art IR analyzers for at-line and in-line process monitoring.

On the one hand, our solutions are based on the latest developments in photonics in the fields of MEMs, Near Infrared Spectroscopy, Hyperspectral Imaging, and Raman. On the other hand, we leverage ICTs, whereby we apply Machine learning, Artificial Intelligence and Data Mining techniques for predictive and active control, as well as for decision support systems. In addition to photonics-based analyzers, we also work with IoT sensors for enabling equipment and machinery for smart manufacturing and transitioning industry to the Industry4.0 paradigm. Our solutions are cloud-based for mobility, remote monitoring and multi-site applications.

Our Innovation Division which is governed by an Innovation Committee comprising Senior Management, and an Innovation Unit for managing and executing the companies innovation roadmap, has 3 core strategic lines, for shaping the Factory of the Future:

1. New optical photonics-based technologies and PAT systems for production process optimization
2. ICTs (Artificial intelligence / machine learning, IOT & cloud platforms)
3. Circular and Bio-based Economy: digitalization and waste valorization

In addition to running an internal R&D program, as well as carrying our various joint private projects with industry partners, our Innovation Division is highly experienced in working in collaborative R&D projects in H2020 (both as coordinators and partners), principally in the SPIRE, FoF, and BBI programs. The company was established in 2007 and currently employs 60+ staff with a turnover of € 5 M. It is equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is ISO 9001:2008 certified and also is certified under UNE-EN-ISO 166002: Management Systems R&D&i. We are also certified by the local (Catalonian) government as a provider of technological R&D services (Tecnio Certification). IRIS is a member of the following sectorial networks: SPIRE, SPIE and SECPHO, in addition to regional industrial collectives such as Smart Space (association of companies and organizations that specialize in the 'smartification' of industry).

IRIS has a matrix of multidisciplinary knowledge groups in the company with strong expertise in engineering and science. With its team and resources, including electronics, mechatronics, mechanical, optical, industrial design engineers, as well as specialists in system engineering, application programming and web development etc. the group can carry out full system design and engineering, including electronic design and assembly, firmware and software development, electrical harnesses and interconnections, mechanical design for enclosure and prototype integration, as well as installation, validation and industrialization of prototypes and equipment in production plants. As such they have recently delivered advanced plastic sorting devices. In addition to this broad cross-disciplinary engineering team, IRIS' research team consists of data scientists, chemometricians, chemists and physicists, among others, who have the expertise to validate the technologies developed.

IRIS will support WP2-3 where they will design the suitable NIR system for the processes under consideration as well as their chemometrics calibration but also in WP6 for the integration and validation of the different systems in the production lines. Finally, they will contribute to the project dissemination and plan an outmost exploitation of their results.

Role in the project

In the IRIS will support WP2-3 where they will design the suitable NIR system for the processes under consideration as well as their chemometrics calibration but also in WP6 for the integration and validation of the

different systems in the production lines. Finally, they will contribute to the project dissemination and plan an outmost exploitation of their results.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Ing. ELODIE BUGNICOURT (female), Project manager and Leader of the innovation unit.

She has obtained a Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. He also coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: Arkema, Wacker Chemie, Ashland.

Dr. IDOIA MARTÍ ALUJA (female), Senior Scientist specialized in chemometric analysis.

She received her PhD in Chemistry at the Universitat Rovira i Virgili (Spain) and then worked in the Spectroscopic and Kinetics Unit (SKU) at the Institut Català d'Investigació en Química (ICIQ, Spain) and at the Institute of Photonic Sciences (ICFO, Spain), and at Zobele Spain.

Dr. DAVID NETTLETON. (male), Data Mining Analyst.

He has obtained a Ph.D in Artificial Intelligence from the Universitat Politécnica de Catalunya, Barcelona, Spain, M.Sc. In Computer Software and Systems Design from the University of Newcastle upon Tyne, U.K., and B.Sc. (Hons) Computer Science from the University of Wolverhampton, U.K. Has over 20 years' experience of applying data mining algorithms (including neural networks, rule induction) and statistical techniques (including ANOVA, regression, principal components) in a diversity of data analysis and modeling projects, such as medical diagnosis/prognosis, fraud detection, customer loyalty and segmentation, online user web search behavior and data privacy. As well as commercial projects, he has conducted research and lectured at two universities. He has published a diversity of articles in specialized data analysis journals/congresses and three books. He has participated in several competitive multi-partner research projects and has a European patent application for a graph matching algorithm.

JOSE ANTONIO IBARRA FUSTE (male), Scrum Master.

He has a PhD. in Signal Theory and Communications, Universitat Politècnica de Catalunya, Spain (2014), Thesis title: "Design Rules and Optimization of Electro-Optics Modulators Based on Coplanar Waveguides"; the project dealt with the development of new models to design high speed electro-optical modulators based in LiNbO3, where a novel software to design and optimize these modulators was created. Master's degree in Telecommunication Systems from Universidad de Oriente, Cuba (2005).Telecommunication Engineer Degree from Universidad de Oriente, Cuba (2003). Author of several papers in journals of high international impact. Experienced in optical communication softwares as Virtual Photonic (VPI), OptSim, and in electromagnetic structures simulators as High Frequency Structural Simulator (HFSS) and ADS. Experience for more than 5 years in many programming languages.

DR. TIMOTHY KEHOE (male), VISUMPalm Product Manager and Technical Lead at Dublin Facilities.

He has received a PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	Relevant product lines: Monitoring devices for real-time quality control.	-VisumPALM: NIR technology for incoming material quality control and detection of adulteration. -VisumNIR: In-Line NIR monitoring for product standardization and increase of production efficiency. -Hypera: Hyperspectral imaging solutions for the industry	Development and commercialization of devices

		-Services: Chemiometrics model development and training	
2	EP18198458, 03/10/2018, Rosales Alejandro, Kehoe Timothy, Garrick Meabh	An integrated optical system for the examination of sample materials	Property owner
3	EP19382101,14/02/2019, Alejandro Rosales, Adrián Manotas, Joan Puig	Device for detecting foreign matter in an edible substance or in a container containing the edible substance and method employing the device	Property owner
4	Rosales, O. Mc Nerney. Edited by Stefan Krämer and Sebastian Engell. Wiley-Vch. Print ISBN: 978-3-527-34074-3.	Sensing Technology. Chapter 6. Resource Efficiency of Processing Plants. Monitoring and Improvement	Publication Owner
5	A.A. Gowen, E. Gaston, J. Burger (2014). C.P. O´Donnell, C. Fagan, P.J.Cullen (Eds.), Process Analytical Technology for the Food Industry, 199-216, Springer, ISBN 978-1-4939-0311-5.	Hyperspectral imaging	Publication Owner.

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Objective	Role of your organization
1	**Visum. SMEI-Phase 2, GA 731043.**	Cloud-enabled IR spectroscopy platform for the rapid integration and customization of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries.	Partner
2	Hypera. H2020- SMEInst, 2016-2017.	Affordable hyperspectral Imaging System for on-line foreign body detection and chemical composition analysis	Partner-Coordinator
3	**ProPAT. SPIRE, GA 637232**	Robust and affordable process control technologies for improving standards and optimizing industrial operations	Partner-Coordinator
4	**SPECTRAFISH. FP7, GA 605399**	Hyperspectral imaging technology for the quality inspection of fish products.	Partner-Coordinator

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.

Our optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the

designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Food and wet chemistry lab equipped with a DSC, a Surface Energy analyzer, an HPLC, a Cobb tester, microbiological equipments, microscope and microtome for structural analysis, migration cells, colorimeter (for color and opacity measurement), Ford cup viscometers, water vapor permeation cells, turbidimeter, climatic chambers, manual and automatic coaters, electrospray, different drying technologies, plasma surface activation, etc.

With a patent-pending optical design, IRIS' VISUM InLine NIR analyzer that has been developed to be easily integrated in production lines, e.g. in pipes, on belts, or in reactors via a customized interfacing with the process. Specific optical fibers probes can also be used for in situ process monitoring as has been proved for controlling melt plastic streams during injection. It integrates miniaturized MEMS-NIR spectroscopic sensors which can be customized with different wavelength ranges depending on the requirements of each application (conventional VISUM InLine unit uses the SWIR spectral region 900-1700nm). It allows fast and continuous quality control (QC) of a wide variety of products and materials. It helps analyzing the impact of the process on the product's critical quality parameters and can provide early warning for corrective actions or be coupled with sorting processes to automatically classify products, reject those out of specifications, etc. Moreover, the VISUM Cloud leverages the Internet-of-Things towards the Industry 4.0 paradigm.

Figure 36. Visum InLine device, examples for integration on a belt (left) or in a pipe (right)

Within IRIS' portfolio, HYPERA is an HSI system with a novel combination of imaging cameras in the VIS- NIR and/or SWIR-spectral ranges coupled with powerful chemometrics and artificial intelligence data processing tools to maximise its resolution and penetration within the sample matrix. HYPERA is designed as a robust industrial equipment suitable for integration into existing processing lines, most typically above a belt where the materials to analyse are fed. Like VISUM, it is thought to be used for automatic QC in critical points along the process line and product classification.

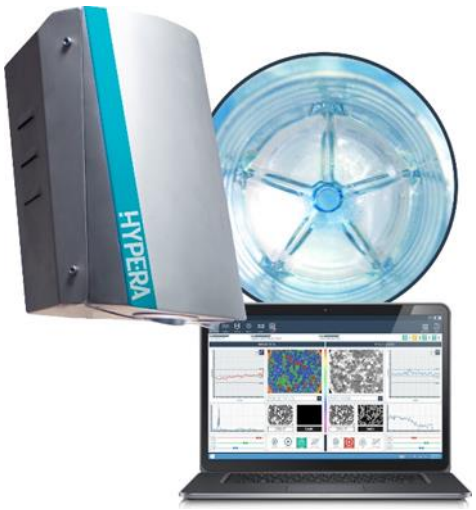


Figure 37. Hypera device

Partner 4 Interuniversitair Micro-Electronica Centrum – (IMEC), Belgium

Acronym of the project: MULTIPLE

Participant Nº: 4	**IMEC**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

IIMEC, founded in 1984, is a world-leading independent Research Institute in nanoelectronics and nanotechnology. Its research focuses on the next generations of chips and systems, and on the enabling technologies for ambient intelligence. IMEC's research bridges the gap between fundamental research at universities and technology development in industry. With headquarters in Belgium, IMEC has research centers in the Netherlands (so called Holst Centre in collaboration with TNO), China, Taiwan, India, USA and Japan. Its staff of around 3500 people includes more than 700 industrial residents, visiting scientists & PhD students. Also, IMEC has ongoing collaboration with over 600 companies and over 200 universities. In 2016, its estimated revenue (P&L) was more than 500M€.

At its Headquarters in Leuven, IMEC has 3 cleanrooms that run a semi-industrial operation, 24 hours a day, 7 days a week, all year round. The 300mm pilot line is used for R&D on (sub-)22nm process technology. This line is being extended for the next-generation of 450mm wafers. A new biolab, new solar cell labs, and the Intel Exascience Lab have been opened. In the 200mm cleanroom, heterogeneous integration is being explored which supports manufacturing with added functionality, such as sensors, actuators, and MEMS and NEMS. This activity is called the CMORE offering. IMEC has an ISO9001 quality certification.

Research and Development at IMEC is organized under the following seven domains and platforms:

- Core CMOS: new processing technologies, materials, devices, and integration and design methods to create future chips and systems.
- Sensor technology: hyperspectral sensors and ion detectors
- Flexible technology: thin film electronics and displays on flexible substrates
- Networking: broad spectrum wireless networking chips
- Digital privacy and security: data encryption and cloud solutions
- Software and data management: machine learning and IoT
- Application domains: smart health, smart cities, smart industries, smart energy, smart mobility

Role in the project

Specifically, for the proposed project, IMEC offers an in-house developed integrated spectral filter technology to be directly deposited on commercially available or custom developed imagers. IMEC leverages experience from and builds on its existing image sensor manufacturing platform for the development and low-volume production of customer specific specialty imagers. The base line technology is a 0.13 µm CMOS platform including a pinned photodiode module processed on 200 mm Si wafers. This detailed imager knowledge is combined with the large flexibility in terms of processes deviating from standard CMOS such as custom filter layers, special (epi) substrates, pixel trench isolation, stitching, backside illumination and others. In developing CMOS Image Sensors, IMEC has more than 15 year of experience for various applications, including space.

The project team proposed offers all guarantees for the successful completion of the work from both the technical and from the management points of view. The IMEC technical and project management team is formed by experienced people coming from a combined imager, hyperspectral systems and process technology backgrounds, with demonstrated experience in EU and ESA R&I projects. With a track record of more than 100 running projects, IMEC ensures a mature and professional effort in achieving project results.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Andy Lambrechts (male), Function: Program Manger-Integrated Imaging team

PhD in Electronics from University of Leuven in 2009. Function: Program Manger & Team Leader of the Integrated Imaging team, with a focus on integrated hyperspectral imaging. Other responsibilities include lens free microscopy, micro mirror based lenses, diffractive optics. Doctoral Researcher on embedded computing, with a focus on energy efficient processor architectures for mobile devices and their compiler technology. Researcher in the hyperspectral imaging team at imec since 2009, responsible for building this activity building on imec's expertise in image sensor, image processing and IC process technology. Senior researcher in the hyperspectral imaging team of imec since 2011, operating as the daily activity lead for hyperspectral projects and the development of IMEC's first integrated hyperspectral image sensor and camera system.

Dr. Murali Jayapala (male), Principal Member of Technical Staff-Integrated Imaging team

PhD in Electronics from University of Leuven in 2005. Principal Member of Technical Staff in the Integrated Imaging team, with a focus on integrated hyperspectral imaging. Other responsibilities include lens free microscopy, micro mirror based lenses, diffractive optics. Senior researcher in hyperspectral imaging team since 2010, responsible to build this activity together with the team. Exploring & coupling imec's silicon processing facilities/capabilities with imaging systems. Hyperspectral application development & exploration for IMEC's hyperspectral technology. Establishing industrial contacts for industrial bi-lateral collaboration. Exploring & coupling IMEC's silicon processing facilities/capabilities with micro mirror based zoom lens application for compact camera systems.

Dr. Klaas Tack (male), Principal Member of Technical Staff-Integrated Imaging team

PhD. degree in Electronics from the University of Leuven in 2006. Principal Member of Technical Staff in the Integrated Imaging team, with a focus on integrated hyperspectral imaging. He started his career at imec as a doctoral Researcher from 2001 until 2006 and researcher from 2006 until 2009 on embedded computing, with a focus on content management for 3D graphics rendering on mobile devices. Since 2009, he is a researcher in the integrated imaging team of imec, responsible for starting the activity on hyperspectral imaging together with Andy Lambrechts. His responsibilities in the team are: Responsible for the design of the HSI sensor prototypes**,** Responsible for building the full system demonstrator and Responsible for building the testing infrastructure for the HSI sensor.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	**B. Geelen, K. Tack, and A. Lambrechts**	A compact snapshot multispectral imager with a monolithically integrated, per-pixel filter mosaic	Proc. SPIE 8974, Advanced Fabrication Technologies for Micro/Nano Optics and Photonics VII, 89740L (March 7, 2014)
2	**B. Geelen, M. Jayapala, K. Tack and A. Lambrechts**	Low-complexity image processing for a high-throughput low-latency snapshot multispectral imager with integrated tiled filters	Proc. SPIE 8743, May 2013.
3	**M. Jayapala, A. Lambrechts, K. Tack, B. Geelen, B. Masschelein and P. Soussan**	Monolithic Integration of Flexible Spectral Filters with CMOS Image Sensors at Wafer Level for Low Cost Hyperspectral Imaging	International Image Sensors Workshop (IISW), Jun 2013
4	**B. Geelen, K. Tack, and A. Lambrechts**	A snapshot multispectral imager with integrated tiled filters and optical duplication	Proc. SPIE 8613, Advanced Fabrication Technologies for Micro/Nano Optics and Photonics VI, 861314 (March 5, 2013)
5	**K. Tack, A. Lambrechts, P. Soussan, and L.**	A compact, high-speed and low-cost hyperspectral imager	Proc. SPIE 8266, Silicon Photonics VII, 82660Q (February 9, 2012)

	Haspeslagh		

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**I-ALLOW**	H2020-ICT-2014-1	Imaging analysis in all lighting and off weather conditions. Develop and demonstrate a low cost imaging solution based on a novel multifunctional approach camera system integrated with a high performance processing unit addressing a vast variety of outdoor scenarios for safety and security applications.	Partner-Hyperspectral filter deposition
2	**RESYNTEX**	WASTE-1-2014	A new circular economy concept: from textile waste towards chemical and textile industries feedstock. Designing, developing and demonstrating a new high environmental impact industrial symbiosis between the unwearable blends (wool, cotton, polymers – representing 93% of all components) of textile waste and the chemical industry.	Partner-Evaluation the use of hyperspectral cameras in textile sorting.
3	**PHYSIS**	EC H2020-COMPET-06-2014	Sparse Signal Processing Technologies for HyperSpectral Systems. Investigate extensions of compressive signal representations and sparsity-enforcing recovery technologies for the acquisition, compression, restoration, and understanding of hyperspectral data.	Partner
4	**CHIEM**	ESA	Compact Hyperspectral Imager Engineering Model. Design a Compact Hyperspectral Instrument and to complete the end-to-end performance analysis.	Partner- Improvement of the Linear Variable Filter (LVF) design to increase its performance, match requirements and increase the usability of the instrument.
5	**Bilateral development of spectrally resolved linear detector array for fuel quality monitoring**	Flanders Government	Bilateral development of spectrally resolved linear detector array for fuel quality monitoring	Partner

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Significant infrastructure

- 5200 m^2 ultra-clean processing area in cleanroom I
- 3200 m^2 ultra-clean processing area in cleanroom II
- 1,750 m^2 class 1 silicon pilot line (200 mm pilot line),

- 6,710 m^2 computer room and utilities
- 16,400 m^2 offices and supporting laboratory space
- 810 m^2 cafeteria and training facilities
- ISO 9001 qualification
- Pilot lines
- 200 mm silicon pilot line
- solar cell pilot line
- MCM-D pilot line
- 300 mm silicon pilot line

Advanced Vision Lab

- State of the art, hyperspectral cameras & illumination setups
- Compute infrastructure (desktops, servers & imaging software)
- Optical benches, sensor calibration and characterization equipment

Partner 5 Airoptic Spolka Z Ograniczona Odpowiedzialnoscia – (AIROPTIC), Poland

Acronym of the project: MULTIPLE

Participant Nº: 5	**AIROPTIC**	AIROPTIC REAL TIME GAS ANALYZERS

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company Profile

AIROPTIC Sp. z o.o. is an ISO 9001:2015 certified SME founded in 2010. AIROPTIC offers high performance laser analyzers for process control and emission monitoring as well as safety and security monitoring and has a global network of distributors. AIROPTIC provides in-situ, extractive and open path systems configuration and today have pioneered in-situ detection multi-gas technology utilizing multiple Mid-IR DFB lasers. The people behind AIROPTIC have been involved in the developing of tunable diode laser gas monitoring instrumentation and spectrometers for more than two decades, including the development of a wide range of gas monitoring instruments for applications in industrial process analytics and in medical analysis. A substantial focus and resources are put on R&D activities concerning development of novel analyzer products. AIROPTIC has participated in several international and national research projects which allowed the company to gain substantial experience in the field of novel Mid-IR sources for gas sensing applications.

Role in the project

In the MULTIPLE Project AIROPTIC will participate in the design of architecture and data collection systems during T1.2 and T1.3 of WP1. In WP2, during T2.3, AIROPTIC will be dedicated to development of multi-gas analyzer technology for real time gas analysis in steel industry and wood based panel industry as well as carrying out the field tests at two facilities (WP6). In WP3, the company will AIROPTIC will develop neural network models and analytical models for robust estimation of gas concentration ratios of different compounds. During WP4, AIROPTIC will support T4.1 and T4.2 as partners in charge of the development of the edge computing devices and the monitoring models producing data. In T4.3,the company will participate the optimization of the chemometric algorithms. During WP5 AIRPOTIC will support multimodal monitoring system integration will all the partners.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Pawel Kluczynski (male), CEO.

He has obtained his master's degree in physics from Poznan University (Poland) in 1997 and his Ph.D. degree from Umea University (Sweden) in 2002 both in a field of applied laser spectroscopy. He started his research on tunable laser spectroscopy in 1997 developing theoretical models for wavelength modulation spectroscopy. In 2001-2010 he has been working with design and architecture of tunable laser gas analyzers at Siemens including spectroscopy, optics and signal processing schemes. He founded AIROPTIC in 2010 and has been the CEO of the company since 2010. He is an author of more than 20 scientific papers and has been cited above 650 times (without self-citations). His H-index is 12. He is also an author of 14 granted patents in the field of tunable laser spectroscopy and inventor of several pending applications in the field of gas analyzers.

Dr. Małgorzata Pokrzywa (female), Project Manager.

She is responsible for the control and coordination of substantive tasks in R&D and commercial projects. She has profound experience (2001-2013) and scientific achievements in the field of genetic engineering and modeling human diseases using transgenic organisms and cell cultures. She is experienced in pre-clinical phase of drug development. She has experience in conducting and coordinating international projects, including supervision of R&D work and R&D staff, i.e. graduate students, PhD students and technical staff. She has been involved in interdisciplinary projects to develop implementation solutions Besides she has education in management and entrepreneurship. She graduated from a mentoring program Mentor4Reserach - Commercialization of scientific achievements for scientists organized by IVA. She has been involved in managing FP7 WideLase, H2020 programmes iCSPEC and MIREGAS as well as 2 national grants, SENSE and HydroC, all devoted to development

of laser based spectrometers for process analytics applications.

Mr. Krzysztof Siembab (male), Electrical Engineering Manager.

He has obtained his Master of Science 2010, from Automatics and Robotics, Poznan university of Technology, Poznan, Poland and he works as an electronics engineering department manager at AIROPTIC. . He and his team are involved in design of electronics hardware an firmware as well as development of signal processing schemes and algorithms for the laser analyzers. He is an author of 7 scientific papers and has been participated in various national research projects with main focus on signal processing schemes and embedded system design for various industrial applications. He has been working towards his PhD in of He has been involved in H2020 programmes iCSPEC and MIREGAS as well as 2 national grants, SENSE and HydroC, both devoted to development of laser based spectrometers for process analytics applications.

Mr. Mateusz Straszewski (male), Spectroscopic Applications Manager

He has received Master of Science 2014, Technical Physics, University of Technology , Poznan, Poland. He is responsible for development of spectroscopic applications, calibration and verifications techniques at AIROPTIC. He has also been working towards his PhD in a field of optical design. He has been involved in H2020 programmes iCSPEC and MIREGAS as well as 2 national grants, SENSE and HydroC, bothh devoted to development of laser based spectrometers for process analytics applications.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	**GasEye Cross Duct CO**	GasEye CO is a high performance in-situ carbon monoxide analyzer. Suitable for safety applications and process control. 24/7 continuous operation. No sample preparation. No zero drift. No field calibration. Low cost of ownership.	Owner
2	**GasEye Cross Duct HCHO**	GasEye HCHO is the world's first in-situ formaldehyde vapor analyzer. Suitable for emission monitoring and process control. 24/7 continuous operation. No sample preparation. No zero drift. No field calibration. Low cost of ownership.	Owner
3	**GasEye Cross Duct NH3 HT**	GasEye NH3 HT is a high performance in-situ ammonia analyzer. Suitable for DeNOX processes and process control. 24/7 continuous operation. No sample preparation. No zero drift. No field calibration. Low cost of ownership.	Owner
4	**GasEye Open Path**	Tunable laser analyzers for air quality monitoring	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**iCspec**	RIA	In-line Cascade Laser Spectrometer for Process Control	Partner – Technology provider
2	**H2020-**	RIA	Programmable multi-wavelength Mid-IR	Partner – Technology provider

	MIREGAS		source for gas sensing	
3	**WideLase**	STREP	Monolithic Widely Tunable Interband Cascade Lasers for Safety and Security	Partner – Technology provider
4	**SensHy**	STREP	Photonic sensing of hydrocarbons based on innovative mid infrared lasers	Subcontractor

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Accreditations

- Quality Management System
  - ISO 9001: 2015
- Confirmation of Conformity to European standards for emission control
  - EN 14181:2015
  - EN 15267-3:2007

Partner 6 SCM – (SCM), Italy

Acronym of the project: MULTIPLE		
Participant Nº: 6	**SCM**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

SCM Group is an industrial group which is leader in the creation, production and distribution of technologically advanced solutions to process a vast range of materials (wood, glass, stone, plastic, metals and composite materials) with specialist brands for specific technologies and centers of excellence expert in industrial components, present in all 5 continents for almost 70 years. SCM employs roughly 4.000 people and has an annual turnover (including subsidiaries) of about 750M Euros.

SCM Group designs systems integrating different technologies and advanced production plants, both for the woodworking sector and for complementary sectors. The SCM Group stand out in its reference market for the ability to supply high tech solutions for the entire production process.

SCM Group has roughly 500 engineers involved in R&D activities, and 400 registered patents. Every year SCM Group invests at least 7% of its turnover in research and development, technology and know-how.

Recently, SCM group has re-arranged its business organization, in particular the R&D department, by shifting some operations concerned research projects to CSR Consorzio Studi e Ricerche Srl. Therefore, in order to implement the activities foreseen in the project SCM Group needs the support of its linked third party CSR Consorzio Studi e Ricerche Srl, which will provide high qualified research competences and all the necessary equipment.

Role in the project

SCM will preside over the furniture manufacturing use case, in collaboration with ROYO. More specifically SCM will lead the data collection task (T1.3) within WP1, as well as strongly contributing to WP5 (Production Optimization) with particular emphasis on furniture manufacturing. Finally, SCM will lead the Integration task (T6.1) in the Demonstration WP (WP6), and will strongly contribute to T6.3 (Demonstration in woodworking).

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Enrico Callegati (Male), Innovation funding manager.

Graduated in International Studies with specialization in Intellectual Property Management. He had his first experience at the Italian Institute of Industrial Promotion, designing and implementing public funding instruments to support manufacturing innovation. After working for 8 years as an expert of public funding for research and innovation in the Italian innovation company CRIT (a network of large manufacturers from Northern Italy including Ferrari, ALSTOM, FIAT Group), he entered in the SCM staff in 2018 with the role of Innovation Funding Manager. He has a long experience in managing and coordinating financed projects both in National and European (Fp7, H2020, etc.) context.

Federico Ratti (Male), Technical & Innovation manager.

Coordinates the whole product development with effort on the new modular product platform. He has 20+ years of experience in product/process design and process optimization.

Michele Mazza (Male), Director of CSR R&D laboratory.

He has 30 years of experience in Machine design and construction, CFD modelling, Acoustic and vibration. He also developed several working tools for wood manufacturing process.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services…	Objective	Role of your organization
1	**Drilling Machines**	The SCM drilling machines are the products of continuing in-house development complementing the sum of the finest skills and know-how acquired when Morbidelli joined the SCM Group. This brand, with its rich tradition, is known today as a model name.	Research & Development and Manufacturer
2	**Edge-Banding Machines**	The widest range of production technologies for customers operating in the furniture and housing fields, either industrial giants or small workshops.	Research & Development and Manufacturer
3	**Belt-Sanding Machines**	The SCM wide-belt sanders are the products of continuing in-house development complementing the sum of the finest skills and know-how acquired when DMC joined the SCM Group. This brand, with its rich tradition, is known today as a model name.	Research & Development and Manufacturer

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**SYMBIOPTIMA**	EU-H2020-RIA SPIRE-6 action	Human-mimetic approach to the integrated monitoring, management and optimization of a symbiotic cluster of smart production units. Role: sustainability assessment of industrial symbiosis flows. http://www.symbioptima.eu/	Partner- Sustainability assessment of industrial symbiosis flows.
2	**FACTORY ECOMATION**	EU-FP7- CP-IP FP7-2012-NMP-ICT-FoF	Factory ECO-friendly, energy efficient technologies and adaptive automation solutions.	Partner- Administrative coordination
3	**CTC**	EU-FP7	Advanced control and supervision techniques for a highly automated shop-floor to realize “close-to-the-customer” woodworking mini-factories.	Partner-Control solution developer. (technical coordination)
4	**INCLUSIVE**	EU-H2020-RIA	Smart and adaptive interfaces for INCLUSIVE work environment	Partner

		FOF-2016		

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

CSR Lab is a structure owned 100% by SCM Group with many acknowledgements and certifications. It is registered to the Italian register of highly qualified laboratories provided by MIUR (Ministry of Education and Scientific Research) authorized to carry out research of an applicative nature in favor of small and medium industries. CSR can provide acoustic and dust tests and has all necessary equipment in order to work on the study, development and modelling of innovative components in the woodworking machinery sector.

Moreover, SCM owns a 8.000 m2., laid out to maximize the customer's product awareness. The machines are on permanent show for demonstration purposes, where potential customers can view the equipment before they make a purchase commitment. The showroom contains some of the most sophisticated CNC technology in the woodworking and panel processing industry. The SCM showroom will host the dissemination events dedicated to woodworking in the MULTIPLE project.

Partner 7 Royo – (ROYO), Spain

Acronym of the project: MULTIPLE

Participant Nº: 7	**ROYO**	Royo

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile of the group

Royo Group International Bathroom (RGIB) is one of the leading European companies in the field of bathroom furniture and shower. RGIB results from the union of two leaders who teamed up to win in this equation: they join the trajectory and experience of Royo Group and the financial strength that, as a platform for growth and consolidation, is provided by HIG.

Company profile of the unit involved in the project

Royo Group is a company born in Valencia and is a leader in Spain, Poland and Mexico, as well as being the third largest European group in its sector. Royo Group is an international company specializing in bathroom furniture and shower equipment, which has 900 employees in its plants in Quart de Poblet (Valencia), Poland and Mexico City and is in an international consolidation phase. Currently, Royo Group produces more than one million pieces of furniture every year.

Role in the project

The tasks of ROYO in the project will focus on the validation of MULTIPLE's multimodal solution performance in its woodworking facilities during WP6. In WP1 will define industrial use case requirements and specifications.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

César Taboas (male), Chief Technology and Innovation officer

Mechanical engineering in the Polytechnic University of Valencia in 1996. He did a master's in quality management in 1997 and a MBA in 2006. He also coursed the Executive development program from IESE. He has worked as a process engineer at Ford for two years and later he became production manager in ROYO Group for ten years. Nowadays he is the Chief Technology and Innovation officer at ROYO. During his professional experience, he has coordinated various CDTI and H2020 projects that have been developed in ROYO Group.

César Martín (male), Designer

Design engineering in the Jaume I Univertiy of Castellón in 2014. He also had the opportunity to study at the Polytechnic University of Turin for one year, with an ERASMUS scholarship. He has worked at ROYO from 2015 in the product development department. Nowadays is designer in the Innovation and design department. He has working on the last H2020 project of ROYO like end user with César Taboas.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services…	Objective	Role of your organization
1	**Royo Click System**	First bath furniture with 15-munites no-tool assembly.	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**TELLME**	EU- FP7	Development and improving learning and training in smaller manufacturing environments with the use of technologies. The aim is to bring innovative training methods to SME manufacturers so they can better supply the needs of their markets, which include larger industries.	Subcontractor
2	**NEW LINE OF POLYLAMINATE COATING**	CDTI	New line of polylaminate coating	Partner
3	**OPTIMIZATION OF THE PRODUCTION PROCESS OF BATHROOM FURNITURE THROUGH SENSORIZATION**	CDTI	Optimization of the production process of bathroom furniture through sensorization	Partner
4	**SAIN4**	IVACE and FEDER program	The objective of the project is to increase the efficiency of the main manufacturing processes, through the construction of an Advanced Management System that will support in the decision making for continuous improvement of availability, performance and quality	Partner
5	**HUMAN**	H2020-	Define and demonstrate workplaces where automation and human workers operate in harmony to improve the productivity, quality, performance of the factory as well as the worker satisfaction and safety.	Partner- Development of tools demonstration in furniture manufacturing.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

ROYO brings the facilities of one or more of its production lines, considering the workers involved in the different tasks of the whole process and the corresponding physical equipment and the activities performed at each point of the production chain.

Partner 8 Photonfocus. AG. – (PHF), Switzerland

Acronym of the project: MULTIPLE		
Participant Nº: 8	PHF	photon focus

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

Photonfocus was founded in 2001 as Spinoff from the Swiss research centre CSEM. Since its foundation, the company is focused exclusively on CMOS vision technology. Photonfocus develops high speed HDR CMOS image sensors with multi tap outputs and global shutter, based on its own patented LinLog® technology. With the LinLog® CMOS image sensors and third-party Low-Light CMOS image sensors, Photonfocus industrial CMOS cameras cover a wide application field in machine vision and in optical metrology. Offering its customers a wide range of cameras to fulfil the needs of the most demanding needs in ultra-low light, HDR, hyperspectral, SWIR and high speed imaging.

Photonfocus AG headquarters is located in Lachen SZ (Switzerland) which consists in 4 offices of total 950 square meters respectively for the design of all camera parts (software, firmware & hardware including mechanics) and to mount and test the cameras.

Role in the project

Photonfocus will develop the camera electronic and cameras for the project in WP2. A SWIR HSI sensor will be implemented in the SWIR camera platform from Photonfocus. This sensor solution will be combined with a VIS/NIR HSI sensor which is available as standard Photonfocus product in a two-aperture camera for hyperspectral imaging.

In the working packages WP3 to WP6 Photonfocus will support the other project partners in the system integration. PHF will participate as project partner on the overall project management and dissemination and exploitation wp7.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Daniel Hofmann (male), Head of Software

Daniel Hofmann is Head of Software at Photonfocus (Switzerland) and CEO of Soluciones de procesado de imagen, SL (Spain). He received his degree in Electrical Engineering in 2005 from the University of Applied Sciences Rapperswil, Switzerland. From 1999 to 2001 he worked as Electronic technician at Feller AG in Horgen, Switzerland. Since 2005 he is working at Photonfocus AG in Lachen (Switzerland) first as software developer then as Head of Software. He is co-founder and CEO of Soluciones de procesado de imagen, SL in Barcelona (Spain) since 2015.

Daniel Gubser (male), Junior Designer Software.

Daniel Gubser joined Photonfocus in 2017 as a junior software developer after he graduated in electrical engineering at University of Applied Science Rapperswil, Switzerland. He is active in the field of embedded software development.

Reto Lienhardt (male), Head of Firmware.

Reto Lienhardt received his degree in Electrical Engineering in 1998 from the University of Applied Science Rapperswil, Switzerland. Before he started his work at Photonfocus, he had been working 2 years as an HW/SW development engineer at Mikrap and 6 years as a digital design engineer in the ASIC & FPGA design centre Memec Design. He joined the Photonfocus team in 2008 and worked as an FPGA engineer. In 2016 he was promoted to Head of Firmware. Reto Lienhardt is responsible for the Firmware design team and the FPGA development.

Cyril Arnould (male), Electrical Engineer

Cyril Arnould is member of the FPGA development team at Photonfocus AG. After getting his Master of Science in Electrical Engineering and Information Technology at ETH Zurich in 2015, he joined Photonfocus AG later in

the same year to start gaining his first experiences working in the field.

Ulf Muehl (male), Mechanic Engineer

Ulf Muehl has a long track record in designing camera bodies for professional cameras in SINAR. Since 2003 he supported Photonfocus in the design of camera cases. In 2018 the business of SINAR was transferred to Leica in Germany. Since this time, he is with Photonfocus and member of the Photonfocus hardware team.

Marc Nyffenegger (male), (FAE, Support)

Marc Nyffenegger is member of the Software development team at Photonfocus AG. After getting his Master of Science in Electrical Engineering and Information Technology in 2015, he joined Photonfocus AG. Starting with embedded software development and software development he focusses on the support of customers in the applications.

Dr. Peter Mario Schwider (male), CTO, Head of Hardware

He is cofounder of Photonfocus AG in 2001. As CTO he is responsible for the CMOS sensor and camera developments. He studied physics in Berlin and Karlsruhe. In 1995 he founded an engineering office working in the field of machine vision system design and developing high-speed vision processors and CCD cameras for industrial and medical applications. In 1998 he joined the CSEM SA Zürich, where he was responsible for the initial operation of CMOS sensors and the design of test equipment's and cameras.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Publications

- P. M. Schwider, Following the CMOS Track, Optik@Photonik 4/2016, pp. 40-43

#	Product/services…	Objective	Role of your organization
1	**HyperSpectral SDK**	Hyperspectral sensors are designed to work in a specific spectral range. IMEC HSI sensors are calibrated with previous knowledge of the filters used in order to compensate the possible errors or mismatches due to the filter effect. The SDK is able to parse the information of the XML calibration file provided by IMEC and use it for the spectral correction. Spectral correction recomposes a raw spectrum into a corrected spectrum, removing higher-order and cross-talk signals from the spectrum and avoiding spectral contributions from unwanted bands. Once the spectral correction is applied to the reflectance hypercube, the available bands change in frequency and number.	Developer with IMEC
2	**2D Cameras**	CMOS area scan cameras for standard applications in machine vision and optical metrology.	Developer with IMEC
3	**3D Cameras**	High speed CMOS cameras with internal sub-pixel calculation of the laser line position for laser triangulation setups.	Developer with IMEC
4	**HyperSpectral Cameras**	High speed CMOS cameras for hyperspectral imaging.	Developer
5	**LineScan**	CMOS line scan cameras based on high speed CMOS area scan sensors.	Developer

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

Photonfocus has many experiences in machine vision and has a large customer basis. Photonfocus HDR cameras and 3D cameras for laser triangulation are widely used in metal industry and in wood industry. In food industry 3D cameras are used in sorting and packaging applications. The Photonfocus HDR sensors are often used in the welding applications. Photonfocus was member of the HQ Tubes project (Grant Nr. 222289) “High-quality Tube and Pipe Production by Hybrid Laser-Arc Welding”.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

AEON Camera Calibrator model ACC2 for EMVA1288 measurements in VIS and NIR. Largest sensor diagonal 80mm, lens mounts: CS-, C-, F- and M72-Mount. For SWIR sensors an additional test unit for the Camera Calibrator with 1050nm, 1300 and 1550nm wavelength with smaller diagonal is available.

Photonfocus has several scanner setups which are used in 3D scanning applications and to check high-speed cameras. These units are equipped with state of the art trigger units and shaft encoders with extraordinary high resolution. This will enable a detailed testing of the HSI cameras in the laboratory.

Partner 9 Russula – (RUSSULA), Spain

Acronym of the project: MULTIPLE		
Participant Nº: 9	**RUSSULA**	RUSSULA

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

RUSSULA is a global engineering company serving the largest steel producers in more than 30 countries. Developing engineering solutions to help steel producers around the world become high-performance businesses.

Since its founding, RUSSULA has supported the modernization of the industry during all economic cycles by deliv-ering practical, cost-effective solutions. Our core values are to understand our clients' needs and to deliver the best engineering service.

RUSSULA has different solutions for Rolling Mills plants: automation and control systems, electrical and mechanical equipment, optimization and tracking for process. As part of the steelmaking process, RUSSULA has expertise for control and building in reheating furnaces, melt shops and water treatment plants.

The RUSSULA automation solution is a unique combination of long product process and electrical system knowledge to ensure our technology solutions increase mill productivity, provide consistent production and premium product quality.

RUSSULA's system makes possible to roll products with ease, taking advantage of today's latest advancements in rolling process technology in combination of RUSSULA R&D.

Role in the project

In MULTIPLE, RUSSULA will design system architecture and the production modelling and parametrization during WP1 and WP5, respectively.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Jose Ramón González (male), Electric department chief.

M.Sc. in Industrial Engineering (electronics). Experience of more than 25 years in steelmaking process control, drives and electrical supervision. His developments are also focused on designing new human/machine interfaces and special interest in new communication protocols. A wide experience in PLC and high-level programing for control process and production managing.

Luciano Guerra (male), R&D director.

M.Sc. in Industrial Engineering (electronics). Experience of more than 13 years in pro-cess control and optimization for steelmaking process. His actual role is focused on being the technological look-out of the company. Researches on applications of artificial vision and robotics for the steel plants.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Publications

- WIPO Patent Application WO/2016/118422: FIORUCCI, Keith. Coil forming laying head system and method of using. U.S. Application No US2016/013656, 2016.
- FERNÁNDEZ, Silvia; SÁNCHEZ, Daniel. Using automation to improve productivity and quality in long products mills. Steel Times International, 2012, vol. 36, no 8, p. 13.
- RIZEK, A. Schultz; VIEIRA, FC Dutra. A high-speed wire-rod mill for Votorantim Metals. Steel Times
- International, 2011, vol. 35, no 3, p. 46.
- Steel's Eco-friendly Water Treatment Plant – May 17, 2016 - AISTech USA

- Improving Rolling Mill Quality and Productivity through Modern Techniques in Automation and Operational Control - May 16, 2016 - AISTech USA

#	Product/services	Objective	Role of your organization
1	**Rolling Mill Control**	The RUSSULA automation team has a unique combination of long product process knowledge and electrical system knowledge to ensure our technology solutions increase mill productivity, provide consistent production and premium product quality. Our high customer retention rate is our best indication of superior technology performance	Owner
2	**Reheat Furnace Control**	Our expertise in reheat furnaces automation has exceeded our clients' expectations in fuel efficiency optimization and scale reduction. Since the company founding, we have automated new and existing, analog and digital reheat furnaces for all kinds of products	Owner
3	**High Speed Data Acquisition Systems (HSDA) systems**	Designed to capture and record signals from the process, providing a real time status of your process, keeping track and storing signal data for days, months or even years.	Owner
4	**Roll Pass Design**	he perfect fusion of art and technology that ensures flexibility, stability and quality in your rolling mill.	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**New integrated technologies for configuration and control of rolling mills**	Program CDTI - FEDER INN-TERCONECTA	New integrated technologies for configuration and control of rolling mills	Coordinator partner
2	**FLEXITREN**	Program CDTI - FEDER INN-TERCONECTA	New Adaptive Design of the Hot Rolling Process for the Flexible Manufacturing of Corrugated and Wire Products	Partner
3	**Design of a pre-heating process free of scale for continuous rolling mills of long products**	Program CDTI - FEDER INN-TERCONECTA	Design of a pre-heating process free of scale for continuous rolling mills of long products	Partner
4	**Simplified modeling of combustion phenomena in billet furnaces**	RUSSULA Chair of Research Studies, University of Vigo	Simplified modeling of combustion phenomena in billet furnaces	Own project

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

RUSSULA has offices all around the globe, offering an immediate and personalized attention to all our customers.

Depending on your current equipment configuration and preferences we propose the best combination of equipment to upgrade your rolling mill.

- Deep knowledge of the most widely used OEM automation platform
- Flexibility to work simultaneously with different brands
- Standard hardware
- Easy integration with existing systems
- Same know-how applied to all platforms
- Open standard programming

We use standard, non-proprietary hardware in all our system designs to ensure multi-sourced, long-term service, and enhancement capabilities.

Partner 10 Laminoirs des Landes– (LDL), France

Acronym of the project: MULTIPLE

Participant Nº: 10	**LDL**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

LDL factory site is located directly across the Bayonne harbor on the Bay of Biscay in Aquitaine, in the historic region of Gascony in Southwest France. LDL rolls heavy plates and deliver reliable results that meet international quality standards and the demands of the European market, producing 500,000 tons of heavy plates/year.

Their flexible solutions meet the individual needs of the customers and the requirements of many industries, such as construction, steel fabrication, wind energy, shipbuilding and mechanical engineering.

The latest European rolling factory is a greenfield project in Tarnos, France. The rolling plant commenced operations in the first quarter of 2018 with the production of heavy plates (Quarto). LDL stands for flexible production designed to meet customer needs, while also providing excellent customer service. This is matched with fast, continuous and reliable deliveries of heavy plates. LDL represents a new generation of steel-rerollers, integrating the business partners into the value chain using the most achievable level of automation. As an industrial fabricator processing the heavy plates, the clients will benefit from an innovative business partner.

Role in the project

The tasks of LDL in the project will focus on the validation of MULTIPLE's multimodal solution performance in its steelworking facilities.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Inazio Fachado (Male), Site director.

BTS M.A.I Mécanique Automatisme Industriel.

Yann Duclos (Male), Maintenance methods technician.

MSc International Logistics.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	**Steel heavy plates**	-On-and offshore wind structures. -Construction and infrastructure projects. -Heavy machinery and yellow goods. -Shipbuilding and offshore platforms. -Pressure equipment. -Mechanical engineering.	Manufacturer
2	**Different steel grades**	-Structural steels for general purposes. -For pressure vessels, boilers, pipes and	Manufacturer

		tubes. -Steel for shipbuilding.	

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Plate mill with production of 500.000 tons per year of heavy quarto steel plates with the next characteristics:

- Thickness : 8-150 mm
- Width: max 3400 mm
- Length : commercial 6 and 12 meters , max 24

LDL has the next infrastructure to meet the demands of the market:

- Reheating Furnace**:** The heavy plate mill is composed of pusher type reheating furnace with 80 ton/h capacity, working on three rows of slabs with max length 3150mm each. The furnace is complete of slab charging device pusher type and discharging system with independent trolleys. The slab dimensions
    - Thickness 200-250-300mm / Width 850-2000-2500 mm /Weight 30 ton
- Discaling System: Slabs discaling system composed of discaling box equipped with heavy pressure nozzles and high pressure water pumps (200-250 bar).
- Rolling line 1: stand ‘4HI’, back up rolls :1600 mm dia x 3600 mm barrel length, work rolls ; 950 mm dia x 3600 mm barrel length, separating force : 4000 ton Thickness adjusting system by hydraulic screw downs with automatic control (HAGC) Stand is equipped in the entry and the exit with side guides and slab turner devices.
- Rolling Line 2: Composed by:
    - Hot Plate Leveller Back Up type with separating force 1500 Ton complete of mechanical screw downs and hydraulic caps. On Line Plate Oxicut located at the entry of cooling bed complete of thickness measuring system and plate marking device.
    - Cooling bed “Walking Beam “ type overall dimensions: 26x30 meters.
    - Out Of Line Plate Oxicut + Plasma.

Partner 11 European Photonics Industry Consortium– (EPIC), France

Acronym of the project: MULTIPLE		
Participant Nº: 11	**EPIC**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

The European Photonics Industry Consortium (EPIC) is an association with 420 leading industry members from the photonics value chain. It is a non-profit, industry driven, small enterprise registered in France and operating from offices in Brussels and Eindhoven, where it can leverage its close ties to other connected organizations such as SEMI and Digital Europe. EPIC members encompass the entire value chain from LED lighting, PV solar energy, Silicon photonics, Optical components, Lasers, Sensors, Displays, Projectors, Optic fibers, and other photonic related technologies. EPIC fosters a vibrant photonics ecosystem by maintaining a strong network and acting as a catalyst and facilitator for technological and commercial advancement. EPIC works closely with related industries, universities, and public authorities to build a more competitive photonics industrial sector, capable of both economic and technological growth in a highly competitive world-wide marketplace.

Role in the project

The role of EPIC in this project is the dissemination and communication and the technology transfer activities. The development of a dissemination plan includes the design of different dissemination tools such as a project website, social media channels (Linkedin, Twitter, Youtube…), a general visual entity and production of flyers, banners and other communication material. Dissemination activities include the organization of international training events & workshops, participation in targeted congresses, press-releases, publication of articles in international journals, aimed at lowering the barrier to entry for all users from all sectors. The second EPIC's role is the design of activities to transfer MULTIPLE results to other potential end-users.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Ana Gonzalez (female), Project leader.

Her expertise relies on the development of systems based on integrated photonic circuits, packaging and assembly, and the investigation of applications such as chemical/biological sensing and Datacom. In addition, she has been involved in several technology transfer and business development processes. She received her Bachelor's degree in Chemistry from the University Autonomous of Barcelona (UAB) and her PhD degree from the Catalan Institute of Nanoscience and Nanotechnology (ICN2).

Dr. Jose Pozo (male), Director of technology and innovation.

He holds a PhD in Electrical Engineering from the University of Bristol (UK), and a M.Sc. and B.Eng. in Telecom Engineering from universities in Spain and Belgium. He has 15 years' worth of experience in photonics technology, strong market knowledge, and a large network within the industrial and academic photonics landscape. He is a member of the board of the IEEE Photonics Society, Benelux.

Dr. Elena Beletkaia (female), Project leader.

She has a vast experience in multiple micro- and spectroscopic techniques. She expertise relies on investigation of application of the non-invasive spectroscopic/multiphoton methods for biomedical applications, e.g. intraoperative resection margins assessment during tutor excisions. She graduated from Lomonosov Moscow State University (MSU) specializing in biophysics and acquired her PhD in Leiden University in Physics of life science.

Mr. Carlos Lee (male), General Director.

He has a BBA in Finance and an MBA in Leadership & Change Management from United Business Institutes (Belgium). Recognized as one of the most energetic and pragmatic sector leaders in the industry. He has a firm grip on the technological aspects of the photonics sector, with over 20 year's senior management experience. His background and core skills are in business, finance and organizational leadership, and it is this which makes him a key driver in the business exploitation of photonics technology. He maintains personal relations with several

hundred industry leaders and is capable of assembling ad hoc teams to evaluate and pursue project business opportunities as they arise. Prior to heading EPIC, he was a key player of SEMI Europe (an association of micro- and nano-electronics industries), having spent fifteen years in roles covering outreach, standardization, technology roadmaps, and organizational development.

Ms. Auri Ripoll (female), Marketing manager.

She received a bachelor's degree in Biology from the University of Barcelona and a master's in marketing management from EAE Business School. She started her career as a scientific professional working for over nine years in several companies. She has worked as a marketing manager in a chemical distribution company where she was responsible for the marketing strategy, including the branding and digital marketing, where she developed online advertisement and social networks campaigns. Auri has also organized the participation of companies in international exhibitions.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	**Europe Drives Fiber Sensor Development for Industrial Applis**	Communication Dissemination	Research & Development. Published in EuroPhotonics, 2018
2	**The Merits of Single-Mode vs. Multimode Fiber Optics**	Communication Dissemination	Research & Development. Published in EuroPhotonics, 2018
3	**Photonic Design, Verification, and Implementation**	Photonic technologies	Research & Development. Published in Optik&Photonik, 2018.
4	**Integration of Photonic Systems**	Photonic technologies	Research & Development. Published in Optik&Photonik, 2018.
5	**The Worldwide Market for Lasers**	Photonic technologies	Research & Development. Laser Technik Journal, 2018

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/ Activities	Type	Objective	Role of your organization
1	**"InPulse"**	H2020-IA	Indium-Phosphide Pilot Line for up-scaled, low-barrier, self-sustained, PIC ecosystem.	Dissemination and Communication and Business Plan Activities
2	**"PASSION"**	H2020-RIA	Development of application driven photonic technologies supporting an innovative transceiver and nodes in metro connections.	Support all the activities regarding the exploitation and the dissemination of the results.
3	**"PIXAPP"**	H2020-IA	Creation of a packaging pilot that have the potential to bring the photonic methods and technologies to mass markets in Communications, Healthcare and Security	Support all the activities regarding the exploitation and the dissemination of the results.

			sectors.	
4	**"MIRPHAB"**	H2020-IA	All-services integrated Pilot Line for the development of MID-IR photonics sensors in Europe, a single-access point to the best technology provided by a consortium of leading companies in the field of photonics.	Support all the activities regarding the exploitation and the dissemination of the results.
5	**"PICS4ALL"**	H2020-CSA	Stimulation of new business for open access InP photonic integrated circuit.	Support all the activities regarding the exploitation and the dissemination of the results.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Tools

- A highly dynamic LinkedIn profile in top 1% most viewed profiles.
- An extensive network of over 4500 personal contacts.
- Management of 15 LinkedIn groups related to photonics (www.epic-assoc.com/linkedin).
- A channel on YouTube with videos cumulating 20.000 views (www.youtube.com/user/CarlosLeeEPIC)

Partner 12 Tematys – (TEMATYS), France

Acronym of the project: MULTIPLE

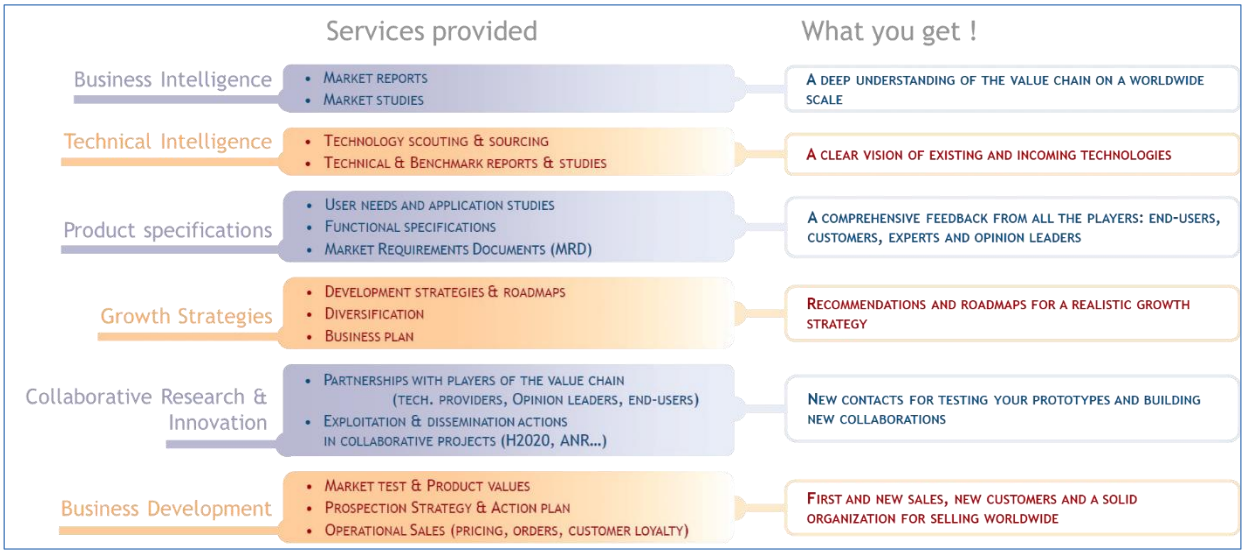
Participant Nº: 12	**TEMATYS**	Tematys

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

TEMATYS is an independent SME dedicated to studies and strategy in Optics, Photonics, Sensors, Imaging, Materials engineering and their application markets. The company has developed a special expertise in Technology Transfer, R&D valorization and marketing of emerging technologies coming from Research Organizations, but also from private industrial groups, SME and start-ups.

TEMATYS provides studies and sourcing services to end-users in application markets (life science, security, aerospace…). TEMATYS also provides strategic views on optical, photonic and sensor markets for clusters and public agencies. TEMATYS clients are Companies and start-ups, Research Organizations and Public institutions. **TEMATYS has more than 150 customers in 18 Countries** (Europe, US, Asia …).At critical steps of the product development process, TEMATYS provides services far beyond market studies. Tematys helps understand the end-user requirements and design products that will meet closely the demand.



Along with custom studies and missions, TEMATYS publishes Market Reports on Optics or Sensors technologies and their applications.

Spectral Imaging: End-user needs, Markets and Trends	2018
Photoacoustic Imaging: Technology, Systems, Market and Trends	2017
Miniature and Micro spectrometers: End-users needs, Market and Trends	2016
Terahertz Components & Systems, Technology and Market Trends (Update of the Report released in 2013)	2016
Cascade Laser Components & Systems: Technology and Market trends	2015
Infrared Imaging Photodetectors and Systems: Technology and Market Trends	2015

Role in the project

In the project TEMATYS will use its expertise to take a role in WP7. In this WP, TEMATYS will collaborate with EPIC for the Dissemination and Exploitation of the project. TEMATYS will be Task Leader for: Task 7.2 - Exploitation and IP management.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Thierry Robin (Male), Founder and Partner.

He has over 20 years' experience in both technical innovation and marketing of emerging technologies, running more than fifty projects in optics, photonics and sensors with prominent players in Security, Healthcare and Industry. He graduated from the Ecole Polytechnique, Paris (France). He holds a master's degree from Ecole Normale Supérieure, Paris (France) and a PhD from the Physics Center of Ecole Polytechnique.

Jacques Cochard (Male), Founder and Partner.

After a degree received from the Ecole des Mines of Nancy (France), he started his career as a consultant for the European Commission. In 2000, he joined Opticsvalley, the photonics cluster of Paris as Project Manager. In 2004, he became Senior Consultant at Tykya before being one of the founders of TEMATYS. Jacques has run more than 70 marketing and technological studies for photonics companies (Cilas, Genoptics / JobinYvon Horiba…), photonics clusters and R&D laboratories (CEA, CNRS, Paris VI Pierre & Maris Curie, GeorgiaTech, Institute of telecom Paris…). Jacques has published more than 20 articles in business press dedicated to laser and photonics (laser focus world, Biophotonics, Optik & Photonik…)

Benoît d'Humières (Male), Gerente,

Benoît d'Humières graduated in Physics and Chemistry from the ESPCI-Paristech and the University P. & M. Curie of Paris. He is an expert in Optics, Electronics and Instrumentation for industrial and research applications. In 1991, after some years of research at the University of Florida, he became an innovation consultant. He has run many marketing and technological studies for companies of any size and Research institutions. Benoît teaches also the marketing of technologies at the Institut d'Optique-Graduate School.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services…	Objective	Role of your organization
1	**Hyperspectral and Multispectral Imaging: Applications and Market Trends, T. Robin.**	Application and market research	Market research. EPIC Meeting at VISION (Stuttgart), November 2018.
2	**Application of Multispectral Imaging in Industry, Biology and beyond, T. Robin**	Application and market research	Market research. CHII (Graz), June 2018.
3	**Spectral Imaging Market and Applications, T. Robin.**	Application and market research	Market research. SPIE DCS (Orlando), April 2018.
4	**Spectral imaging spreads into new industrial and on-field applications, C. Bouyé, B. d'Humières.**	Application and market research	Market research. Proc. SPIE 10539, Photonic Instrumentation Engineering V, February 2018.

5	**Spectral Imaging: Ready for on-field and industry 4.0, C. Bouyé, B. d'Humières**	Application and market research	Market research. Novus Light, December 2017.

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/ Activities	Type	Objective	Role of your organization
1	**"MIR-BOSE"**	H2020-RIA	Mid-and far-IR optoelectronic devices based on Bose-Einstein condensation	Partner-Exploitation tasks
2	**"MIRPHAB"**	H2020-IA	All-services integrated Pilot Line for the development of MID-IR photonics sensors in Europe, a single-access point to the best technology provided by a consortium of leading companies in the field of photonics.	Partner-Exploitation tasks
3	**"C3PO"**	H2020-RIA	Advanced concept for laser uplink/downlink CommuniCation with Space Objects.	Partner-Exploitation tasks
4	**"COM'TONIQ"**	FR-ANR	Ultra high data rate quasi-optic phoTONIC-based COMmunications.	Partner-Exploitation tasks
5	**COST Action NQO**	H2020	The Action aims at promoting and coordinating forefront research in nanoscale quantum optics (NQO)	Subcontractor-Providing market perspectives and feedback from end-users
6	**FET-OPEN PETACom (Petahertz Optoelectronics)**	H2020	PETACom proposes to create a new paradigm in ultrafast electronics.	Subcontractor-Providing market perspectives and feedback from end-users

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Partner 13 Yiotis SA – (JOTIS), Greece

Acronym of the project: MULTIPLE		
Participant Nº: 13	JOTIS	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

JOTIS S.A. is a food industry. Products manufactured in its Athens-based facilities include baby foods, powder mixes for baking and cooking, chocolates, syrups and chilled desserts. Its large product range results to a wide plastic packaging material range necessary for the production of its finished products, along with strong know how in area such as Process Science, Structural Design and Material Science.

The company was founded in 1930 and today occupies 315 employees, exports to the 5 continents, has an annual turnover exceeding 70 M€ and, thus, is included among large enterprises according to pertinent EU legislation. The company's financial data display a highly respected Greek food industry as annual turnover is increasing and gross profit averages more than 50% of the annual turnover. Exports account for approximately 3.5%. The company's staff is highly educated and constantly trained. Disciplines employed lie in the field of Chemistry, Agricultural Engineering, Food Technology and Chemical Engineering. It must be noted that JOTIS annual R&D expenditures account for 3% of the company's annual turnover, indicating the company's orientation towards research. For the past 20 years, JOTIS S.A. has participated in more than 30 National and European Research Programs mostly as coordinator. The company's organizational scheme combines with its highly educated workforce ensure the necessary administrative and technical support of any large-scale project.

Role in the project

The tasks of JOTIS in the project will focus on the validation of MULTIPLE's multimodal solution performance in its food processing facilities.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Ladikos Dimitrios (male), Ph. D, R&D and Quality Assurance Director of JOTIS S.A.

He received his PhD in Food Biochemistry/Science from Procter Department of food Science, Leeds University and he joined JOTIS in 1989 as head of R&D Department. During period 1991-1996 he served the Greek National Supreme Chemical Council as a member of the opinion committee on food legislation approvals. As a member of the Board of Directors, he is actively participating in the company's formulating decisions on innovation strategy and he is overviewing research in the development of novel food products, patents application, participation in EU or National Funded Research programs and innovation partnerships.

Tavantzis George (male) Msc. Head of Reasearch & Innovation Department of JOTIS S.A.

He graduated in 2006 as a "Chemical Engineer" from National Technical University of Athens and received his Master of Science in "Engineering Business Management" from the University of Warwick in 2009. He joined the company's Research and Development Department and is currently employed in the fields of Development and Improving Production Lines, Waste Management, Auditing. His research activity in the fields of Dairy Products and High Hydrostatic Pressure Technology resulted to a number of publications and participation to various conferences. He has attended numerous training seminars focused on Food Engineering, Waste Treatment and Food Legislation.

Antoniou John (male), Food Scientist for the Research & Innovation Department of JOTIS S.A.

Ph.D, graduated in 2006 as a "Food scientist & Technologist" from Agricultural University of Athens, Greece. His Ms.c Degree was specialized in Food Engineering and received from Food Technology and Human Nutrition Department of Agricultural University of Athens, Greece (2008). His Doctorate Degree was received in 2014 from Jiangnan University, Wuxi, China and specialized in Food Science & Technology and resulted in a number of scientific publications (13) and participation to various conferences.

Katerina Pissaridi (female), PhD. Research Project Supervisor.

She is a Chemical Engineer, graduated in 2007 from the National Technical University of Athens (NTUA). She holds a PhD (2012) in biomaterials from NTUA.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organization
1	**Development of nano-emulsifiers**	Manufacturing of in new Vegetable Cream and Glaze Coating	Owner
2	**Using the Stevia rebaudiana plant in manufacturing process.**	Production of high added value products, with applications in the food industry	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**LIFE13 ENV FOODPRINT**	LIFE	Development of an integrated strategy for reducing the carbon footprint in the food industry sector	Coordinator
2	**RETEX I**	RETEX I	Equipment for dealing with industrial waste and scrap (RETEX I)	Partner
3	**SYNERGASIA 09**	SYNERGASIA 09	Development of novel bioprocesses towards exploiting Food Industry's Residues for Biofuel Production	Partner

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

In terms of the company's available lab and research equipment it should be noted that the company has an Independent Research Institute in a surface area of 1305 m^2, equipped with state-of-the-art analytical equipment towards carrying out quality control of its raw materials, packaging materials and finished products. The laboratories are equipped with high resolution instruments covering a wide range of analytical measurements including spectrometric, chromatographic and combined techniques such as ICP-MS, GC-MS/MS, LC-MS/MS, GC-MS/ECD and HPLC towards performing migration studies from packaging materials to food and determination of heavy metals.

Partner 14 Senorics GMBH I.G– (SENORICS), Germany

Acronym of the project: MULTIPLE

Participant Nº: 14	**SENORICS**	SENORICS

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile

SENORICS is a start-up originating from the University of Technology Dresden commercializing an optical sensor solution for near-infrared (NIR) spectroscopy and imaging based on organic electronics. This proprietary technology allows small, robust, wavelength selective and fully customizable detectors and imaging arrays enabling NIR-spectroscopy and NIR-imaging on a chip level. Thus, SENORICS can deliver unique solutions for a multitude of measurement and detection problems in industrial applications. Examples are determining ingredients or compositions of raw materials, consumables or end products in the food, chemistry, packaging or automotive industry and many other branches.

Role in the project

In the MULTIPLE project, SENORICS will participate in the WP1 through task 1.2 for design a system architecture. In WP2, the company will perform the development of cost-effective versatile spectrometers specifically tailored to the target applications. In WP5 will participate with all the partners in multi modal monitoring system integration.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Dr. Robert Brueckner (male), CTO bei Senorics.

He studied in Physics with competences in electronic engineering and semiconductor physics and received his Diploma in Physics specialized in Optoelectronics and Photonics. Optical microcavities as now utilized for Senorics main technology have been investigated in detail in 2009. In 2013, Robert achieved his PhD in Physics (summa cum laude) on metal organic microcavities: the basis of electrically contacting a cavity device. Between 2012-2016 He led the Organic and Polymerpath of the excellence cluster of TU Dresden (cfaed) and in 2016 received the Carl-Zeiss-Award for Young Researchers for the outstanding publication performance.

Dr. Matthias Jahnel (male), Head of Production.

In 2008, Matthias received his diploma degree in electrical engineering from University of Applied Sciences Lausitz and in 2012 the master's degree in physics from Brandenburg University of Technology Cottbus. From 2008 to 2016 he performed as leading technologist & researcher at Fraunhofer Society working on 2008: Conceptual design and development of production plan, Development of semiconductor process technology (lithography, evaporation, sputtering, etching), Integration of OLED and OPD to silicon backplane (CMOS) and Upscaling to manufacturing level. In 2016 he obtained his PhD degree in physics with the topic Organic photosensors with spectral adaptation and the development of organic devices for narrow-band light detection and light emission.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Publications

- Nature Communications 8, 15421 (2017); https://doi.org/10.1038/ncomms15421
- Patent: PCT/EP2016/069256

#	Product/services…	Objective	Role of your

			organization
1	**Design and manufacturing optical spectroscopic sensor solutions for industrial applications**	Quality control and process monitoring.	Provider
2	**Customized sensors, corresponding software and calibration models**	Sensoring and measuring key parameters of client production process	Provider
3	**Developing of proprietary new sensor technology**	Detecting visible and near-infrared light.	Provider
4	**Senorics miniaturized spectrometer chip**	Near-infrared spectroscopy to measure Ingredients %: water, sugar, cotton, carbs, protein, alcohol, plastics, cotton and drugs.	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project	Type	Objective	Role of your organization
1	**ORGADETECT H2020-EU.1.1.**	ERC	Combining (OPBTs) with the novel photodetector technology invented in TU Dresden to create photodetectors as well as fully integrated chip-sized NIR-spectrometers with high detectivity, high resolution, and with all the advantages of organic electronics such as flexibility and low cost mass production.	Coordinator
2	**PLASMONIR**	BMBF	Organic infrared detectors with in-cavity inflated plasmonic wave length	Coordinator

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

SENORICS has access to a manufacturing tool for producing detectors and imaging chips based on the proprietary SENORICS technology. A long-term cooperation and permission for use contract with the owner of the tool exists.

Partner 15 MRA Instrumentação para Medição Registo e Análises, S.A.– (MRA), Portugal

Acronym of the project: MULTIPLE

Participant Nº: 15	**MRA**	

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile of the group

Grupo Álava is an entirely privately owned group which has been providing high technology solutions in the Testing, Measurement, Communications, Security, Defense and Preventive Maintenance fields since it was first founded in 1973.The group offers consultancy, engineering, distribution, training and technical services, providing turn-key projects for several sectors including Aerospace, Automotive, Security, Defense, Communications and Finance, as well as Testing and Research Centers, Universities, Public Services and Industry in general. With a team of over 175 professional (70% of whom have an engineering degree) and a turnover of more than € 40 million, Grupo Alava Ingenieros has experienced an average annual growth of 15% throughout the last decade thanks the its ability to adapt to the specific needs of its customers and its responsible attitude towards supplies and services carried out, all of which is backed by - solid international partnerships - in-house resources for integration, installation and after-sales technical service.

Company profile of the unit involved in the project

MRA Instrumentaçao (MRA), which is placed in Portugal, is a company leader in high-tech solutions. MRA belongs to Alava Group. Its mission is to offer its clients a wide offer of products and systems based on cutting-edge technologies for related areas to Industry 4.0 Solutions, Sensors and Measurement, Communication, Predictive Maintenance, Monitoring, Test, Computer Vision or Thermography for multiple sectors. Additionally, MRA Instrumentaçao provides high added-value services such as:

- End-to-end projects management.
- Technological integration and technical advising.
- Predictive maintenance of assets.
- Specialized technical consultancy.

Thanks to the combined offer of products and services, MRA can propose comprehensive approaches to the needs of its clients, covering the whole aspects of the projects: diagnostic, design, implementation, monitoring, training and technical after-sales services. Thanks to its wide product portfolio, MRA is present in most of the sectors of the industry, including the process industries. MRA has signed more than 50 distribution agreements with world-class companies. With more than 45 years of experience, MRA has carried out more than 350 international projects around the World. In them, MRA has demonstrated solid technical management capabilities and financial solvency. MRA based its business strategy in implementing continuous innovations to its product portfolio and incorporating to it the latest technologies. Thanks to its R&D strategy, MRA was awarded in 2016 as “Innovative SME” as part of Alava Group.

Role in the project

MRA will collaborate in WP1 (Requirements, specifications and data collection), WP4(Smart edge computing), WP5(Integration and production optimization), WP6 (Integration and industrial demonstration), WP7(Exploitation and dissemination) and WP8 associated to project management.

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

Rafael Ramos(male) senior manager at MRA Instrumentaçao

He brings more than 20 years of experience coordinating technical and commercial teams. He demonstrates wide experience in the development of projects in the area of recording interactions, quality management and Speech Analytics. He developed several projects in the Security Telecommunications Recording Area. In

addition, he also managed the coordination of the technical teams in the implementation, training and maintenance of the developed solutions.

Pedro Rebelo (male) Key account manager at MRA Instrumentaçao

He brings more than 18 year of experience as sales engineer, corporate development, research and development and product engineer.

Nelson Dias (male) Key account manager at MRA Instrumentaçao,

He brings more than 17 year of experience as product engineer. He has a wide experience working on signal processing, digital signal processors, numerical analysis, sensors, automotive engineering and electronics.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services	Objective	Role of your organisation
1	**Sales and Installation Services of Sensors and Data Capture Systems**	Wide product portfolio that covers from OEM applications to high-precision application to measure magnitudes as: shifting, velocity, acceleration, torque, pressure, flow, temperature, humidity, etc. Great variety of data capture and analysis systems together with calibration equipment.	Owner
2	**Sales and Installation Services of Vibrations**	Equipment and service to data capture and monitor vibrations on industrial equipment, real-time and post-processing analysis of the measured signal.	Owner
3	**Sales and Installation Services of Non-destructive Tests**	Making possible to contactless examine the quality of industrial products with technologies such as: active thermography; X-rays; shearography; objects digitalizations with laser technologies; or deformation analysis using video-correlation technologies.	Owner
4	**Sales and Installation Services of Vision Technologies**	Ad-hoc design and implementation of computer vision solutions for industrial applications. Distribution of wide range of cameras for industrial purposes: computer vision cameras; high speed cameras; multi-spectral and hiper-spectral cameras; or thermography cameras.	Owner
5	**Predictive Maintenance Services**	Monitoring of industrial equipment by highly trained technical experts, using cutting-edge measuring technologies and analysis software. The objective of this service is to foresee problems on the machines through a preventive diagnosis. As a result, reliability on the plant is increased, while improving the security and availability of the assets avoiding unexpected breakdowns.	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project /Activities	Type	Objective	Role of your organisation
1	**FANDANGO**	CDTI-CIEN	Advance Fabrication of Automotive components using	Partner

			trustworthy and secure digital twins. FANDANGO Project explores Digital Twins as the hybridation of models or virtual simulations with real data captured from the assets, processes or products with the goal of standardize data exchange; improve digital twin abilities through machine vision and other technologies; and manage data reliability and data ownership.	
2	**ESCAPHIB**	CDTI-CIEN	Structures and Systems for a Hybrid Propulsion Passenger Aircraft) its ultimate goal aims the research of four new key blocks of electrical technology to be tested on a hybrid aircraft demonstrator (E-FAN X) developed by the AIRBUS Group. The development of these new blocks will focus on the high power cooling system, the intake and exhaust of a high power generation system, the battery protection system and the development of new configurations in tail cones, leading to the validation and verification of these systems on the ground and in some flights.	Partner
3	**INSPECTOR**	CDTI-CIEN	Industrial inspection and maintenance of complexor unattended facilities: Machine Learning and big data analysis project, which has been applied to predictive maintenance associated to AEGs and hydroelectric generators. The project has improved the semiautomatic signal process, significantly reducing the workload of the analysts of predictive analysis and diagnosis of machines in perfect working order.	Partner as Preditec
4	**VODI**	CDTI-CIEN	Video operation of installations: The VODI project uses Artificial Vision techniques to observe the open/closed state of electrical disconnectors during the complete solution supply operation: Specific Hardware, Monitoring Software, Artificial Vision Algorithms, Connectivity. It is capable of detecting temperature and it is electromagnetically shielded.	Partner
5	**ENJAMBRE**	CDTI-CIEN	Critical emergency missions with manned and unmanned aerial means in cooperative flight. The works to be carried out were: Definition of the technical specifications of the mission management system. Analysis of the sensors that meet the defined technical and operational characteristics. Execution of the integration of the different sensors required for each of the missions with the intelligent mission control system, carrying out its functional validation.	Partner
6	**ONTIME**	Retos-Colaboración	Remote operation of information transmission in emergency missions. The work to be carried out was: Develop a new information capture system that makes an intelligent selection of sensors based on mission conditions, synchronizing and integrating heterogeneous multisource information. Develop new advanced visualization and interaction technologies to facilitate decision-making in emergency missions, based on Mixed Reality technologies.	Partner
7	**INTEFIX**	7PM-UE	Intelligent Fixtures for the machine of low rigidity components. The main result of this project has been the integration of new and cutting-edge technologies (sensors, actuators, control algorithms, simulation tools, etc.) applied	Partner

			to workpiece handling systems in order to develop intelligent and modular devices capable of modifying the behavior and interactions between process and systems in machining operations. These systems reduce time and costs with better performance and capabilities.	

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Following facilities of MRA (Grupo Álava) will be at disposal of the project:

- Two laboratories equipped for electronic repair of industrial equipment. One located in Leganes, of 70 m^2, and the other in Zaragoza, of 35 m^2.
- CMDT center (Condition Monitoring, Diagnostic & Training Center): center specialized on remote monitoring of industrial critical equipment, where predictive maintenance analyst foresees breakdowns by monitoring key elements of industrial assets.

Partner 16 ThingsO2– (TiO2), Spain

Acronym of the project: MULTIPLE		
Participant Nº: 16	**ThingsO2**	THINGSO2

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Description of the entity

ThingsO2 is a company specialized in the digitalization of industrial environments through cutting-edge IoT and Big Data technologies. Its solutions are designed to increase the knowledge about the plant industrial assets and processes, knowing their actual working-state together with real-time visualization and deep analysis of the parameters. Therefore, our solutions contribute to increase the reliability, efficiency, effectiveness and quality of industrial processes.

The company is formed mainly by senior specialists in computer science who are developing an IoT platform based on open source technologies in order to satisfy their clients that are mainly coming from industrial sector.

Company profile of the unit involved in the project

ThingsO2, which is placed in Madrid (Spain), will be a key partner of the project in order to support the activities which are specialists:

- Edge Computing
- Technological integration and technical advising
- Cloud applications based on big data technologies
- Front End development.
- Machine learning integration

The company develops the IoT platform using Agile methodologies and gathering in it all the knowledge from its clients and its stakeholders which convert ThingsO2 in one of the most prominent start-ups in Spain at the moment.

Role in the project

ThingsO2 will collaborate in WP1 (Requirements, specifications and data collection), WP4(Smart edge computing), WP5(Integration and production optimization), WP6 (Integration and industrial demonstration),

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Tomás Ciudad Sobrino (male) – Project Manager

He is an industrial engineer with more than 10 years of experience in the industrial sector. Their role has evolved from I&C specialist to project manager. He has participated in several projects of the digitalization in the energy industry and now I am responsible for all the projects in ThingsO2.

Marcos Bejarano (male) – Product Owner

He is a Computer Science engineer with more than 15 years of experience in software development. He has the role of the product owner of the ThingO2 platform, which oversees gathering all the requirements from the business unit, prioritizing and sharing them with the development team.

Rafael Garrote (male) – Technical Lead

He is a Computer Science engineer specialized in big data technologies. He has participated in several projects related to big data applications and he has wide experience coding. He is the ThingsO2 technical lead which supports the development teams in their tasks.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

- N/A

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project /Activities	Type	Objective	Role of your organisation
1	**FANDANGO**	CDTI-CIEN	Advance Fabrication of Automotive components using trustworthy and secure digital twins. FANDANGO Project explores Digital Twins as the hybridation of models or virtual simulations with real data captured from the assets, processes or products with the goal of standardize data exchange; improve digital twin abilities through machine vision and other technologies; and manage data reliability and data ownership.	Subcontractor
2	**HEATLAND**	PRIVATE	The project consists in testing a new type of pavement in order to reduce the heat that is transmitted to the environment. The ThingsO2 platform is in charge of monitoring the main characteristics of the pavement installed.	Main contractor
3	**ALTRAN**	PRIVATE	Smart PDM, the project consists of monitoring a rotatory machine in order to achieve its best performance and minimizing the maintenance time.	Subcontractor

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

ThingsO2 has developed its own services and front-end interface based on open source libraries which are hosted in the cloud and is the basis of the big data IoT platform. In addition, the company has developed its own software for the edge including connectors with the industrial equipment using standard protocols as OPC UA and Modbus among others.

4.2 Third parties involved in the project (including use of third-party resources)

01 – AIMEN

No third parties involved.

02 – ABRAIA

No third parties involved.

03 – IRIS

No third parties involved.

04 – IMEC

No third parties involved.

05 – AIROPTIC

No third parties involved.

06 – SCM

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	**N**
Does the participant envisage that part of its work is performed by linked third parties	**Y**

SCM is going to involve its related companies CSR and SCM España in order to develop part of the work in various WPs.

Acronym of the project: MULTIPLE		
Participant Nº: [Linked third party of n.] 6	**CSR Srl (Linked thid party of SCM Group S.p.A.)**	
1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal		

Description of the entity

CSR was born within the SCM group, world leader in the field of woodworking machines, with the aim of bringing together and developing all the research activities of the companies of the Group. Today CSR is an autonomous and independent reality from the managerial and operational point of view which, thanks to the experience gained and the potential of the structures, approaches in the areas of pertinence the problems of manufacturers and users operating in these sectors. The industry needs more and more specialized research for the determination of design hypotheses, for the evaluation of the performance parameters of the prototypes and for carrying out controls to verify the quality standards as well as their compliance with the regulations. In this context CSR is a qualified partner accredited at European level that offers solutions to industrial research problems operating in the following areas: Acoustic studies, Vibrational studies, Dust studies, Laboratory checks and Structural analysis.

Role in the project and in the WPs

SR will work in close collaboration with beneficiary SCM Group, taking specific care of the following tasks:

- In WP1 (Requirements), supporting Industrial use case requirements and specifications, System architecture and Data collection activities
- In WP6 (Industial Demonstration), overseeing WP management activities and benchmarking
- In WP7 (Exploitation and dissemination), carrying out innovation management activities for SCM
- In WP8 (Project Management) coordinating the reporting and costs accounting activity within SCM

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Enrico Callegati (Male). Graduated in International Relations. He has gained experiences at the Italian embassy at Warsaw, the EU, and the Institute of Industrial Promotion. After working for 8 years as an expert of public funding for research and innovation in an Italian SME named CRIT, he entered in the SCM staff in 2018 with the role of Innovation Manager. He has a long experience in managing and coordinating financed projects both in National and European (Fp7, H2020, etc) context.

Michele Mazza (Male). Director of CSR R&D laboratory. He has 30 years of experience in Machine design and construction, CFD modelling, Acoustic and vibration. He also developed several working tools for wood manufacturing process.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Not applicable

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

REEMAIN - Resource and Energy Efficient ManufacturINg (FoF.NMP.2013-1).

REEMAIN combines cutting edge knowledge and experience from production processes, professional energy simulation, energy and resource planning, and renewable energy and storage to develop and demonstrate a methodology and platform for the boost, of efficiency resources.

CTC - Close To the Customer (FoF.NMP.2013-6).

The CTC project aims to support European Industry to adapt to global competitive pressures by developing methods and innovative enabling technologies towards local flexible manufacturing of green personalized products close to the customer in terms of features offered, place of fabrication, time to deliver, and cost.

Smart and adaptive interfaces for INCLUSIVE work (FoF.NMP.2016)

Modern manufacturing systems are becoming more and more complex because of higher demands for fast production rate with high quality and flexibility, that is the ability to customize the production lines to the constantly changing market requests. Despite high levels of automation of machines and robots, humans remain central to manufacturing operations since they take charge of control and supervision of manufacturing activities. Human operators interact with machines and robots by means of human-machine interfaces (HMIs), which are unavoidably becoming very complex as new functions are implemented by the production system and include a wide range of possible operational modes and commands. In this scenario, human operators experience many difficulties to interact efficiently with the machine; this is particularly true for middle age workers who feel uncomfortable in the interaction with a complex computerized system and young inexperienced or disabled people who cannot effectively manage such complex production systems. To overcome this gap, INCLUSIVE aims to develop a new concept of interaction between the user and the machines in which the behavior of the automation system adapts to human operator capabilities.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

CSR Lab is a structure owned 100% by SCM Group with many acknowledgements and certifications. It is registered to the Italian register of highly qualified laboratories provided by MIUR (Ministry of Education and Scientific Research) authorized to carry out research of an applicative nature in favour of small and medium industries.

CSR can provide acoustic and dust tests, and has all necessary equipment in order to work on the study, development and modelling of innovative components in the woodworking machinery sector.

6. Resources to be committed

CSR will allocate 10 PMs distributed in WP1(2), WP5(4), WP7(2) and WP8(2), and 10,000€ in 'other direct cost':

CSR	Cost (€)	Justification
Travel	10,000€	Attending to project meetings (5,000€). Attending to dissemination events (5,000€)
Total	**10,000€**	

Acronym of the project: MULTIPLE		
Participant Nº: [Linked third party of n.] 6	**SCMGROUP ESPANA S.A.U. (Linked thid party of SCM Group S.p.A.)**	scmgroup Industrial Machinery and Components

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Description of the entity

SCM España is the Spanish subsidiary of the SCM group. It hosts roughly 30 employees, mainly technicians and engineers, devoted to customer assistance for the machinery maintenance and aftersales services. SCM Group is an industrial group which is leader in the creation, production and distribution of technologically advanced solutions to process a vast range of materials (wood, glass, stone, plastic, metals and composite materials) with specialist brands for specific technologies and centers of excellence expert in industrial components, present in all 5 continents for almost 70 years.

Role in the project and in the WPs

SCM España will work in close collaboration with beneficiary SCM Group, taking specific care of the following tasks:

- In WP6 (Industial Demonstration), supporting ROYO in the deployment of Furniture manufacturing demo

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Pier Alpañez (Male). Managing Director. Managing Director, CEO. Spanish, Italian, French. Master in marketing and sales management. He has worked at SCM SPain (previously NOVOREX IBERICA) since 1989. Previously, freelance.

Montserrat Ventura (Female). Sales Support, International Trade Specialist. Spanish, Italian. She has worked at SCM Spain (NOVOREX) since 1999. Previously in IQV, Expeditions Manager for 7 years.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

Not applicable

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

None

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Our headquarters in Sant Cugat del Vallés, Barcelona has offices of 200 SQ.M. approx, and a warehouse of approx 1000 SQ.M. The workforce is made up of 33 people, of which 15 based at our headquarters, the rest are dispersed across the country. In the offices, we have a local data server for greater privacy, as well as a 1GB local network; 2 WIFI networks; Windows Server, VMware virtualization software, ERP, CRM, IService for remote management and technical support, CAD for technical drawings. In the warehouse, we have brand new SCM machines on display worth approximately € 500k and € 300k in spare parts in order to provide the best possible service to our domestic customers.

6. Resources to be committed

SCM España will allocate 4 PMs in WP6 and 2,000€ in 'other direct costs':

SCM España	Cost (€)	Justification
Travel	2,000€	Attending to project meetings 2,000€
Total	**2,000€**	

Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	**N**

07 – ROYO

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	**N**
Does the participant envisage that part of its work is performed by linked third parties	**Y**

Acronym of the project: MULTIPLE		
Participant Nº: [Linked third party of n.] 7	**Mobiliario ROYO**	Royo

1. Description of the legal entity and its main tasks, with an explanation of how its profile matches the tasks in the proposal

Company profile of the group

Royo Group International Bathroom (RGIB) is one of the leading European companies in the field of bathroom furniture and shower. RGIB results from the union of two leaders who teamed up to win in this equation: they join the trajectory and experience of Royo Group and the financial strength that, as a platform for growth and consolidation, is provided by HIG.

Company profile of the unit involved in the project

Mobiliario Royo is the Parent Company of Royo Group, which, among others, includes Royo Spain SL.

Mobiliario Royo provides balanced resources to ensure the expected development of all its affiliated companies by offering complete corporate services that include: Strategic Plans definition and execution; Financial services (including accounting, planning, analysis, taxes, treasury and financing); Procurement with a base of common suppliers; Research and Innovation in materials, procedures and designs; IT infrastructure and tools; Communication and Brand Equity build up; etc.

Role in the project

The tasks of Mobiliario ROYO in the project will focus on the validation of MULTIPLE's multimodal solution performance to the production processes performed under the Royo Group Internationl Bathroom associated companies.. Mobiliario Royo will help to define the overall requisements and specifications regarding RGIB processes and catalogue of products, aiming to widening the MULTIPLE projects results not only to Royo Spain but the whole Group. Mobiliario Royo will assist to Royo Spain in defining the production needs (WP5), the KPIs to be achieved and to assess on the integration of MULTIPLE results onto Royo Spain Management Systems (WP6).

2. Curriculum vitae or description of the profile of the persons, including their gender, who will be primarily responsible for carrying out the proposed research and/or innovation activities]

Key personnel

César Taboas (male), Chief Technology and Innovation officer

Mechanical engineering in the Polytechnic University of Valencia in 1996. He did a master's in quality management in 1997 and a MBA in 2006. He also coursed the Executive development program from IESE. He has worked as a process engineer at Ford for two years and later he became production manager in ROYO Group for ten years. Nowadays he is the Chief Technology and Innovation officer at Mobiliario Royo (group company providing central and common services to all group companies, i.e. knowledge management and exploitation) . During his professional experience, he has coordinated various national (CDTI) and H2020

projects that have been developed in ROYO Group.

3. List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

#	Product/services…	Objective	Role of your organization
1	**Royo Click System**	First bath furniture with 15-minutes no-tool assembly.	Owner

4. List of up to 5 relevant previous projects or activities, connected to the subject of this proposal

#	Project/Activities	Type	Objective	Role of your organization
1	**TELLME**	EU- FP7	Development and improving learning and training in smaller manufacturing environments with the use of technologies. The aim is to bring innovative training methods to SME manufacturers so they can better supply the needs of their markets, which include larger industries.	Subcontractor
2	**NEW LINE OF POLYLAMINATE COATING**	CDTI	New line of polylaminate coating	Partner
3	**OPTIMIZATION OF THE PRODUCTION PROCESS OF BATHROOM FURNITURE THROUGH SENSORIZATION**	CDTI	Optimization of the production process of bathroom furniture through sensorization	Partner
4	**SAIN4**	IVACE and FEDER program	The objective of the project is to increase the efficiency of the main manufacturing processes, through the construction of an Advanced Management System that will support in the decision making for continuous improvement of availability, performance and quality	Partner
5	**HUMAN**	H2020-	Define and demonstrate workplaces where automation and human workers operate in harmony to improve the productivity, quality, performance of the factory as well as the worker satisfaction	Partner-Development of tools demonstration in furniture manufacturing.

			and safety.

5. Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Mobiliario ROYO brings the IPR and knowledge of the RGIB to be used in the MULTIPLE project.

6. Resources to be committed

Mobiliario ROYO will allocate 9 PMs distributed in WP1(3), WP5(6) and WP6(6), and 6,000€ in 'other direct costs':

MOB. ROYO	Cost (€)	Justification
Travel	6,000€	**WP6**: Attending to project technical meetings (1,000€); **WP8**: Attending to project meetings (5,000€)
Total	**6,000€**	

Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	**N**

08 – PHF

No third parties involved.

09 – RUSSULA

No third parties involved.

10 – LDL

No third parties involved.

11 – EPIC

No third parties involved.

12 – TEMATYS

No third parties involved.

13 – JOTIS

No third parties involved.

14 – SENORICS

No third parties involved.

15 – MRA

No third parties involved.

16 – TiO2

No third parties involved.

5 Ethics and Security

5.1 Ethics

Human Embryos/Foetuses

Not applicable.

Humans

Not applicable.

Human Cells/Tissues

Not applicable.

Protection of Personal Data

Not applicable.

Animals

Not applicable.

Non-EU countries

Not applicable.

Environment Protection

Not applicable.

Dual Use

The research conducted in MULTIPLE does not have any potential for military applications.

Misuse

The research conducted in MULTIPLE does not have the potential for malevolent/criminal/terrorist abuse.

Other Ethics Issues

There are no other ethics issues currently identified beyond those discussed above

5.2 Security

Please indicate if your project will involve:

• Activities or results raising security issues: NO

• 'EU-classified information' as background or results: NO